



2021 K-Innovation ODA Program with Indonesia

Policy Consultation on the Arrangement of Science, Technology and Innovation (STI) Governance: S&T Think Tank, R&D Program Evaluation, and S&T Laws and Regulations

2021 K-Innovation Partnership Program with Indonesia

Title	Policy Consultation on the Arrangement Of Science, Technology and Innovation(STI) Governance: S&T Think Tank, R&D Program Evaluation, and S&T Laws and Regulations
Institute	Science and Technology Policy Institute (STEP I)
Partner country	Republic of Indonesia
In cooperation with	National Research and Innovation Agency (BRIN)
Program investigators	Ji Hyun Kim, Head, Office of Development Cooperation Research, STEP I Wangdong Kim, Senior Research Fellow, Office of Development Cooperation Research, STEP I
Project manager	Deok Soon Yim, Senior Research Fellow, Office of Development Cooperation Research, STEP I
Researchers	Eun Joo Kim, Principal Researcher, Division of Global Innovation Strategy, STEP I Shinae Kang, Researcher, Division of Global Innovation Strategy, STEP I Hyanghee Lee, Senior Researcher, Division of Global Innovation Strategy, STEP I Seong-soo Kim, Professor of Hankuk University of Foreign Studies (HUFS) Young Joo Ko, President, Daejeon Institute of Science and Technology for Enterprise and People (DISTEP) Ki-Seok Kwon, Professor, Hanbat National University Dudi Hidayat, Director, Directorate for Research, Technology & Innovation Policy Formulation(DRTIPF), BRIN Changyul Lee, Researcher, Hankuk University of Foreign Studies (HUFS) Jung-Hun Kim, Researcher, IBS Consulting Company Anugerah Yuka Asmara, Researcher, DRTIPF, BRIN Aditya Wisnu Pradana, Researcher, DRTIPF, BRIN Fithriana Munawaroh, Policy analyst, DRTIPF, BRIN Siti Annisa Silvia Rosa, Policy analyst, DRTIPF, BRIN Kusnandar, Researcher, DRTIPF, BRIN Ria Hardiyati, Researcher, DRTIPF, BRIN Karlina Sari, Researcher, DRTIPF, BRIN Galuh Syahbana Indraprahasta, Researcher, DRTIPF, BRIN Sofia Yuniar Sani, Policy analyst, DRTIPF, BRIN Dadang Ramdhan, Researcher, DRTIPF, BRIN Anggini Dinaseviani, Researcher, DRTIPF, BRIN Maulana Akbar, Researcher, DRTIPF, BRIN Sri Supadmi, Policy analyst, DRTIPF, BRIN
Project Coordinator	Eun Joo Kim, Principal Researcher, Division of Global Innovation Strategy, STEP I

Policy Research 2021-38-03

Policy Consultation on the Arrangement of Science, Technology and Innovation (STI) Governance: S&T Think Tank, R&D Program Evaluation, and S&T Laws and Regulations

Published on 31 December 2021

Copyright © 2021 by Science and Technology Policy Institute (STEP I). All Rights Reserved.

5th~7th Floors, Building B, Sejong National Research Complex, 370 Sicheong-daero, Sejong-si,
30147 Republic of Korea

Tel.: +82-44-287-2000 Fax: +82-44-287-2068

www.stepi.re.kr

For more information, visit <https://iicc.stepi.re.kr>

Printed by Mirae Media

Tel.: +82-2-815-0407 Fax: +82-2-822-1269

ISBN 978-89-6112-809-4 93300



Executive Summary

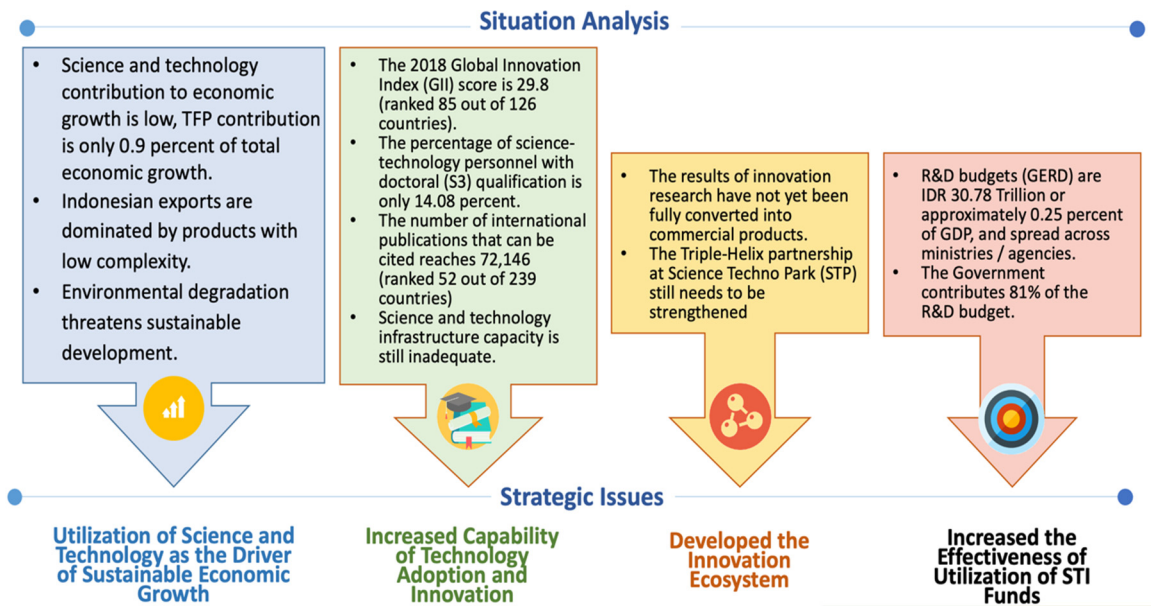
1. Background

In the advent of the Fourth Industrial Revolution, Indonesia has been restructuring and strengthening its National Innovation System (NIS). However, since the Indonesian NIS was set-up decades ago and there are many stakeholders, more concerted efforts have to be made in order to have an effectively working NIS in order to improve Indonesia's global competitiveness. The latest assessment of Indonesia's condition of science & technology (S&T) development has been formally incorporated into its National Mid-Term Planning 2020-2024. In 2021, the Indonesian government made a big change in its National Innovation System by making BRIN (National Research & Innovation Agency) the mega-integrated national research institutes which has incorporated various research institutes.

According to the findings of the situational analysis, there are four significant concerns in the area of S&T development that need to be addressed in Indonesia's NIS:

- How can the utilization of science and technology as a driver of sustainable economic growth be increased?
- How can the capability for technology adoption and innovation be increased?
- How can the innovation ecosystem be developed?
- How can the effectiveness of the Science, Technology, & Innovation (STI) Fund be increased?

[Figure 1] Situational Analysis of S&T Policy in Indonesia



Source: Bappenas (2019), p. 5.

Considering the mentioned problems and development objective of the Indonesian STI system, it is very timely to address the following important questions:

- How a think tank can promote STI development, what type of STI think tank to establish, and how to operate it?
- How can good governance among STI related stakeholders be set-up to have an effective and sustainable STI system in Indonesia?
- How can national research & innovation programs (R&D programs) be effectively planned and managed; what are the structure and mechanisms to do it, including the actors and process of national R&D programs?
- What kinds of STI law and regulation are necessary? And which contents have to be inserted?

2. 2021 K-Innovation with Indonesia

The 2021 K-Innovation ODA Program with Indonesia has three objectives. The first objective is to support Indonesian policy-makers and researchers in implementing policy recommendations on the establishment of a new design of an STI think tank in Indonesia, to design an evaluation mechanism for national research programs (PRN), and restructure S&T regulations in Indonesia. The second objective is to increase policy research capacity building through collaboration research between Korean experts and Indonesian researchers & workshops. The fourth objective is to strengthen STI cooperation between Korea and Indonesia as part of Korea's New Southern Policy.

The 2021 K-Innovation ODA Program consists of four activities as follows: The kick-off seminar, joint research to produce a policy consulting report, and capacity-building & dissemination workshops (in a webinar format due to COVID-19 restrictions). The first phase is to hold a kick-off seminar between the STEPI team and BRIN (formerly Indonesian Institute of Sciences, LIPI) to discuss the details of activities for 2021 and evaluate last year's activities. At the kick-off seminar, Korean experts give presentations on Korean practices based on the themes in cooperation with Indonesian experts from BRIN.

The Korean experts collaborate with Indonesian researchers for joint research on three themes of building a new STI think tank in Indonesia; evaluation mechanism for national research program; and restructuring S&T regulations in Indonesia. Three teams are supervised by Korean experts assigning focal points for Indonesian researchers. The capacity-building program for Indonesian researchers in Korea is an important part of K-Innovation. Regardless of the COVID-19 restrictions hindering flexibility to organize a capacity-building program for Indonesian researchers in Korea, K-Innovation warrants that Korean specialists share Korean methods and exchange ideas on the topics of their responsibility via videoconferencing. The last activity is a dissemination seminar by which the research teams can present and receive feedback on the policy consulting outcomes from the Indonesia's STI community. In addition to the dissemination seminar, Korean and Indonesian researchers work to publish academic papers based on the outcomes of 2021 K-Innovation.

3. Outcomes of Policy Consulting

The 2021 K-Innovation with Indonesia focused on consulting to address three key issues for Indonesia's National Innovation System (NIS):

- Building an STI think tank in Indonesia
- Designing an evaluation mechanism for national research programs (PRN) 2020-2024
- Restructuring law and regulations on S&T in Indonesia

To address the concerns above, the research teams of Korea and Indonesia conducted joint researches with four major activities—such as conceptualization, focused group discussions (FGD), in-depth interviews, and workshops—to disseminate the outcomes of research with the stakeholders from both countries.

3.1 Policy Recommendations for Building a New Think Tank for Science & Technology Innovation in Indonesia

The establishment of a think tank in Indonesia should take into account existing legislation, norms, and institutions in order to achieve feasible STI policy governance. Nonetheless, establishing a new think tank unit focused on STI policy in the global arena would be difficult, and may even conflict with other legislation or agency goals. Indeed, the intent of top management, as well as political will, has a role in the establishment of an STI policy think tank unit. This research suggests four different STI policy think tank units that could be established and built in Indonesia.

[Table 1] Suggested STI Think Tank Governance in Indonesia

Option	Funding source	Governance	Status of the members	Functions
1	Government budget (Ministry of Research & Technology/ BRIN)	Government-affiliated	Government officers	Planning, Implementing, Evaluating
2	Budget of non-government organizations	Autonomous & independent body	Private sector	Planning, Implementing, Evaluating
3	Budget of government agencies & donors	Quasi-independent	Private sector	Planning, Implementing, Evaluating
4	Government budget under the Ministry of Finance (LPDP scheme)	Quasi-state-run	Contracted civil service officers	Planning, Implementing, Evaluating

Source: Developed by author

The ideal condition cannot be reached due to the current obstacles in Indonesia. Before one of the options can be implemented, there are important issues that must be addressed. Firstly, there is a need to reform the rigidity to enable a more flexible bureaucracy apparatus. A less flexible system may negatively impact finance, law, and administrative procedures, for instance. Managing funds from both government and other sources requires creating a new form of think tank such as quasi-independent and independent think tanks in Indonesia. Secondly, a favorable coordination system should be structured among state-run agencies. In addition, an STI policy think tank needs coordination with a wide range of stakeholders. Thirdly, Indonesia needs to increase the number and pool of technical human resources in STI policy. The limited number of researchers performing functions in STI policy studies and lack of data on STI also result in a shortage of technical human resources in STI policy. Fourthly, evidence-based policymaking on STI should be strengthened. In most cases, policymaking without evidence tends to lead to trial judgment & errors. Fifthly, it is necessary to develop an enabling environment for promoting innovation. In Indonesia's case, state-run bodies should have the same frequency for synergy to create an innovation system. The Ministry of Research & Technology (MoRT)/ National Research & Innovation Agency (BRIN) can be the main actor for constructing NIS among the institutions.

3.2 Policy Recommendations for Management & Evaluation of R&D Programs

The new Law No. 11 Year 2019 has mandated the integration of all public R&D institutions in Indonesia under the auspices of the Ministry of Research & Technology/ BRIN to ensure the sustainability of one-way R&D activities in order to increase R&D capacity in national development. Nonetheless, the definition of ‘integration’ under this rule has sparked some debate over how it should be accomplished. The STEPI-LIPI team came up with three key governance solutions for the Research Endowment Fund.

[Table 2] Governance Options for the Research Endowment Fund

Governance option	Option 1	Option 2	Option 3
Investment manager	Ministry of Finance	Ministry of Finance	Ministry of Research & Technology/ BRIN
R&D program manager	Ministry of Research & Technology/ BRIN	Ministry of Research & Technology/ BRIN	
Selection & review process		Independent Institution	
Opinion by author*	-	Most recommended	Least recommended

Source: developed by Author.

*This does not reflect or represent the opinion of other parties.

The Research Endowment Fund can be evaluated in two ways. Option 1: The Ministry of Research & Technology/ BRIN can carry out the review process. The Ministry of Research & Technology/ BRIN has the role of ensuring that the different programs and policies enacted may support R&D activities, leading to the fulfillment of national development goals, as an actor in charge of managing R&D matters under the government. The Ministry of Research & Technology/ BRIN has sufficient capacity and capability to undertake evaluations, which are backed up by skilled personnel. This alternative is based on the belief that evaluations conducted by an independent organization other than policy-makers can ensure a more objective evaluation process. This assessment will be used by the Ministry of Research & Technology/ BRIN to plan the next program. In

such an instance, the Academy of Indonesian Science (AIPI) may be entrusted with evaluating the research endowment fund's programs and projects. This option would be preferable to the preceding one.

3.3 Policy Recommendations for the Management & Evaluation System of the National R&D Program

The Indonesian government has a long- and mid-term R&D policy such as the National Research Master Plan (RIRN) (2017~2045), which is the Master Plan on the national R&D program and National Research Priority (PRN) to be updated every five years. This implies that Indonesia has infrastructure in place for the formation, implementation, and evaluation of an R&D program and a strong political will to implement its R&D policy. Even so, it appears that Indonesia remains at the early stages of implementing national R&D programs, without a clear ex-ante and ex-post evaluation processes. In addition to this research, the investment target for R&D announced by the Indonesian government is not insufficient to sustain R&D programs, while financing sources for implementing R&D programs are not determined. The cooperation mechanism between industry and national research institutes is still ineffective. Due to their shortage of R&D capabilities, industries are not focused on innovation, and some state-owned enterprises in Indonesia are primarily responsible for R&D.

Policy recommendations for Indonesia include as follows: 1) R&D investment should be increased as long as there are clear indications of R&D funding sources; while programs should be developed to induce the private sector to make R&D investment; and 2) the governance for R&D policy and programs could be restructured. By doing so, ex-ante and ex-post evaluation and implementation of R&D programs and projects can be efficiently carried out, and cooperation and coordination among research actors (national research institutes & universities) can be effectively facilitated.

3.4 Laws & Regulations of Science & Technology

For starters, STI governance and law in Indonesia is in the process of going through massive

changes but there are also difficulties due to the large-scale changes.

- Law 11-year 2019 is a law with a significant meaning to reorganize Indonesia's science & technology governance.
- The legislation's primary purpose is to integrate public research institutes under the newly launched National Research & Innovation Agency (BRIN).

In this background, Korean STI laws were reviewed historically to see the reasons for the development of each law. Especially those laws closely related to the formation of S&T governance were reviewed. Those laws are related to 1) policy/ budget coordination, 2) the governance structure of public research institutes, 3) policy instruments, especially on national R&D programs, and 4) establishment of professional R&D management agencies.

This study introduced the structure and function of BRIN, which was launched in 2021 and translated its Indonesian name into Korean for the first time.

- In this process, four functional areas of BRIN were identified, which are as follows:
1) Policy functions related to the planning, budgeting, & evaluation of R&D programs, 2) management of the Research Endowment Fund, 3) balancing the principles of accountability and autonomy for integrated public research institutes, and 4) supervision of BRIN

The Indonesian government intends to innovate its science & technology system by Law No. 11-year 2019.

- The current change in Indonesia can be seen as a situation similar to when Korea reformed its structure of state-funded research institutes in 1999 and enacted the Framework Act on Science & Technology in 2001.
- It is recommended to benchmark Korean Acts which are the Science & Technology Promotion Act, the Science & Technology Framework Act, Act on the Establishment,

Operation, & Fostering of Government-Funded Research Institutes, and Industrial Technology Promotion Act.

4. Conclusion

In the advent of the Fourth Industrial Revolution, Indonesia has put its national priority on building STI capacity, aiming to achieve a knowledge-based economy in the mid and long term. The 2021 K-Innovation program contributes to Indonesia's STI development by providing policy consultation on the national innovation system of Indonesia, which is one of the key partner countries in Korea's diplomatic policies. Policy suggestions are drafted from K-Innovation's activities led by Korean experts with Indonesian researchers to improve the STI system in Indonesia. More specifically, Koreans shared insights and policy directions based on Korean and Indonesian practices regarding the STI think tank, STI law and regulation, and management and evaluation system of the national R&D program. STEPI has been supporting LIPI – which became part of BRIN over the last several years through the K-Innovation program. It is believed that the STI cooperation at the institutional level would contribute to strengthening cooperation at a national level as well.

CONTENTS

Chapter 1. Project Overview.....	3
1. Introduction	3
2. Objectives	9
3. Project Framework	10
4. Project Team.....	11
5. Project Activities	13
6. Project Schedule	25
7. Project Outputs	26

Chapter 2. Overview of Policy Consulting on the Arrangement of STI Governance: Institutional Framework of STI Policy Think Tank, Evaluation Mechanism & Implementation of Research National Program, & Restructuring S&T Regulations in Indonesia	29
1. Overview of Policy Consulting.....	29

Chapter 3. The Evolution of STI Think Tanks in South Korea & Implications for Indonesia & Current Status	41
1. Introduction	41
2. Theories and Framework	42
3. STI Policy System & Think Tanks in Korea: Structure & Process	49
4. Analysis of the Korean STI Think Tank	58
5. Definition of Think Tank in Indonesian Context	63
6. Overview of S&T Governance in Indonesia.....	69
7. Ideal STI Policy Think Tank & Practice in Indonesia	77

8. Ideal STI Policy Think Tank: Concept & Early Design	81
9. Proposed Designs for Indonesian STI Policy Think Tank.....	89

Chapter 4. Building Monitoring & Evaluation Systems for National Research & Development Programme in Indonesia..... 97

1. Introduction	97
2. Research Evaluation: Basic Concept & Principles	100
3. Methodology.....	106
4. Results	110
5. Conclusion	130
6. Follow-up Work	132

Chapter 5. Laws & Regulations of Science & Technology 137

1. Introduction	137
2. Typology of Laws of Science & Technology.....	139
3. Laws of Science & Technology in Korea	146
4. Laws of Science & Technology in Indonesia	159
5. Major Regulation Issues Related to Laws of Science & Technology in Korea	186
6. Summary and Implications.....	193

Chapter 6. Conclusion 211

1. Summary.....	211
2. Policy Recommendations.....	216

ABBREVIATIONS 221

REFERENCES..... 225

List of Tables

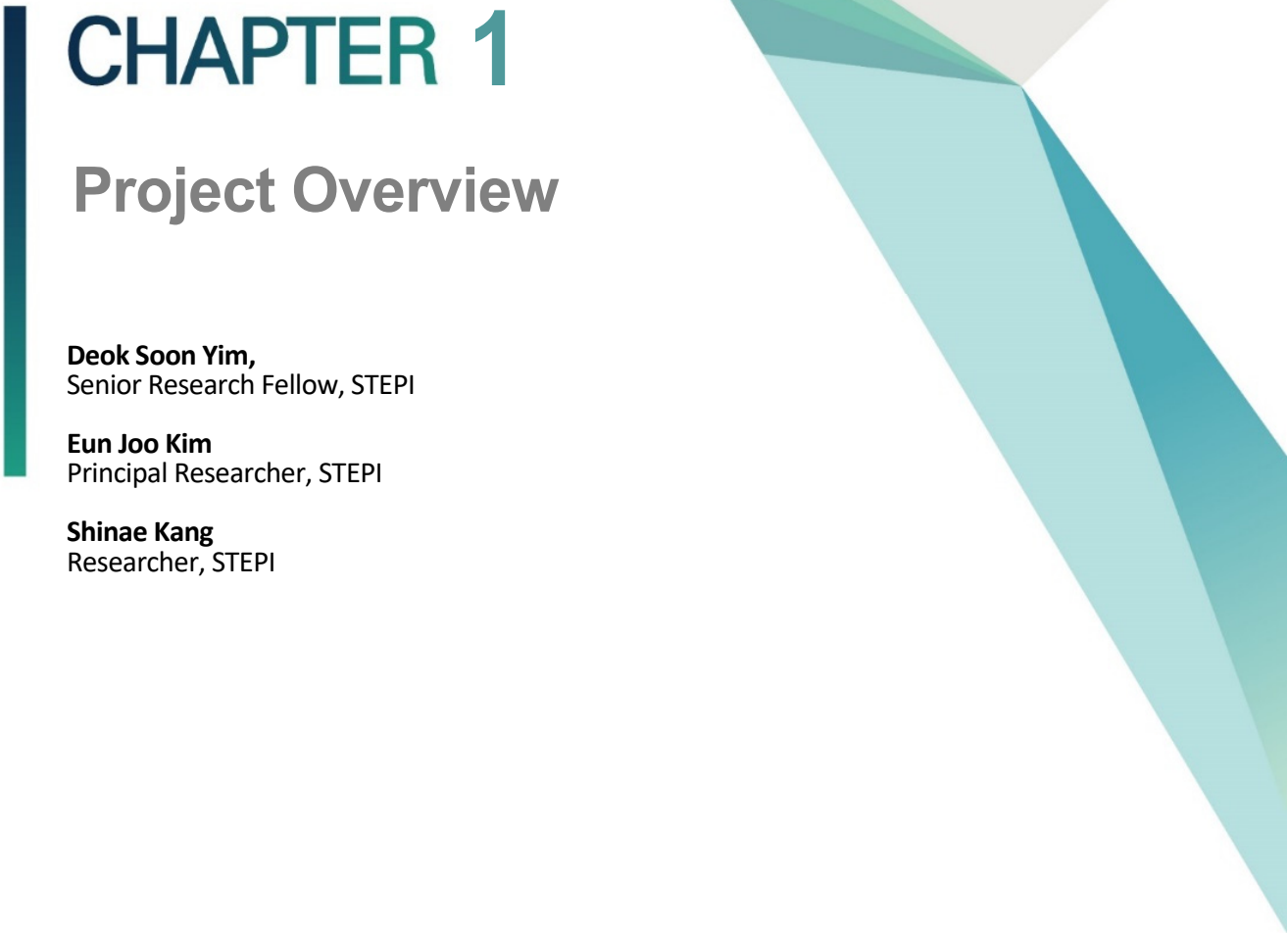
[Table 1] Suggested STI Think Tank Governance in Indonesia	v
[Table 2] Governance Options for the Research Endowment Fund	vi
[Table 1-1] STEPI's Program & Support for Indonesian Policy-makers & Researchers	3
[Table 1-2] Project Team of 2021 K-Innovation with Indonesia	12
[Table 1-3] Kick-Off Seminar Program of 2021 K-Innovation with Indonesia...	13
[Table 1-4] Online Training Program	17
[Table 1-5] Feedback Survey Result of the Online Training Workshop	20
[Table 1-6] Indonesian Participants of the Online Training Workshop.....	20
[Table 1-7] STEPI Special Session Program.....	22
[Table 1-8] Project Schedule	25
[Table 2-1] Categorization of Participants for FGDs	33
[Table 2-2] Timeline for Joint Research.....	36
[Table 3-1] Mapping STI Policy Process & Required Role of Think Tanks.....	61
[Table 3-2] Type of Researches Based on Institutional Arrangement: Case of BRIN-Indonesia	75
[Table 3-3] Characteristics of State-Run Institutions on Evidence-Based Policymaking	83
[Table 3-4] Characteristics of Public R&D Institutions Referring to Evidence-Based STI Policymaking Flowchart	87
[Table 3-5] STI Policy in Frame of Micro, Meso, and Macro-Policy	92
[Table 4-1] Five Annual Targets as Stated in the RIRN.....	98
[Table 4-2] Indicators of Research Monitoring & Evaluation for Variables of Input & Process	104
[Table 4-3] PRN 2020-2024 Focus & Examples of Product Development Target	116
[Table 4-4] Proposed Indicators for Master & Project Proposal Selections of PRN	126
[Table 4-5] Proposed Indicators for PRN Monitoring and Evaluation	129

[Table 4-6] Design of PRN 2020-2024	132
[Table 4-7] Examples of Designing Ongoing Research Program	133
[Table 5-1] Method & Standards for Analyses of Science & Technology Laws	143
[Table 5-2] History of S&T Laws in Korea	156
[Table 5-3] History of S&T Laws in Indonesia	170
[Table 5-4] Research Permit Related Sanctions According to Law No. 18-year 2002 & Law No. 11-year 2019.	184
[Table 5-5] Governance Elements in Korean S&T Laws	194
[Table 5-6] Structure of Law No.11-year 2019 (Related Korean S&T Laws).	197
[Table 5-7] Quasi-State-Run Sector in Korea	199

List of Figures

[Figure 1] Situational Analysis of S&T Policy in Indonesia	ii
[Figure 1-1] Situational Analysis of S&T Policy in Indonesia	7
[Figure 3-1] System Approach in Politics.....	42
[Figure 3-2] Ideal Function of Think Tanks in Political System	44
[Figure 3-3] R&D Governance System in South Korea	52
[Figure 3-4] Characteristics of STI Policy Process in Korea	57
[Figure 3-5] Mapping STI Policy Process & Think Tank	60
[Figure 3-6] STI Governance and Think Tank	62
[Figure 3-7] Governance of R&D Institution Before BRIN Era	70
[Figure 3-8] Draft of Organization Structure of BRIN	74
[Figure 3-9] Ideal STI Policy Think Tank Relationship before Integration within BRIN	78
[Figure 3-10] The Practice of STI Policy Think Tank Relationship before Integration within BRIN	79
[Figure 3-11] Ideal Flowchart of Evidence –Based Policymaking	81
[Figure 3-12] Elements of Evidence-Based STI Policymaking in Indonesia	85
[Figure 3-13] STI Policy Think Tank as Upstream & Downstream Policies	89
[Figure 3-14] STI Policy Think Tank as Micro, Meso, Macro-Policy	90
[Figure 3-15] STI Policy Think Tank as an Integrated Process	93
[Figure 4-1] Elements in Monitoring and Evaluation	101
[Figure 4-2] Steps in Designing Monitoring and Evaluation.....	102
[Figure 4-3] Framework for a Research Evaluation System	103
[Figure 4-4] Process of National R&D Budget Allocation & Coordination	111
[Figure 4-5] Process of Selecting National R&D Projects.....	113
[Figure 4-6] PRN Structure.....	116
[Figure 4-7] First Scheme of PRN Proposal Selection.....	117
[Figure 4-8] Second Scheme of PRN Proposal Selection	118
[Figure 4-9] Identified Problems in PRN Proposal Selection	119
[Figure 4-10] PRN's Current Mechanism of Internal & External M&E.....	120

[Figure 4-11] Identified Problems in PRN M&E	123
[Figure 4-12] Proposed Structure & Mechanism for PRN Proposal Selection	124
[Figure 4-13] Proposed Structure & Mechanism for Monitoring & Evaluation for PRN Research.....	128
[Figure 5-1] Three Dimensions of Laws of Science & Technology	145
[Figure 5-2] Source of R&D Funding in Indonesia (in IDR trillion)	179
[Figure 5-3] Structure of the National R&D Innovation Act (2020).....	187
[Figure 5-4] Structure and Function of BRIN	202

A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a large light grey triangle pointing downwards, and several overlapping triangles in various shades of teal and green, also pointing downwards.

CHAPTER 1

Project Overview

Deok Soon Yim,
Senior Research Fellow, STEPI

Eun Joo Kim
Principal Researcher, STEPI

Shinae Kang
Researcher, STEPI

Chapter 1. Project Overview

1. Introduction

1.1 STEPI's Support for Indonesia on STI Policy

Since 2014, STEPI has collaborated with Indonesia to provide policy advice on STI policy and governance, as well as support capacity building for Indonesian government officials and researchers from LIPI and BPTT who are in charge of STI policymaking. STEPI's K-Innovation initiative began as a capacity-building program for Indonesian researchers and policy-makers who wanted to improve their research and STI policymaking capabilities. However, as of 2018, the K-Innovation ODA Program's policy consulting includes evaluating the 2015-2019 Research Strategy Planning and developing the National Research Priority 2020-2024.

[Table 1-1] STEPI's Program & Support for Indonesian Policy-makers & Researchers

Year	Target	Activities
2014	STI policy-related government officials & researchers from Asian countries including Indonesia	- Operated the capacity-building program on STI policy
2015	Ministry of Research, Technology & Higher Education (RISTEK-DIKTI), Agency for the Assessment & Application of Technology (BPPT), UNESCO office in Jakarta	- Operated the capacity-building program for Asian researchers including Indonesia, Malaysia, and Sri Lanka, identifying policy consulting themes
	RISTEK- DIKTI and BPPT	- Operated the capacity-building program on Korea's policy for STI human resources, educational system, and education based on demand analysis of education, development, & evaluation

Year	Target	Activities
2016	BPPT	- Operated the capacity-building program to increase STI capacity for Indonesian researchers and enhance the competency of officials as part of reforming bureaucracy
	BPPT	- Operated the capacity-building program to increase the capacity of developing and managing human resources on planning, design, & education, monitoring and evaluation of the education and training program
2018	LIPI	- Operated the capacity-building program and consulted on the R&D strategy plan, Science & Technology Park
2019	LIPI	- Operated the capacity-building program and consulted on the evaluation of STI policy and R&D governance

STEPI developed the two-year program, 'Policy Consulting on the Evaluation of LIPI R&D Strategic Planning 2015-2019 & Development of LIPI R&D Strategic Planning 2020-2025 and Science & Technology Park' through a joint workshop with LIPI in Jakarta in December 2017.

- The 2018 K-Innovation ODA Program provided policy consulting on the 2020-2025 Research & Development Strategy, technopark operation and maintenance, STI human resource management, and research capacity strengthening.
- The K-Innovation ODA Program in 2019 focused on STI policy review and R&D governance consulting in Indonesia. With the establishment of the new Ministry of Research & Technology and National Research & Innovation Agency (BRIN), STI governance has emerged as a major issue in the Indonesian research community.

LIPI requested STEPI to share Korean experiences and provide policy counseling on the management of the research endowment fund and national research & development program in the face of such changes in STI governance. The arrangement of STI governance: Coordination mechanism, institutional framework, financial system, and investment in Indonesia were formulated.

1.2 Status of Indonesia's National Innovation System

Since the late 1970s, Indonesia's government has been concerned with the development of science and technology (S&T), which has been demonstrated by the rapid expansion of scientific infrastructure in Indonesia. In 1978, the Science & Technology Park in Serpong, a Jakarta satellite city, was completed. The government's dedication to technology use for development was reflected in the creation of the Agency for Assessment & Application of Technology (BPPT), while the aircraft industry was established as a serious attempt to master high technology. To promote the growth of an S&T workforce, the government initiated a program in 1984 to send a large number of students studying science and engineering degrees overseas. This was the first systematic and long-term effort to expand the supply of high-quality S&T workforces.

Since the late 1970s, S&T policy has mostly focused on S&T infrastructure development, and the link between S&T and the economy has not been officially declared as a policy goal. When a World Bank staff report on the Indonesian science & technology system found a severe gap between publicly supported research & development organizations' research and the demands of the Indonesian economy in 1995, policy orientation began to shift. To solve this gap, the government established the Industrial Technology Development Project, which attempted to improve the management of a number of government research institutes, so that they could provide services to industries. Since then, the government has devised and executed a number of policies and programs aimed at bridging the gap between government research institutes, universities and the private sector. In this regard, the extensive application of the NIS framework in innovation policy coincides with this new policy perspective.

Despite the fact that the NIS framework was widely employed in innovation policy in many countries during the 1990s, it was seldom formally discussed and used as a policy framework in Indonesian policymaking circles until 2008. This was the first systematic and sustained program to expand the supply of high-quality S&T workforces. Similarly, rather than using the term 'National Innovation System', the S&T law (2002) to promote linkages

between government research institutes, universities, and industries was named the 'National System of Research, Development, and Application of Science and Technology'. The name is still centered on science and technology.

The NIS framework was only officially endorsed in the scientific and technology community in 2008 at the National S&T Coordinating Meeting hosted by the Ministry of Research & Technology (MoRT). However, the context is S&T policy, about which S&T officials must continue to explain the concept to policy-makers in other fields, such as industrial and education policies. In 2008, the S&T community's communication efforts became more concentrated. However, the lack of a national coordination organization for innovation policy has made communication and cooperation difficult, while the NIS paradigm has gradually faded from Indonesian STI Policy debates in recent years. The new Law No. 11, which superseded the 2002 S&T law and is titled the Law of Science & Technology System, scarcely mentions the national innovation system.

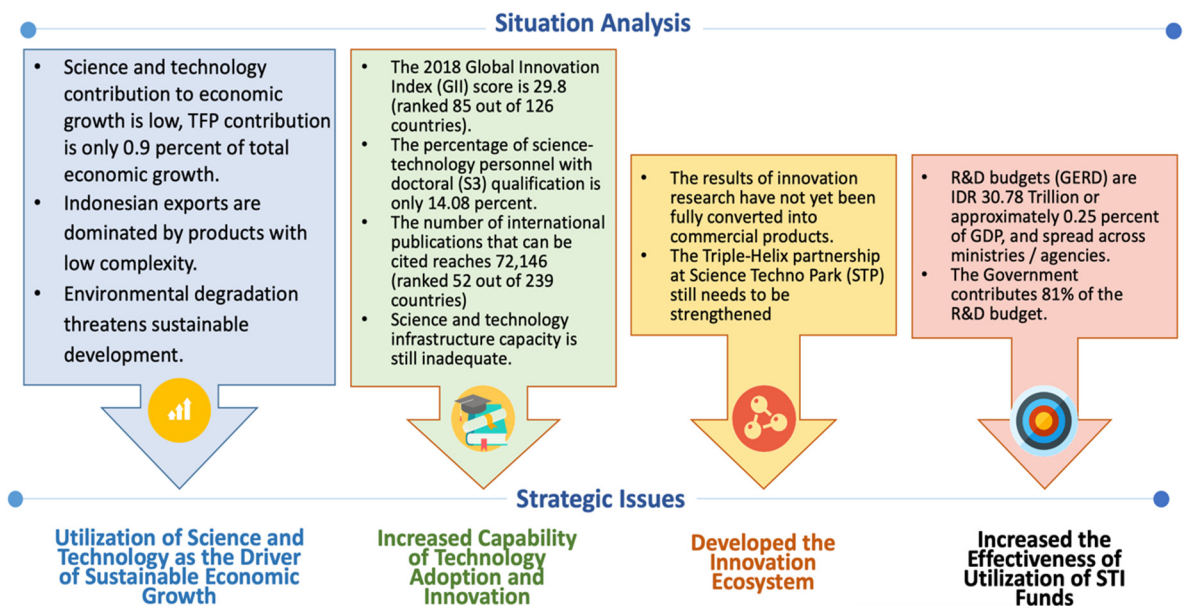
The elements of NIS are available and functioning in Indonesia, but they are not yet fully interconnected. Therefore, the country's economic competitiveness has not been maximized, and the interconnection and interrelationship between the elements of NIS were hampered by policy inconsistency. R&D, industry, education, and finance policies are mostly based on sectoral approaches. The interconnection issues among the pieces of Indonesia's NIS are not easy. There are differences in how people perceive technological advancement and how interested they are in it. The success of an innovation system is determined not only by the performance of individual players but also by how they interact with one another. This shows that Indonesia's innovation system requires concentrated efforts, such as requiring a powerful think tank capable of formulating and promoting a strategic approach to the development of an innovation system. Meanwhile, if the government wants to increase ties between government research institutes and business, and foster innovation system growth, it needs to overhaul its R&D funding process. Since researchers and research institutes cannot 'make a profit' from their research contracts, the current funding process is exceedingly inflexible, operating as an obstacle to them. Research institutes should be given more autonomy in managing their research funding. The line item budgeting system, in particular, needs to be replaced by a performance-based budgeting

approach that allows research institutes more autonomy and flexibility in managing research funding.

In conclusion, Indonesia has made a concerted effort to strengthen its NIS. Nevertheless, more determined efforts are needed to increase Indonesia's global competitiveness by having a better functioning Indonesian NIS. The latest assessment of Indonesia's status of science and technology development has been officially incorporated into its National Mid-Term Planning 2020-2014. According to the scenario analysis, there are four significant concerns in the area of S&T development that must be addressed as follows:

- How can the utilization of science and technology as a driver of sustainable economic growth be increased?
- How can the capability for technology adoption and innovation be increased?
- How can the innovation ecosystem be developed?
- How can the effectiveness of the Science, Technology, & Innovation (STI) Fund be increased?

[Figure 1-1] Situational Analysis of S&T Policy in Indonesia



Source: BAPPENAS (2019), p. 5.

Given the aforementioned concerns and the Indonesian STI system's development goals, it is critical to address three key issues as follows:

- How can a think tank promote STI development, what type of STI think tank should be founded, and how should it be run?
- How should a research trust fund be handled as part of funding mechanism reform, and what is the optimal institutional arrangement in an Indonesian-specific context?
- What is the structure and processes for effectively planning and managing the national research & innovation program, who are the actors, and what is the business process?

Three issues of STEPI 2021 are as follows:

- Improving the legal framework and institutional arrangements of mission-oriented organizations such as think tanks in Indonesia
- Designing an evaluation mechanism for national research programs (PRN) 2020-2024
- Restructuring regulations on S&T in Indonesia

2. Objectives

The 2021 K-Innovation ODA Program with Indonesia has three main objectives as follows. The first is to support Indonesian policy-makers and researchers in implementing policy recommendations on the establishment of a new STI think tank in Indonesia, designing an evaluation mechanism for PRN 2020-2024, implementing the first year of PRN 2020-2024, and restructuring STI related laws and regulations in Indonesia. The second one is to enhance STI policy research capability through diverse activities such as joint research, and training & dissemination workshops. The last objective is to lay a firm foundation for STI cooperation between Korea and Indonesia.

3. Project Framework

The 2021 K-Innovation ODA Program consists of four activities as follows: The kick-off seminar, joint research to produce a policy consulting report, and capacity-building & dissemination workshops. The meetings and workshops were changed to a webinar format due to COVID-19 restrictions. The first phase is to hold a kick-off seminar between the STEPI team and LIPI to discuss the details of activities and evaluate last year's activities. At the kick-off seminar, Korean experts give presentations on Korean practices and cases based on the themes in cooperation with Indonesian experts and government officials from the National Research & Innovation Agency (BRIN) and Ministry of National Development Planning (Bappenas).

Korean experts collaborate with Indonesian researchers for joint research on three themes as follows: 1) Improving the legal framework and institutional arrangements of mission-oriented organizations such as think tanks in Indonesia, 2) Designing an evaluation mechanism for PRN 2020-2024 and 3) Restructuring regulations on S&T in Indonesia. Three teams supervised by the Korean experts assign focal points for the three groups of Indonesian researchers. Regardless of COVID-19 restrictions hindering organizing vis-à-vis meetings and workshops since 2020, K-Innovation warrants that three research teams utilize videoconferencing instead to share Indonesian situations and Korean practices, and exchange their policy ideas and suggestions. The last K-Innovation activity is to hold a dissemination workshop by which the research teams can present and receive feedback on the results of their joint researches from the STI research community of Indonesia and Korea. After the dissemination workshop, Korean and Indonesian researchers will work to publish academic papers based on their joint research results.

4. Project Team

More than 20 researchers from Korea and Indonesia participated in the implementation of the 2021 K-Innovation Project. Dr. Deok Soon Yim of STEPI took the lead on this project and Dr. Eun Joo Kim acted as project coordinator of the 2021 K-Innovation Project with the assistance of STEPI researchers, Ms. Shinae Kang and Ms. Hyanghee Lee. The Korean team invited three Korean experts – Prof. Seong-soo Kim (Hankuk University of Foreign Studies), Dr. Young Joo Ko (President, Daejeon Institute of Science & Technology for Enterprise & People), and Prof. Ki-Seok Kwon (Hanbat National University) – all of whom possess a great deal of expertise in STI policy, for policy consulting on improving the legal framework and institutional arrangements of mission-oriented organizations such as think tanks in Indonesia, designing an evaluation mechanism for PRN 2020-2024, conducting the implementation of the first year of PRN 2020-2024, and restructuring regulations on S&T in Indonesia. With an in-depth understanding of the Indonesian STI environment and situation, the Korean experts have facilitated collaborations with BRIN as they have a good understanding of the STI governance and policy in Indonesia. Korean experts headed the Indonesian research teams, each of which assigned a focal point for better communication and coordination.

Dr. Dudi Hidayat served as the Indonesian counterpart, and Mr. Anugerah Yuka Asmara assumed the role as coordinator of the 2021 K-Innovation on top of heading the Indonesian team on the research of an STI think tank. Dr. Kusnandar served as focal point for the research on management and evaluation of the National Research Program. For the research on the STI law and institution, Mr. Dadang Ramdhan acted as focal point under the guidance of a Korean team leader. There have been 3 to 5 core Indonesian researchers involved in this project from the National Research & Innovation Agency (BRIN) or previous: Indonesian Institute of Sciences (LIPI) from various backgrounds (public administration, economy, engineering, geography, etc.)

Korean advisers headed the research teams, each of which assigned a focal point for better communication and coordination among Indonesian researchers.

[Table 1-2] Project Team of 2021 K-Innovation with Indonesia

Name	Position/ organization	Roles
Dr. Deok Soon Yim	Senior Research Fellow, STEPI	Project manager
Dr. Eun Joo Kim	Principal Researcher, STEPI	Project coordinator
Ms. Shinae Kang	Researcher, STEPI	Participant
Ms. Hyanghee Lee	Researcher, STEPI	Participant
Dr. Dudi Hydayat	Director, Directorate for Research, Technology & Innovation Policy Formulation(DRTIPF), BRIN (Ex PPKMI- LIPI)	Counterpart
Team 1. Think Tank for Science-Technology-Innovation (STI) Policy		
Prof. Ki-Seok Kwon	Professor, Hanbat National University	Team leader
Anugerah Yuka Asmara	Researcher, BRIN (Ex PPKMI- LIPI)	Joint research coordinator & focal point for Indonesian team
Aditya Wisnu Pradana	Researcher, BRIN (Ex PPKMI- LIPI)	Participant
Fithriana Munawaroh	Policy analyst, BRIN (Ex PPKMI- LIPI)	Participant
Siti Annisa Silvia Rosa	Policy analyst, BRIN (Ex PPKMI- LIPI)	Participant
Team 2. National Research & Development Program		
Dr. Youngjoo Ko	President, Daejeon Institute of Science & Technology for Enterprise & People (DISTEP)	Team leader
Dr. Kusnandar	Researcher, BRIN (Ex PPKMI- LIPI)	Focal point for Indonesian team
Karlina Sari	Researcher, BRIN (Ex PPKMI- LIPI)	Participant
Dr. Galuh Syahbana Indraprahasta	Researcher, BRIN (Ex PPKMI- LIPI)	Participant
Ria Hardiyati	Researcher, BRIN, (Ex PPKMI- LIPI)	Participant
Team 3. Science & Technology Law and Institution		
Prof. Seong-soo Kim	Professor, Hankuk University of Foreign Studies	Team leader
Dadang Ramdhan	Researcher, BRIN, (Ex PPKMI- LIPI)	Focal point for Indonesian team 3
Maulana Akbar	Researcher, BRIN, (Ex PPKMI- LIPI)	Participant
Anggini Dinaseviani	Researcher, BRIN, (Ex PPKMI- LIPI)	Participant
Sri Supadmi	Policy analyst, BRIN, (Ex PPKMI- LIPI)	Participant

5. Project Activities

5.1 Activity 1: Project Preparation Meetings

The Korean team has been communicating with their Indonesian counterparts to elaborate on project activities. In addition to detailing the 2021 K-Innovation, the composition of research teams for joint research and joint research itself were also determined for mutual understanding between STEPI and BRIN. BRIN played an active role in planning joint researches for the three themes to be dealt in 2021 K-Innovation by presenting the research background and current situations of Indonesia.

5.2 Activity 2: Kick-Off Seminar

The 2021 K-Innovation Partnership Program Kick-off Meeting was held in April 15, 2021 between the two parties. For the STEPI side, the meeting was conducted at the Amber Room, Millennium Hilton Seoul at 11.00 EST, while for the LIPI side, the meeting was conducted through Zoom Meetings (<https://us02web.zoom.us/j/88368696253?pwd=QkZUWDIVVk1vUHFCbEVxbGZMRkZhZz09>, ID: 883 6869 6253, Password: 791647).

5.2.1 Invited Participants

Aside from the internal participants within BRIN, government officials and experts from BAPPENAS, RISTEK, related Universities, NGOs, and media outlets joined this kick-off seminar, attended by a total of 82 participants.

[Table 1-3] Kick-Off Seminar Program of 2021 K-Innovation with Indonesia

11:00-11:05	Opening Remarks	Dr. Mi Ock Mun, President, STEPI	Pre- recorded
11:05-11:15	Welcoming Remarks	Dr. Dudi Hidayat, Acting Deputy Chairman, LIPI	Pre- recorded

11:15-11:20	Congratulatory Remarks	Dr. Mego Pinandito, Executive Secretary, Indonesian Ministry of Research & Technology/ National Research & Innovation Agency	Pre- recorded
11:20-11:30	Presentation 1. Work Plan of 2021 K-Innovation Partnership Program with Indonesia & Major Outcomes of 2020 K- Innovation	Dr. Deok Soon Yim, Senior Research Fellow, STEPI	
11:30-11:45	Presentation 2. The STI think tank, the process of the STI policy and work plan for joint research on improving the legal framework & institutional arrangements of mission-oriented organizations such as think tanks in Indonesia	Prof. Kiseok Kwon, Hanbat National University	
11:45-12:00	Presentation 3. Improving the legal framework & institutional arrangements of mission-oriented organizations such as think tanks in Indonesia	Anugerah Yuka Asmara Research Center for Policy & Management of Science, Technology and Innovation LIPI	Pre- recorded
12:00-12:15	Presentation 4. All-stage- evaluation issues of national R&D programs and work plan for joint research on designing an evaluation mechanism for PRN 2020- 2024 and conducting the implementation of the first year of PRN 2020-2024	Dr. Youngjoo Ko, President, Daejeon Institute of Science & Technology for Enterprise & People	Pre- recorded

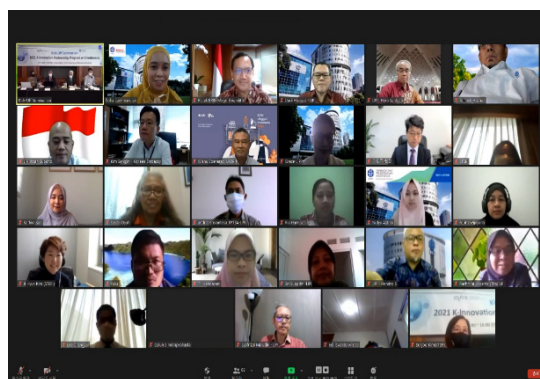
12:15-12:30	Presentation 5. Designing an Evaluation Mechanism for PRN 2020-2024	Dr. Kusnandar Research Center for Policy & Management of Science, Technology and Innovation LIPI	Pre-recorded
12:30-12:45	Presentation 6. S&T Laws, issues of STI regulations and work plan for joint research on restructuring regulations on S&T in Indonesia	Prof. Seongsoo Kim, Hankuk University of Foreign Studies	
12:45-13:00	Presentation 7. Status of regulations on S&T in Indonesia	Dadang Ramdhan, M.Sc Research Center for Policy & Management of Science, Technology and Innovation LIPI	Pre-recorded
13:00-13:25	Open discussion: Q & A	Ms. Kalina Sari, Researcher, STEPI	
13:25-13:30	Closing remarks	Dr. Deok Soon Yim, Senior Research Fellow, STEPI	

The Kick-Off Seminar K-Innovation Partnership Program with Indonesia is the beginning of the 2nd annual STEPI K-Innovation Partnership Program.

Dr. Mi Ock Mun, President of STEPI delivered the opening remarks by recalling the consolidated partnership with Indonesia on STI starting from 2014. She highlighted the importance of STI for economic transformation and extended her appreciation to the Korean experts and Indonesian researchers for implementing the project, as well as to the RISTEK-BRIN for providing all the necessary support in making this project a success.

Dr. Dudi Hidayat, Acting Deputy Chairman of LIPI said that STI was expected as a strategic tool in enhancing competitiveness in the country's sustainable development. He emphasized that learning from the experiences and best practices of South Korea is quite reasonable for Indonesia in spite of the two countries' different contexts. In his welcome remarks, he added that for this reason, learning the lessons from South Korea should be carried out with a complex understanding of mechanism and corresponding context, particularly the institutional and regulatory context.

Dr. Mego Pinandito, Executive Secretary of the Indonesian Ministry of Research & Technology briefly explained the urgency of development of STI in Indonesia. He elaborated the conditions and factors that Indonesia is facing such as limited budget, the urgency in transforming the STI issue by an institutional mechanism and structures, the formation of the National Innovation & Research Agency (BRIN), uncoordinated research projects with limited budgets have been perennial problems in Indonesia. To solve such problems, he suggested the necessity to review and arrange the STI governance of Indonesia including coordination, and mechanism, institutional framework as well as an R&D financial &



investment system.

5.3 Activity 3: Joint Research to Produce a Policy Consulting Report

Three teams were organized to produce a policy consulting report covering the three themes of building a new STI think tank in Indonesia, the management & evaluation system of the National Research Program, and STI laws & regulations. Korean experts led the teams, and LIPI assigned focal points for each team to communicate with the Indonesian researchers. Each team had online meetings to conduct joint research, and STEPI organized thematic seminars by inviting Korean experts.

5.4 Activity 4: Online Training Workshop on STI Policy & Strategy Development for Indonesia

The training on STI Policy & Strategy Development for Indonesia was held on September 7-8, 2021 with the objective to share Korea's experience in science & technology and innovation (STI) governance (coordination mechanisms and legal & financial systems), to reinforce STI policy planning and execution capacity of Indonesia, and enhance scientific and technological competitiveness.

The online workshop consisted of 4 lectures with 39 participants having joined the online training workshop in total. Indonesian participants had discussions with the panels of each lecture.

[Table 1-4] Online Training Program

DAY 1		
11:00-11:05	Opening	
11:05-11:15	Speech from Deputy for Scientific Services BRIN	Dr. Dudi Hidayat Director of Policy Formulation for Research, Technology, & Innovation
11:15-11:25	Introduction of the Program	Dr. Deok Soon Yim Senior Research Fellow, STEPI
11:20-12:15	Lecture 1. STI Policy Instruments in Korea	Prof. Youngrak Choi Former President of STEPI
12:15-12:45	Q&A and discussion	Prof. Youngrak Choi Former President of STEPI
16:00-17:00	Lecture 2. Exploring the Role of the Think Tank in the Open Innovation System	Prof. Veny(ILWON) Seo Chonnam National University
17:00-17:30	Q&A and discussion	Prof. Ki-Seok Kwon Hanbat National University Prof. Veny Seo Chonnam National University

DAY 2		
11:00-12:00	Lecture 3. R&D Program Evaluation in Korea	Dr. Jae Young Kwon Korea Institute of Science & Technology Evaluation and Planning (KISTEP)
12:00-12:30	Q&A and discussion	Dr. Youngjoo Ko President, Daejeon Institute of Science & Technology for Enterprise & People Dr. Jae Young Kwon Korea Institute of Science & Technology Evaluation and Planning (KISTEP)
16:00-17:00	Lecture 4. S&T Law & Regulations	Prof. Bongchul Kim Hankuk University of Foreign Studies
17:00-17:30	Q&A and discussion	Prof. Seong-soo Kim Hankuk University of Foreign Studies Prof. Bongchul Kim, Hankuk University of Foreign Studies
17:30-18:30	Wrap-up	Dr. Deok Soon Yim, Senior Research Fellow, STEPI

The first topic was explained by Professor Youngrak Choi and titled ‘STI Policy Instruments in Korea’, which introduced the Korean government's STI goals and explained implications derived from past STI development experience. In particular, emphasized were the importance of directionality of sectors, topics, and infrastructure for a national master plan as well as policy capabilities and role of the private sector.

The second lecture was on ‘Exploring the Role of the Think Tank in the Open Innovation System’. Firstly, Professor Seo defined the concept, role, and types of think tanks and how in the open innovation system, they play the role of boundary spanners such as devising new ideas in multi-level governance and connecting formal and informal networks among their diverse roles.

The topic of third lecture was titled the ‘National R&D Program Evaluation in Korea’, and lectures were given on the background of R&D in Korea and national evaluation system of R&D performance. It was introduced how performance evaluation is divided into self-,

meta-, final, and in-depth evaluations. Regulations for reports and budgets were also added. The fourth topic was titled ‘Science & Technology Law and Regulations’, where the presenter explained the economic growth context and situation from the 1960s to 1990s, starting with Korea's five-year economic development plan, and introduced the most important laws and institutions in Korea's developmental history. In addition, Korea's corresponding laws related to STI were explained.



1980s: 2,000,000 Houses Project

- New town and City-Infrastructure
- For the Needs of the Houses / Connection among the cities



- The Project
- Building 2,000,000 houses during 1988-1991
- For 25% of the whole Korean population
- New town construction around Seoul

- The budget
- About 45% of the Korean GDP at the time

- Construction Companies
- From 2,610 (1988) to 8,869 (1991)



A feedback survey on the online training workshop with the 39 participants as respondents was conducted. For the results, most of the participants gave high satisfaction scores on all the questions.

[Table 1-5] Feedback Survey Result of the Online Training Workshop

Questions	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
1. The topics covered in the lecture were relevant to me.	37.7%	50%	9.4%	2.8%	37.7%
2. The content was well-organized and easy to follow.	37.7%	57.5%	3.8%	0.9%	37.7%
3. The lecture was helpful for understanding the field	30.2%	60.4%	7.5%	1.9%	30.2%
4. The lecture would be helpful to strengthen the institutional capacity of my organization.	24.5%	62.3%	12.3%	0.9%	24.5%
5. The time allotted for the lecture was sufficient.	15.1%	38.7%	39.6%	6.6%	15.1%
6. Q&A and discussion was adequate to enhance my understanding.	13.2%	57.5%	27.4%	1.9%	13.2%
7. I will be able to apply the knowledge learned in my work/ job practice.	17%	59.4%	17.9%	5.7%	17%
8. I will recommend this lecture to others.	42.5%	46.2%	10.4%	0.9%	42.5%

[Table 1-6] Indonesian Participants of the Online Training Workshop

Name	Position	Affiliation
Anggini Dinaseviani	Researcher	BRIN
Anugerah Yuka Asmara	Researcher	BRIN
Aditya Wisnu Pradana	Researcher	BRIN
Agus Widodo	Director	BRIN
Ayom Widipaminto	Director	BRIN
Badrudin Kurniawan	Lecturer	State University Of Surabaya
Budiati Prasetiamartati	Policy & Innovation Development Lead	Program Knowledge Sector Initiative
Dadang Ramdhan	Researcher	BRIN

Name	Position	Affiliation
Dudi Hidayat	Director of Policy Formulation for Research, Technology, & Innovation	BRIN
Endah Dwi Novianti	Senior engineer	BRIN
Fajri Nursyamsi	Director of Advocation and Central Network	Centre for Policy & Law Studies (PSHK)
Fithriana Munawaroh	Policy analyst	BRIN
Filda Nurria Agustifa Marthaatmadja	Policy analyst	LPDP
Galuh Syahbana Indraprahasta	Researcher	BRIN
Iskhak Fatonie	Enabling environment senior coordinator	Knowledge Sector Initiative
Karlina Sari	Researcher	BRIN
Kurnia Aliyanti	Policy analyst	BRIN
Kusnandar	Researcher	BRIN
Leo Kamilus Julianto Rijadi	Policy analyst	BRIN
Maulana Akbar	Researcher	BRIN
Nur Huda	Research associate	CIPG
Nungki Indrianti	Policy analyst	BRIN
Nunung Nur Chayati	Head of subdivision of research and assessment	Regional Development Planning, Research & Development Agency
Parama Tirta Wulandari Wening Kusuma	Senior planner	BRIN
Rohmat Hidayatuloh	PRN team	LPDP
Rudi Arifiyanto	Senior planner	BAPPENAS
Ria Hardiyati	Researcher	BRIN
Sofia Yuniar Sani	Policy analyst	BRIN
Siti Annisa Silvia Rosa	Policy analyst	BRIN
Sasti Orisa	M & E team for PRN	BRIN
Sri Supadmi	Policy analyst	BRIN

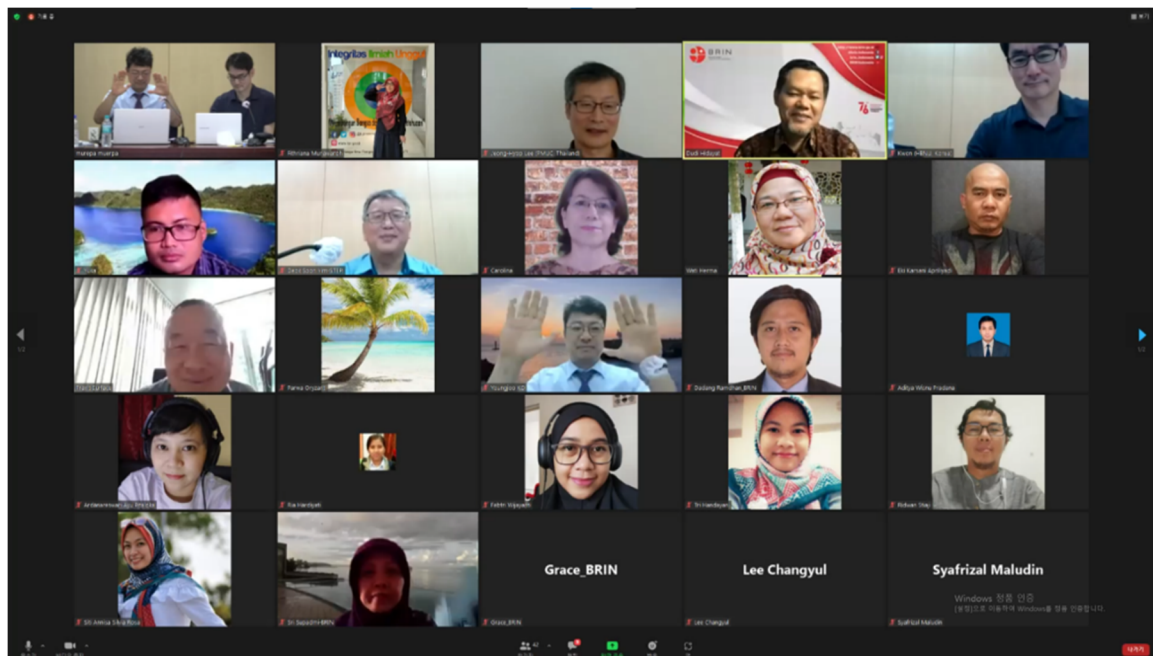
Name	Position	Affiliation
Safendri Komara Ragamustari	Researcher	BRIN
Syarip Hidayat	Legal Drafter	BRIN
Totok Hari Wibowo	Chairman	Indonesian Policy Analyst Association
Tikkyrino Kurniawan	Researcher	Research Centre for Marine and Fisheries Socioeconomics (MMAF)

5.5 Activity 5: STEPI Special Session in the 2021 ASIP Conference

STEPI held a special session titled ‘Strengthening Indonesia’s STI system in Partnership with Korea: Current Issues and Implications for Asian countries’ at the conference of Asian Society for Innovation & Policy (ASIP) on October 9, 2021 at Jeju University and through online zoom. The objective of this STEPI special session was to promote the results of policy advice to the STI community in Asia.

[Table 1-7] STEPI Special Session Program

16:00–16:05	Opening remarks	Dr. Dudi Hidayat National Research & Innovation Agency (BRIN) Dr. Deok Soon Yim, STEPI
16:05– 16:35	STI Think-Tanks and Their Governance	Presenter: Prof. Ki-Seok Kwon, Hanbat National University Discussant 1: Dr. Anugerah Yuka Asmara BRIN Discussant 2: Dr. Jeong Hyop Lee Senior Advisor to Thailand Program Management Unit for Competitiveness
16:35– 17:05	National R&D Program Evaluation	Presenter: Dr. Youngjoo Ko President, Daejeon Institute of Science & Technology for Enterprise & People Discussant 1: Dr. Kusnandar, BRIN Discussant 2: Prof. Mohan Awari Nottingham University Business School
17:05–17:35	S&T Laws & Regulations	Prof. Seong-soo Kim Hankuk University of Foreign Studies
17:35– 18:00	Q&A and closing remarks	



6. Project Schedule

The 2021 K-Innovation with Indonesia was implemented according to the timeline below. The project schedule was adjusted over the project cycle according to internal factors such as delay of the contract for joint research in Indonesia and external factor of the COVID-19 pandemic.

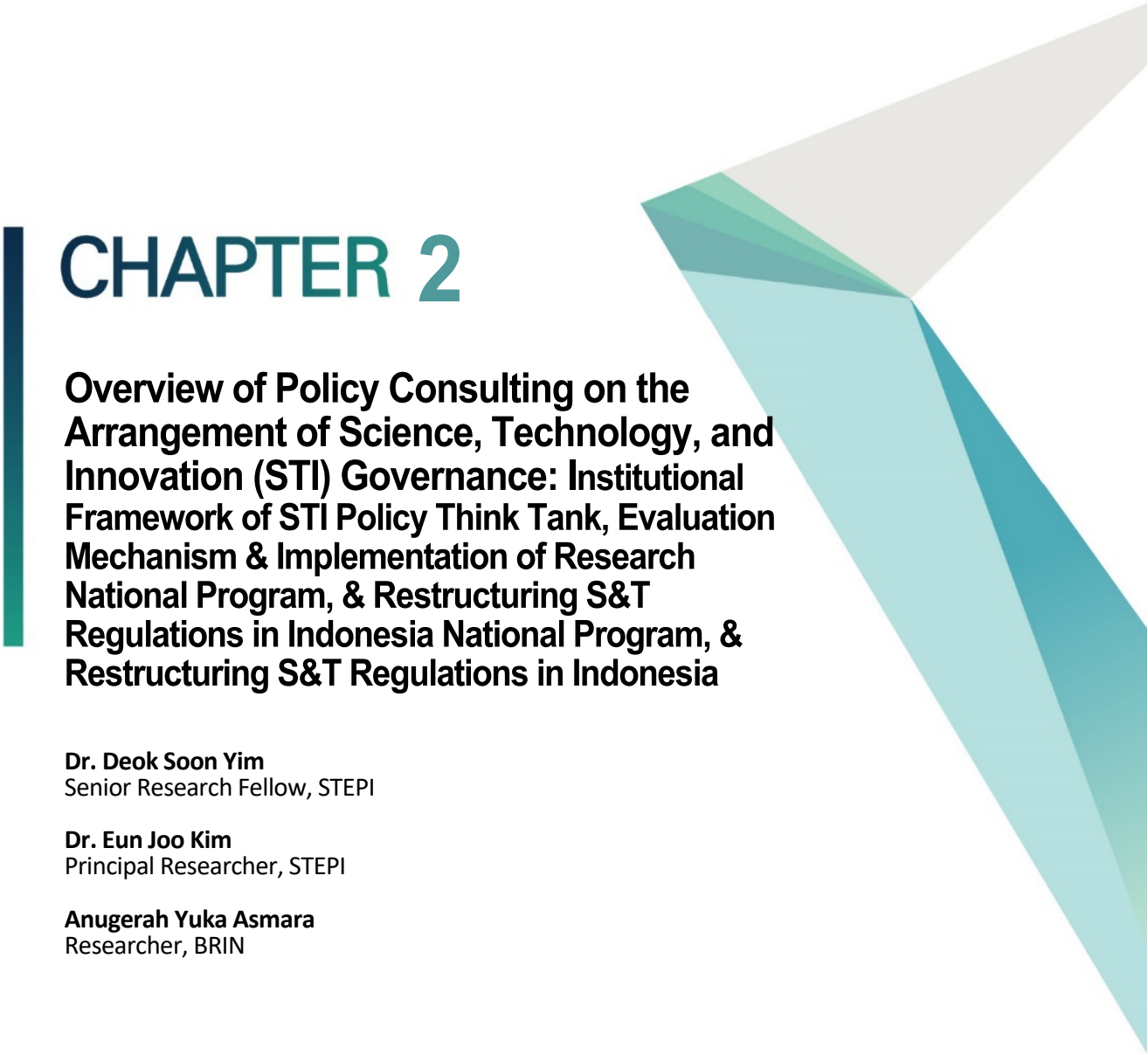
[Table 1-8] Project Schedule

Activities/Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
Consultations on project activities, etc.												
Literature review												
Kick-off workshop												
Online training workshop												
Joint research for policy consulting report												
Mid-term report												
Dissemination seminar (virtually)												
Academic paper												
Final report												

7. Project Outputs

Actionable policy recommendations of STI governance in Indonesia are the key outcomes of the 2021 K-Innovation in the following areas of R&D coordination, mechanism, institutional framework & financial system, and investment in Indonesia. Throughout the project cycle, the project is expected to have the following deliverables:

- Deliverable 1: Presentation materials from the kick-off seminar
- Deliverable 2: Concept paper on joint research on the three themes
- Deliverable 3: Presentation materials for the dissemination seminar
- Deliverable 4: Policy consulting report of the 2021 K-Innovation (in English & Korean)
- Deliverable 5: Academic papers on the three themes

A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a large light grey triangle pointing downwards, and several overlapping triangles in shades of teal and light blue, also pointing downwards.

CHAPTER 2

Overview of Policy Consulting on the Arrangement of Science, Technology, and Innovation (STI) Governance: Institutional Framework of STI Policy Think Tank, Evaluation Mechanism & Implementation of Research National Program, & Restructuring S&T Regulations in Indonesia National Program, & Restructuring S&T Regulations in Indonesia

Dr. Deok Soon Yim
Senior Research Fellow, STEPI

Dr. Eun Joo Kim
Principal Researcher, STEPI

Anugerah Yuka Asmara
Researcher, BRIN



Chapter 2. Overview of Policy Consulting on the Arrangement of STI Governance: Institutional Framework of STI Policy Think Tank, Evaluation Mechanism & Implementation of Research National Program, & Restructuring S&T Regulations in Indonesia

1. Overview of Policy Consulting

1.1 Conceptualization

The 2021 K-Innovation with Indonesia focused on consulting to address three key issues for Indonesia's National Innovation System as follows:

- Framing think tanks or policy institutes aligned with the science-technology-innovation policy at a national level

For designing a workable STI think tank organization in an Indonesian-specific context, Indonesia needs to first understand and learn the Korean-specific context of its legal framework, state-run organization regulations, and institutional arrangement that have made possible a quick and adaptive establishment of an organization designed for particular missions, such as a think tank. The 2021 K-innovation project is designed to facilitate BRIN researchers and policy-makers from the Ministry of Research & Technology, Ministry of State Apparatus & Bureaucratic Reform, and National Planning Agency to learn in more detail the legal framework and institutional arrangement in a Korean-specific context. This activity is aiming at producing a plan to design necessary regulations and institutional arrangement that are most likely supportive for the establishment of mission-oriented organizations, such as a think tank.

- Designing an evaluation mechanism for PRN 2020-2024 and conducting the implementation of the first year of PRN 2020-2024
Indonesia's mid-term R&D policy PRN 2020-2024, which is the guidelines for

national research and development programs was implemented in 2020 but doesn't have an established evaluation mechanism for PRN 2020-2024. Therefore, designing and carrying out an evaluation of PRN implementation during 2020 is one of the important tasks for Indonesian policy & research communities. The evaluation should be formative in nature by aiming to understand the real implementation situation and impediments for successful implementation. The 2021 K-Innovation will assist in designing and carrying out an evaluation process for PRN, which was implemented in 2020. It is also essential for Indonesian researchers and policy-makers to have a deeper understanding and lessons from the evaluation practice in a Korean-specific context.

- Restructuring regulations on S&T in Indonesia

The Indonesian government has formulated a number of government regulations as an important part of the implementation of Law No. 11-year 2019 on the National S&T Systems. The regulations concern many areas such as technology transfers, technology audits, invention & innovation promotion, research code of ethics, research data management etc. In this context, 2021 K-Innovation facilitates the process that will enable Indonesian LIPI researches and policy-makers to have a full understanding about how the Korean government formulates regulations on the above matters. This project will produce design options and actionable policy recommendations for restructuring regulations on S&T in an Indonesian-specific context.

The common theme of the issues above is institutional building of policy-makers and researchers. The Indonesian government needs to design effective organizations and institutions to carry out particular functions, i.e. an STI policy think tank, research planning evaluation, analyzing of existing S&T regulations. To address the concerns above, the Korean and Indonesian teams conducted joint researches with four major activities such as conceptualization of STI policy and a national innovation system, focused group discussions (FGD), in-depth interviews, and workshop to disseminate the outcomes of research to the relevant stakeholders from Indonesia and Korea.

The STEPI team did the conceptualization work based on Korean or other countries' experiences with regard to the following:

- Establishing an effective national STI think tank or policy institute arrangement and suggesting an effective one to be adopted in an Indonesian-specific context;
- Appropriate design of an evaluation mechanism on national research programs that can be conducted in an Indonesian-specific context;
- Providing formulation-and-implementation process structure of Korea's S&T regulations that will be adjusted in an Indonesian-specific context.

The STEPI team prepared a concept paper for joint research to address the three topics, based on the Indonesian context. During the process, the Indonesian-BRIN team also provided assistance in understanding the Indonesian context. The general goals of three activities in collaboration STEPI and BRIN are as follows:

- To make a provision of rationales for the creation of an STI think tank in Indonesia;
- To analyze and identify influencing factors on the establishment of an STI think tank;
- To improve the research outcome of the previous year regarding detailed contents and feasibilities.
- To improve the evaluation mechanism for implementation of the first year of PRN 2020-2024 in a systemic way relating to program formulation.
- To classify standards for science and technology laws;
- To introduce the current status of science and technology laws in Korea and a summary of their developmental history with a focus on the system (policy)-regulation relation; and
- To analyze current issues in the Indonesian regulatory system and implications of the Korean experience

1.2 Methodology

There are three research methods of this study such as: 1) focused group discussions (FGD), 2) in-depth interviews, and 3) seminar dissemination. Each team has a different scope to do the following research activities.

1) Theme 1

- Collection of evidences of the supporting rationales for the creation of the think tank
- Analysis of the collected data through and interviews on our subject
- Collaboration between the participants in the group for the analyzing and drafting reports

2) Theme 2

- Interviews about the evaluation mechanism of PRN 2020-2024 by inviting some relevant experts from both countries:
- Collecting and analyzing materials such as reports, papers etc.
- Having an online workshop or colloquium to discuss more details (including the ASIP conference 2021)

3) Team 3

- Classification of the Korean S&T law system
- Status of the Indonesian regulatory system and issues (by BRIN team or Ex LIPI)
- Exchange of ideas to draw implications

1.2.1 Focused Group Discussions (FGD)

After the desk review, the BRIN research team plans to hold focused group discussions (FGD) virtually due to COVID-19 restrictions. The purpose of FGDs is to capture various perspectives, opinions, and ideas on the three themes from key stakeholders, including to widen and reinforce external networks of national & local STI actors. Key participants for

FGD can be categorized based on the type of participants and their institutions. The key participants listed in the table below were selected actors who are representatives of their affiliations and well versed with the STI concept and practice. All actors are either from academia (A), business (B), government (G), or society (S), and commonly referred to as quadruple helix actors. However, there are particular actors interviewed who play dual roles from either academia, business, government, or society.

[Table 2-1] Categorization of Participants for FGDs

Type of participants	Description	Affiliations/ Institutions
National policy experts	Those who have a capability or authority in influencing national policies	- Coordinating Ministry of Economic Affairs
Key or experienced researchers	Those who are interested in a specialized field of research and involved in STI activities	- Indonesian Institute of Sciences (LIPI) - Agency for Assessment & Application of Technology (BPPT) - National Nuclear Agency (BATAN) - National Space Agency (LAPAN) - Indonesian Researchers Union (Himpenindo) - Local Research & Development Agencies (Balitbangda) - Agriculture Research Institute, Ministry of Agriculture
Decision-makers/ stakeholders	Those who have full authority and resources in decision-making or policymaking at the national or regional level	- Director of Science-Culture-& Higher Education Ministry of National Planning & Development - Director of Research & Innovation Policy at Ministry of Research & Technology/ National Research & Innovation Agency (BRIN) - Vice director at Fiscal Policy Agency at the Ministry of Finance - Ex Minister of Research & Technology (period 2014-2019)
Public officials	Those who are in street-level bureaucrats or frontline staff who execute policies from the top level at a ministry	- Staff or division chiefs at the Ministry of National Planning & Development - Staff or division chiefs at the Ministry of Research & Technology - Staff or division chiefs at the Ministry of Coordinating Economy Affairs

Type of participants	Description	Affiliations/ Institutions
		- Staff or division chiefs at the Ministry of Finance - Staff or division chiefs at the Ministry of Industry
Shareholders/ business actors	Those who are engaged in profit-seeking rather than achieving social goals	- Small and medium enterprises - National business associations of automotive industry and biofuel energy
Funding-related actors	Those who have the capability, authority, and resources to gain/ collect funding sources and spend/ allocate funds	- Governmental finance management agency for national education at the Ministry of Finance (LPDP)
Academic scholar/ lecturer	Those who are capable and expert in academic fields and contribute to a practical field	- Lecturers on public policy, public administration, and social sciences from Gadjah Mada university (UGM), Brawijaya University (UB), Indonesia University (UI), and Diponegoro University (UNDIP)
Policy analyst (PA)	Those who are capable and expert in public and STI policy analysis at each institution	- PA in the Indonesian Institute of Sciences (LIPI) - PA in the Agency for Assessment & Application of Technology (BPPT) - PA in the National Nuclear Agency (BATAN) - PA in the National Space Agency (LAPAN) - PA in the Ministry of Finance - PA in the Ministry of National Planning & Development - PA in the Public Administration Agency - PA in the Ministry of Industry - PA in the Ministry of Education
Non-government organizations (NGOs)	Those who are frequently interested and involved in many discourses in science-technology-innovation issues, notably located inside the Indonesian geography	- Knowledge sector initiative (KSI) - Centre for Law and Policy Studies (PSHK) - Center for Innovation, Policy, & Governance (CIPG) - Independent consultant of public policy

*Note: LIPI, BATAN, BPPT, and LAPAN are public R&D institutes that since September 1, 2021 are integrated to be the National Research & Innovation Agency (BRIN). Since most interviews of them were conducted before September 1, 2021, we use the old name for the sake of this study.

1.2.2 In-Depth Interview

In-depth interviews are aimed at gaining deeper and holistic results based on the view and perspective of several key info. providers such as national policy experts, key or experienced researchers in specialized fields, decision-makers/ stakeholders, policy analysts, politicians, public officials, shareholder/ business actors, funding-related actors, academic scholars dealing with national policy, economy, engineering, and non-government organizations (NGOs). The key info. providers involved in in-depth interviews are included in Table 2-1 above. Those involved as FGD participants are also invited to be key info. providers in in-depth interviews. It is necessary to triangulate research finding fields to reinforce and support robust data analysis.

Each actor will be involved in several discussions during both formal and informal meetings. Each of the key info. providers is expected to share his/her ideas & opinion to enrich the plan & design of the three themes. The average time of discussion of this scheme is at least 1 hour lasting 3 hours depending on the time availability of key info. providers and the BRIN team. Due to Covid-19 restrictions, all interview activities were conducted in Zoom Meetings or Google Meet. Some of the informal discussion were also conducted through Whats App Call and messages.

1.2.3 Dissemination Seminar

The STEPI and BRIN (Ex LIPI) research teams planned to hold national workshops or conferences to communicate the results of the research in order to reach as many stakeholders as possible. Dissemination workshops/ conferences were aimed at creating new and extended networks that are useful in developing new designs of STI policy and institutional structure with regard to the three themes of STI policy think tanks, research program evaluation, and restructuring S&T regulations.

However, due to COVID-19 restrictions, the dissemination workshop was held as a virtual seminar. The special seminar was held with the virtual participation of Indonesian stakeholders during the International Conference of Asian Society for Innovation & Policy (October 8, 2021) via online. Indonesian researchers who are actively engaged in the joint

research presented their paper addressing the three issues to get constructive feedback from the audience via YouTube.

1.3 Timeline

The timeline was arranged starting from ideas of STI studies, conceptualization, in-depth interviews, FGDs, mid-term consulting report, dissemination seminar, final consulting report, and academic paper draft. All activities were conducted sequentially from March to November 2021 and were adjusted depending on the conditions and needs of both parties based on mutual agreement. From April to September 2021, researchers collected qualitative data through FGDs and in-depth interviews as the key methodologies to understand the status of the three topics due to a lack of reliable quantitative sources about the STI policy in Indonesia. Researchers prepared the policy consulting report covering the possible design of an STI think tank policy, research endowment, and management scheme of the national research & development program.

[Table 2-2] Timeline for Joint Research

Month/ Activity	March	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.
Coining ideas									
Conceptualization									
In-depth interviews									
FGDs									
Mid-term consulting report									
Dissemination seminar									
Final consulting report									
Academic paper draft									

In March 2021, the researcher team coined appropriate ideas of three research themes, and then continued with conceptualization with Korea's experts. In-depth interviews were conducted in lockstep with FGD activities in the June-September 2021. Nonetheless, FGDs were continued to October 2021 because the researcher team needed to triangulate research results. An early draft of the consulting report was conducted in September-October 2021 during two months because the researcher team needed intensive discussion with experts from South Korea. Dissemination seminar was also held along with the Conference of Asian Society for Innovation & Policy (October 8, 2021), and continued to the final consulting report during October-November 2021. At the end of November 2021, the teams will prepare an academic paper draft of research results during March-November 2021.



CHAPTER 3

The Evolution of STI Think Tanks in South Korea & Implications for Indonesia & Current Status

Prof. Ki-Seok Kwon

Professor, Hanbat National University

Anugerah Yuka Asmara

Researcher, BRIN

Aditya Wisnu Pradana

Researcher, BRIN

Fithriana Munawaroh

Policy analyst, BRIN

Siti Annisa Silvia Rosa

Policy analyst, BRIN

Chapter 3. The Evolution of STI Think Tanks in South Korea & Implications for Indonesia & Current Status

1. Introduction

Think tanks in the area of science, technology, and innovation (hereinafter 'STI') policy have a critical role in the development of national innovation system (hereinafter 'NIS') in a country. The STI think tank provides information on NIS and policy ideas to develop the S&T system. However, the roles of think tanks are somewhat different according to the idiosyncratic development of NIS. Arguably, in industrialized countries, various established academic communities expanded their rollout of the STI policy regarding researches and policy agenda, whereas newly born think tank agencies in catch-up countries work as state-run organizations to foster the nascent NIS as well as specialized policy institutions as done in industrialized countries.

In the same vein, the experience of Korean think tanks provides a unique reference to other catch-up countries attributed to not only its rapid economic success but also aligned management of the STI system & process and appropriate function of STI institutional elements including STI think tanks. On the other hand, the pitfalls in the evolutionary path of think tanks can provide lessons for others not to choose inappropriate decisions when they create a new STI think tank in a particular national context.

Under this scenario, this chapter attempts to analyze the role of think tank in the STI policy system evolved in South Korea based on the concepts of policy system and think tank proposed herein. Accordingly, we have identified several stylized facts in the evolution of the Korean STI think tank. Based on such findings, we put forward some policy implications for the establishment and operation of an STI think tank in catch-up countries such as Indonesia as well as other industrialized countries.

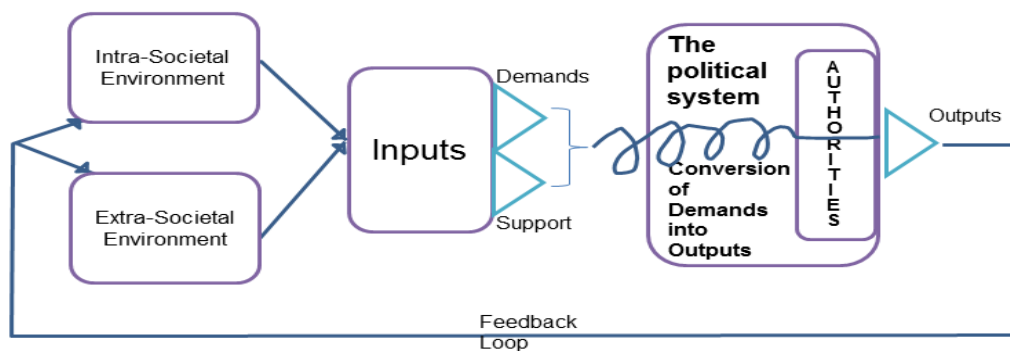
2. Theories and Framework

2.1 Analytical Framework: Political System Approach

Easton (1965) suggests a system approach applying the lenses of the system concept on the political system. As shown in the figure below, the political system maintains its entity by the conversion of demand and support inputs from intra- and extra-societal environments to outputs such as public services and goods including policies. Based on this approach, this study conceptualizes the STI policy system as 1) political structure, 2) policy process, and 3) the interactive patterns between them.

Firstly, political structure consists of political actors, their functions, and institutional rules defining the identities of the actors. Secondly, the policy process includes four stages of agenda setting, decision-making, implementation, and evaluation. These four stages provide different settings for the actors to function in the political structure. Thirdly, we observe the complex relations between the policy actors aligned under the political structure. In particular, the various actors in different stages highlight the identity of think tanks in the policy system. Based on this framework, we discuss the STI think tank's function, its relations to other actors, and identity in the STI policy system.

[Figure 3-1] System Approach in Politics



Source: Easton (1965), p.112.

2.2 Concepts of Think Tanks

Think tanks usually have dynamics that they tend to adapt themselves to the cultural, political, historical, and economical context of a country. Accordingly, each existing form of think tanks in a country possibly could have a significantly different applicability in another countries' context (Urrutia, 2013). In spite of various forms of think tanks in different systems, McGann (2019) maintains that two of the general properties of think tanks are their independency and non-profit organization. In a similar vein, Urrutia (2013) argues that the credibility of a think tank can be dependent on how they can maintain their independency and make contributions for the public interest.

In this study, we review the categorization of think tanks according to the various points of view. In particular, their roles and affiliation provide a classification of think tanks. For example, McGann (2019) categorizes the think tanks into autonomous independent, quasi-independent, and government or corporation-affiliated organizations. The criteria of this classification are the think tanks' source of funds and their autonomy from the funding sources.

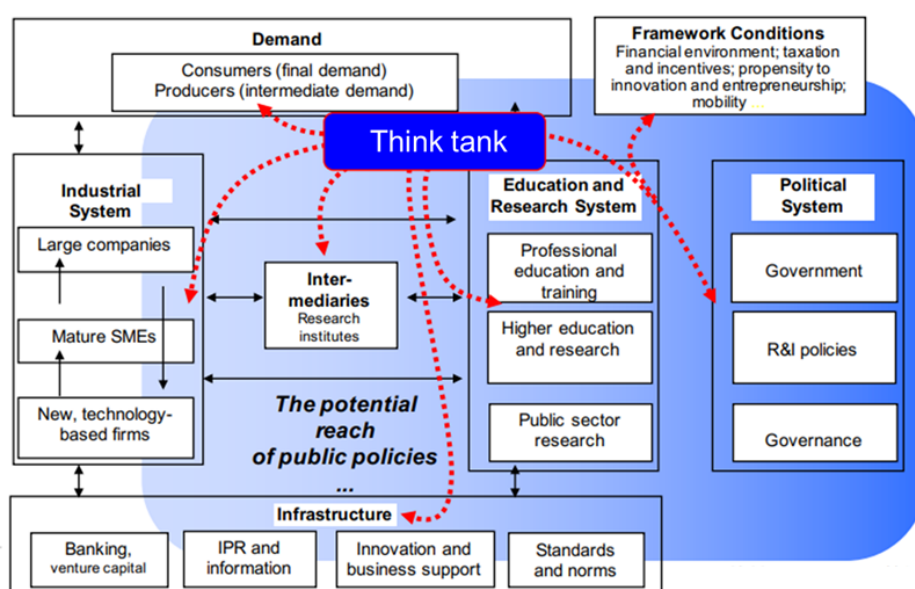
Concerning the different roles of think tanks, Weaver (1989) suggests a classification consisting of three categories. The first type is a university without students. This regards the university as an educational and teaching institution as well as an academic research institution. In other words, a think tank is a research organization for policy research. The second type is a contract research organization. The government generally accredits this type of organization to carry out research on particular policy issues. These think tanks function as consultants to assist their clients to provide policy ideas and recommendations.

The third type is the advocacy think tanks that try to disseminate their own ideas in the policy communities. This type of think tanks is closer to an interest group rather than academic communities.

2.3 The Role of Think Tanks in the STI Policy System

Kuhlman and Arnold (2001) provide an extensive explanation on the operation of the national innovation system based on the co-evolutionary lenses on the various interconnected actors' collaborative functions such as diffusion of knowledge, resource mobilization, and formation of markets. In this regard, we assume the ideal role of think tanks is the alignment and coordination of the interdependent activities of innovation actors in NIS. In other words, a think tank addresses the demand of the NIS system, subsystems such as an industry, education, & research system, the political system, infrastructure, and its correlated activities.

[Figure 3-2] Ideal Function of Think Tanks in Political System



Source: Revised from Kuhlman and Arnold (2001); quoted in Arnold (2016) p. 5.

In particular, the figure above shows the ideal function of think tanks in the political system. Firstly, the think tanks collect and analyze the demands of actors and institutional framework conditions of the STI system such as finance and taxation. Secondly, think tanks monitor the condition of actors and their activities in the STI system consisting of an industrial, education, research, and political systems. Thirdly, concerning the infrastructure, think tanks investigate the banking system, IPR, governmental supports, etc. Fourthly,

regarding the systemic view on NIS, a think tank monitors and diagnosis the proper activities and interaction of the actors as a function of the whole system.

Against this perspective, we attempt to summarize the role of an STI think tank as follows considering the concepts of think tanks. Firstly, we define a think tank as a place with a source of policy ideas. Additionally, the ideas are collected and debated by experts to popularize such ideas to policy-makers. These ideas could be unpopular but may also be strategic to decision-makers. Based on this idea, a think tank can carry out academic or contracted research for clients such as the government.

Secondly, a think tank can play the role as evaluators and planners of the governmental programs and projects. In addition to the R&D planning function, a think tank sometimes assess government programs in how far its programs reach their objectives. The think tank often plans and evaluates independently or working under contracts. Since the government plans and implements these programs under the strategic goals of national R&D visions at the upper levels or from the long-term perspective, the think tank has an appropriate position to monitor and adjust the process of implementation of the programs.

Thirdly, the think tank plays the roles as a reservoir of expert human resources and network hub of expertise including science, technology, and innovation-related studies. As discussed above, a think tank is a kind of academic agency keeping researchers. These researchers spread out into universities, firms, institutes, and other intermediary agencies, who eventually are wiring up the national innovation system. This is particularly important for catch-up countries, which suffer from a shortage of experts in these areas of studies.

2.4 Research Method

This study uses qualitative research method to collect comprehensive data related to a STI policy think tank in Indonesia. Data were derived from the literature review covering think tank, STI policy, STI institution, and governance. Governmental documents are used to complete literature tracing of an STI policy think tank in Indonesia. Additionally, we conducted in-depth interviews with many key info. providers ranging from academia,

business, government, and communities such as non-government organizations. Totally, there were 30 key info. providers with whom we discussed these during interviews through online Zoom media. Nonetheless, we also made onsite visits to a user of research results in Bandung city, and academic experts in Bandung city to get data concerning delivery of research results to the market. It will be one of the considerations for local government to make innovation policy in a region.

The first interview was held on August 23, 2021 with Mr. MK and Mr. I as resource people. Mr. MK from the Research and Development (R&D) Agency in the local government of Sidoarjo (Balitbang Kabupaten Sidoarjo) and Mr. I from the Research and Development (R&D) Agency in East Java province government (Balitbang Provinsi Jatim). Both of them are researchers, while the purpose of the discussion was to find out how the practice of policy think tank in public research & development (R&D) institutes (public policy & regional development) is formed at regency level and provincial level is implemented.

On August 24, 2021 the second interview was done. The resource person was Mr. FP, an expert of public policy/ innovation policy from the University of Brawijaya. The purpose of the discussion is to find out about public policymaking or STI policy in an academic and practical way. The next interview was held on September 1, 2021 with Mr. YM as a resource person. He is the director of the Directorate of Science-Technology-Innovation (STI) Policy and Institution at the Ministry of Research & Technology/ National Innovation & Research Agency. The purpose of the discussion is to find out about a national policy think tank at the ministry level (specific field of STI affairs).

The next interview was held on September 2, 2021, and we had several resource people, who are Mr. TH, Mr. TM, Mr. RA, Mr. AF, Mrs. R, and Mrs. EC. Mr. TH is a policy analyst at the Coordinating Ministry for Economy Affairs (Kemenko-Perekonomian), Mr. TM is the director of the Directorate of Higher Education and Science & Technology at the National Agency for Planning & Development (Bappenas), and the others are key staff from the Directorate of Higher Education and Science & Technology at the National Agency for Planning & Development (Bappenas). The purpose of the discussion is to find out about a national policy think tank at the ministry level (specific field of STI affairs and field of national

economy).

The next interview was held on September 13, 2021, the resource person was Mr. WK, who is an expert of public policy/ innovation policy from the University of Gadjah Mada. The purpose of the discussion is to find out about public policymaking or STI policy in an academic and practical way. Then on September 16, 2021, the next interview was with Mr. YW, who is a policy analyst from the Centre for Specific Technology Region & Innovation System at the Agency for Assessment & Application of Technology (BPPT). The purpose of the discussion is to find out how the practice of a policy think tank in public research & development (R&D) institutes (technology development & engineering) is implemented.

On September 17, 2021, we interviewed Mr. TT, Mr. MN, and Mr. SO as resource people. Mr. TT is a vice director from the Fiscal Policy Agency at the Ministry of Finance (Kementerian Keuangan), Mr. MN is a policy analyst from the Research Centre for National Industry Development at the Ministry of Industry (Kemenperin), and Mr. SO is a key staff from the Research Centre for National Industry Development at the Ministry of Industry (Kemenperin). The purpose of the discussion is to find out about a national policy think tank at the ministry level (field of finance and field of national industry).

The next interview was held on September 22, 2021, the resource person was Mr. TW who is a deputy from Public Administration Agency (LAN). The purpose of the discussion is to find out about a national policy think tank at the special agency for public administration. The next interview was held on September 27, 2021 with several resource people. They are Mrs. MG, Mr. FS, Mr. JH, Mr. RR and Mr. AW. Mrs. MG is a vice director/ policy analyst from the Centre for Policy & Performance at the National Agency for Planning & Development (Bappenas), then Mr. FS, Mr. JH, and Mr. RR are key staffs from the Centre for Policy & Performance at the National Agency for Planning & Development (Bappenas). Mr. AW is a director from the Centre for Aeronautics and space Policy at the National Aeronautics and Space Agency (LAPAN). The purpose of the discussion is to find out about a national policy think tank at the ministry level (field of macro development & planning) and how the practice of a policy think tank in public research & development (R&D) institutes (Aeronautics) is implemented.

The next interview was held on September 28, 2021, the resource person was Mr. FM, who is a policy analyst from the Centre for Partnership & Dissemination at the National Nuclear Agency (BATAN). The purpose of the discussion is to find out how the practice of a policy think tank in public research & development (R&D) institutes (Nuclear) is implemented. The next interview was held on October 14, 2021, the resource person was Mr. MW, who is a policy analyst from Research Centre for Education Policy at the Ministry of Education (Kementerian Pendidikan). The purpose of the discussion is to find out about a national policy think tank at the ministry level (field of education).

On October 27, 2021, we interviewed several resource people, who were Mrs. MU, Mr. FS, Mrs. BP and Mr. FN. Mrs. MU is a researcher from the Centre for Innovation & Governance (CIPG), and Mr. FS is a key staff from the Centre for Innovation & Governance (CIPG). Mrs. BP is a director from the Knowledge Sector Initiative (KSI). Mr. FN is a director/ researcher from the Centre for Law & Policy Studies. The purpose of the discussion is to find out about the NGO agency in science-technology-innovation (STI) policy, international NGO agency in public policymaking and evidence-based policy, and NGO agency in public affairs.

The next interview was held on November 12, 2021, the resource person was Mr. RP who is an expert of public policy/ innovation policy from the University of Padjadjaran. The purpose of the discussion is to find out about public policymaking or STI policy in an academic and practical way. The next interview was held on November 13, 2021, the resource person was Mrs. VW, a Researcher from the Secretariat of the Vice President (Setwapres). The purpose of the discussion is to find out about public policymaking or STI policy. The last interview was held on November 24, 2021, the resource person was Mr. TS, who is a director from CV. Nusantara Jaya (Small-medium enterprise). The purpose of the discussion is to find out about beneficiaries or participants in STI and Industrial Policymaking (on a small-medium enterprise scale). The completed data of interview can be seen in the Appendix 1.

3. STI Policy System & Think Tanks in Korea: Structure & Process

3.1 Evolution of Korean STI think tanks

3.1.1 Creation of STEPI: Late 1980s

STEPI has its origin from the creation of CSTP (Center of S&T Policy) as a subordinated research center of KAIST (Korea Advanced Institute of Science & Technology) in 1987. The CSTP has initiated its early missions as follows. Firstly, CSTP carries out planning and implementation function of science and technology policy for several ministries. Secondly, CSTP provides R&D information to governmental organizations, intermediary agencies, and firms. Thirdly, CSTP creates and supports academic networks for communities of science, technology, and innovation-related studies. Fourthly, CSTP steadily promotes methodological and theoretical foundations for R&D evaluation. Fifthly, CSTP leads and supports international cooperation for R&D as well as STI policy. Sixthly, human resource training is also one of the important missions for Korean NIS due to its lack of experts at the early stage of catch-up.

3.1.2 Enhancing Planning & Evaluation Function: In 1990s

KAIST has separated into two organizations in 1989, which are called KIST as a research-oriented institute and KAIST as an institution for higher education. Accordingly, CSTP has been reorganized under KIST. CSTP has changed its name to STEPI in 1993. STEPI has 77 researchers (50 doctoral degree holders) of total 168 staffs in 1996, which is seven times bigger in size compared to the inception year of 1987. Breaking down the ratios of research staffs' majors, 45% account for science & technology disciplines, and 35% comprise social sciences such as economics & management majors. Based on the 'Act on Creation & Management of Public Research Institutes' in 1999, STEPI became reborn as an independent legal body. Moreover, from MOST (Ministry of Science and Technology), the

jurisdiction of STEPI has shifted to that of the Prime Minister. Accordingly, the autonomy of its activities has been strengthened as STEPI is able to avoid direct control of MOST.

3.1.3 Expansion of STI Policy Studies: In Mid-90s and Mid-00s

In order to enhance the active exchange of expertise and knowledge in the field of STI policy studies, STEPI has played a hub role for innovation communities including university researchers, academic societies, and intermediary agencies. During this period, STEPI became a national think tank from a supporting agency for MOST at the initial stage.

3.1.4 Searching for Upcoming STI Issues: Since the 2010s

Since the 2010s, innovation issues after catching up such as introducing a new product to the global market has resulted in the creation of the new centers in STEPI. Based on the technology foresight techniques, the center for future studies suggests a new national STI agenda to be addressed. In other centers, R&D solutions for social issues have been created and various evaluation programs constructed for that. Additionally, technology regulation, interdisciplinary R&D, incubation of start-ups, and S&T ODA become a critical STI policy issue in the upcoming period. Recently, the location of STEPI has relocated to special self-governed Sejong City. Therefore, it is near to Daedeok Science Town in Daejeon, which creates a good opportunity for an interdisciplinary approach and street-level policy.

3.1.5 Conclusion: Roles of the Korean STI Think Tank

Considering the changing evolutionary patterns of the roles described above and recent divisional settings within STEPI, we can summarize the roles of a Korean STI think tank. At the initial stage, the outstanding scientists' R&D consultancy, planning R&D, and program evaluations are important for the policy practitioners including government officials, as the demands for day-to-day operations of NIS are immense in spite of a shortage of experts in STI policy and management areas besides R&D itself.

As NIS develops into maturity, the STI think tank starts to set up a strong relationship with academic communities. Accordingly, it then becomes a hub of network among the academic

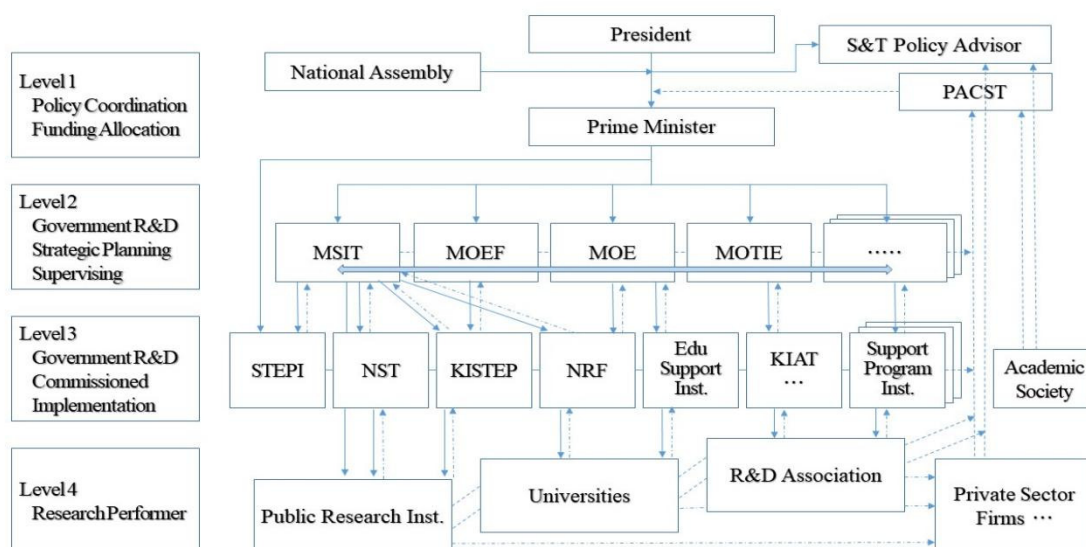
actors, playing a critical role for collaborative STI policy researches. Within an advanced NIS, the STI think tank tends to suggest new STI agenda derived for the near future. However, for the holistic approach is still weak for a Korean STI think tank, which is addressed intensively in the later sections.

3.2 Korean STI Policy Structure

As suggested in Section 2, a political structure includes actors, their functions, and rules defining the actors' activities. Based on this idea, we define the STI policy structure as one governing the relationships of major STI policy actors involved in the STI policy process and their functional relations constraining their activities. For example, in a regular committee, an STI think tank such as STEPI puts policy ideas into MOST's (Ministry of Science and Technology) plans for their own next year's R&D programs every year. This policy process reveals the STI policy structure in which a think tank is involved. In this way, we identify the STI policy structure embedded in the STI policy process.

According to Korean STI policy structure, we allocate various STI policy actors at different levels of 1, 2, 3, and 4, respectively, as shown in the figure. The vertical dimension represents governance according to the hierarchy. For example, the president and presidential offices coordinate and align different ministries' STI policy-planning activities. Moreover, each ministry has developed their own agencies for R&D planning and evaluation such as KISTEP for MSIT (Ministry of Science & ICT), NRF for MOE, and KIAT for MOTIE (Ministry of Trade, Industry & Energy). At the bottom level (i.e. Level 4), R&D performers including universities are allocated.

[Figure 3-3] R&D Governance System in South Korea



Source: Authors and adapted from Yang et al. (2016)

Based on the STI policy structure given above, we discuss several characteristics of the actors involved in STI policy activities according to the different four levels. Firstly, Level 1 represents the scene that the highest actors coordinate and integrate the overall priority setting over a national STI agenda. However, MOEF (Ministry of Economy & Finance) has a strong control over the allocation of the R&D budgets and staffing. Therefore, MOEF tends to weaken the function of coordination and integration, which is usually carried out by PACST (Presidential Advisory Council on Science & Technology) and S&T policy advisor in the presidential office. Conclusively, higher-level actors' STI policy coordination over the whole NIS is strongly restricted (Hong, 2016).

Secondly, at Level 2, in the early catch-up period, MOST was the exclusive ministry implementing science and technology policy in the Korean NIS. As the other ministries have recognized the importance of R&D for carrying out each policy in their respective areas, they have developed their own R&D policy implementation system in a multi-faceted way. Thereafter, increasing overlaps between each ministry's expanding R&D agenda and programs have become main barriers for the efficient operation of the whole NIS. Even so, coordination and integration between ministries' R&D program and funding have been

delayed.

Thirdly, Level 3 includes various R&D agencies carrying out planning, funding, and evaluation functions on behalf of the ministries affiliated. One of the main weaknesses of these organizations is autonomy and expertise issues. They suffer from insufficient delegation, which hampers efficient decision-making based on specialties they have accumulated. Ministries' direct intervention in the implementation process often causes a rent-seeking problem (e.g. funding decision by personal relationships).

Fourthly, the street-level R&D performers such as university researchers and PRIs are located in Level 4. The major bottleneck at this level is arguably the effective university-industry-government relationship. In other words, an institutional framework promoting interdisciplinary research and circulation of researchers are not still sufficient for preparing an upcoming technological paradigm such as AI and bio-health. Moreover, a new division of labor between R&D actors is required, which means that PRIs act as an R&D network hub and university researchers have to produce more creative knowledge at a global level.

3.3 STI Policy Process

3.3.1 General Characteristics of STI Policy Process

■ Policy-agenda setting

The first stage of policy process is to identify social problems and issues, calling for the intervention of governments. As the official attention based on the limited cognitive resources, governments are not able to address all the social problems and issues. The identification and screening public issues chosen by the decision-makers are the main activities of agenda setting.

The policy agenda is generally set by the following: (1) government officials, (2) political parties, (3) media outlets, and (4) interest groups. Government officials such as the president, ministries, and national assembly members are the obvious agenda creators. Regarding the STI policy system, R&D agenda setting involves the political activities dealing with the people's needs and conflicts between expert groups. Usually, the pressure and

needs for applied research and developments have come from outside scientific communities, while scientists' communities are likely to determine the agenda of basic research.

It is also obvious that the policy practitioners act as efficient R&D agenda setters under politically identified environments, based on related legal authority or orders. Moreover, the media deals with, collects, and delivers the social problems and issues. Not only the government officials, but also the scientist groups carry out the role of setting the R&D agendas. For example, the scientist groups set the priority on their R&D agenda, and the success largely depends on the R&D performers including universities and PRIs.

■ Policymaking

Policymaking is to describe, receive, or deny the policy alternatives solving the policy problems. Policymaking signifies the description of policy objectives and the selection of the most appropriate policy means or instruments. Since policymaking greatly relates to the following stage of implementing policy, the distinct boundary between these two stages is often unclear. However, the policymaking stage focuses on the perfection of action alternatives, while the next stage addresses the formal choice of policy actions.

Higher level of R&D actors such as the president and ministers are strongly involved in STI policymaking. Based on the R&D agenda given by people and political forces, and themselves, STI programs are planned and specified. From the perspectives of an STI think tank, this stage highlights the importance of scientific knowledge and evidence of the alternatives. The STI think tank mainly collects the R&D ideas about policy agendas, communities and policy learning.

..Policymaking occurs in government bureaucracies; interest group offices; legislative committee rooms, meetings of special commissions; and policy-planning organizations otherwise known as 'think tanks'. The details of policy proposals are usually formulated by staff members rather than their bosses, but staffs are guided by what they know their leaders want' Thomas R. Dye (2005: 42).

■ Policy implementation

Policy implementation is a stage that converts the new laws and financial programs into actions. The success of policy depends on the government decisions and its formal and informal structures. The implementation stage could look like an automatic extension of the policymaking stage. It is arguably the 'black box' bridging the divide between policymaking and policy results.

STI agencies and R&D performers at the street level are the main actors for STI policy implementation. Usually, STI agencies create more detailed R&D programs and the following projects. Moreover, in order to implement these programs and projects, they choose R&D performers, manage the R&D process, and evaluate the final R&D outcomes. Accordingly, R&D performers hire researchers, buy research equipment, and do research activities. Sometimes R&D performers such as university academic experts join the R&D committee managing larger R&D programs by screening proposals and evaluating research results.

■ Policy evaluation

After governmental officials carried out policies, the evaluation stage follows. The focus of the stage is to evaluate the output of the decision-making and policy goals that are initially targeted. The evaluation creates feedback to the policy results, which empowers the policy decision-makers to monitor and derive lessons from policy implementation. This feedback procedure helps to find new social problems and issues, which are building new policy goals and means at the subsequent period. This makes policy evaluation another influential device for the policymaking process.

Regarding the R&D process, the organized R&D policy evaluation or more specific programs comprise of five areas: (1) the need for specific social problems and issues, (2) the R&D program design, (3) R&D implementation, (4) R&D performances, and (5) R&D efficiency and effectiveness.

According to the levels of R&D, ministries and R&D agencies normally carry out these

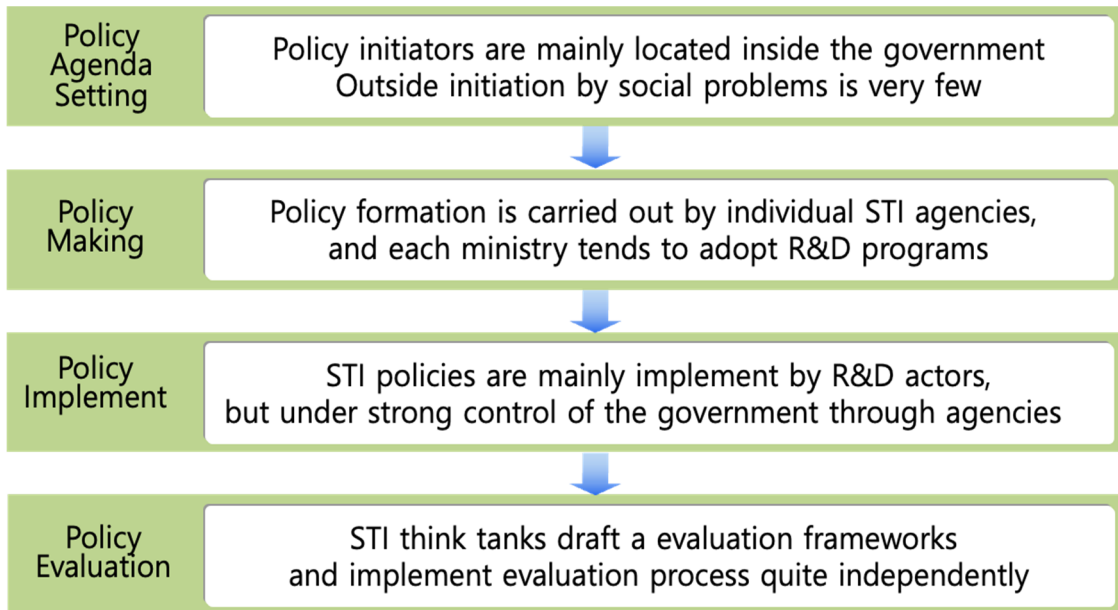
activities at different levels such as system, program, and project ones. At the system level, STI think tanks collect R&D statistics and qualitative information on NIS. They implement professional researches and diagnosis on whether the NIS system works well. At the program level, the results of R&D program feedback onto the next program planning and R&D goal setting. At the project level, R&D agencies accumulate the various profiles on R&D actors, process, and results.

3.3.2 Stylized Facts in STI Process in Korea

In Korea, we identify some stylized facts in the STI process consisting of policy agenda setting, policymaking, policy implementation, and policy evaluation. Firstly, regarding agenda setting in R&D, outside initiation is limited. In other words, policy practitioners such as bureaucrats and scientists dominate R&D agenda setting, rather than the media or people, because R&D tends to be recognized as an area of expertise in Korea. Therefore, scientific communities actually determine the needs and upcoming agenda. Sometimes government officials regard scientists as a means of incorporating public opinion onto urgent social problems and issues.

Secondly, in general, after STI agencies prepare detailed policy alternatives, ministries take over initiatives to determine overall directions and strategies of the alternatives. In some areas dominated by the ideas of powerful political groups, their preferences influence the choice of policy alternatives. Thirdly, R&D policy implementation is the duty of R&D actors such as researchers in universities and public research institutes. However, they are still under the strong control of R&D agencies and ministries. Fourthly, as opposed to project evaluation, program evaluation tends to lack in strong attention from them. What's more, we can seldom find the effective practice of R&D system evaluation or NIS in general.

[Figure 3-4] Characteristics of STI Policy Process in Korea



Source: Authors

4. Analysis of the Korean STI Think Tank

4.1 Overview of the Analytical Framework

In this section, we integrate the STI structure and STI policy process in order to analyze the role of an STI think tank in Korea. For STI policy structure, we introduce various R&D actors at four different levels accordingly, whereas, for STI policy process, we review characteristics of STI process in Korea. By combining the two frameworks, we then analyze how the STI think tank works considering the concept and role of an STI think tank in the national innovation system.

4.2 Mapping STI Policy Process & Think Tank

This figure illustrates the ‘to-be’ positioning of think tank in the process of STI policy, mapping it with Korean R&D governance. The horizontal axis demonstrates the sequential process, while the vertical axis is for the structural governance from the previous framework. Firstly, the main actor in the agenda-setting stage is the P.O. (presidential office) and its advisory bodies. In Korea, the Presidential Committee on the Fourth Industrial Revolution (PCFIR, or 4th IR) and Presidential Advisory Council on Science & Technology (PACST) play the main roles in collecting information about the issues, providing advice for methods of innovation to allow the significant development of S&T policy.

The 4th IR basically focuses on the recent specific technology sectors, coordinating important policies pertaining to the development and acquisition of artificial intelligence (AI) and data technology, as well as new industries and services. The committee’s legal framework is found in the Presidential Decree on the Creation & Management of the PCFIR, promulgated and in effect as of August 22, 2017.

The main mission of this specialized committee includes 1) the deliberation and

coordination of the policy mix submitted by various ministries and committee members, 2) organizations of public campaigns related to the 4th Industrial Revolution 3) encouragement of public participation, 4) preparation of the groundwork for regulatory and institutional reforms in support of public-private partnerships. The committee members consist of governmental and civilian members. The subcommittee specialized on six sectors as follows: Data, smart city, digital healthcare, S&T, industrial economy, and social institutions.

PACST has superseded the function of the preceding National Science & Technology Council, supporting the innovation of national science and technology by deliberating the decision-making process related to science and technology policies. The committee provides advice on the methods of innovation to allow the significant development of our science and technology, plays an important role in the improvement of our society and economy, and deliberates on the policies to put the contents of the advice into practice¹. The legal authority for the committee comes from the amendments to the Presidential Advisory Council on Science & Technology Act in 2018. It consists of professional advisors from various fields, while supporting staff routinely deliver the agenda and set the priorities. The advisory council has up to 20 members with deliberate members from civilian and government authorities.

Next, ministries allocate the fiscal budget in the decision-making stage. In STI policy, there are three R&D related ministries -MSIT, MOTIE, MSS – and the budget controlling MOEF. The R&D related ministries are involved in policy decision-making by distributing the R&D programs to the intermediary agencies and planning the policy measures. The role of think tank in the decision-making stage is to provide the policy research in shaping the policy mix.

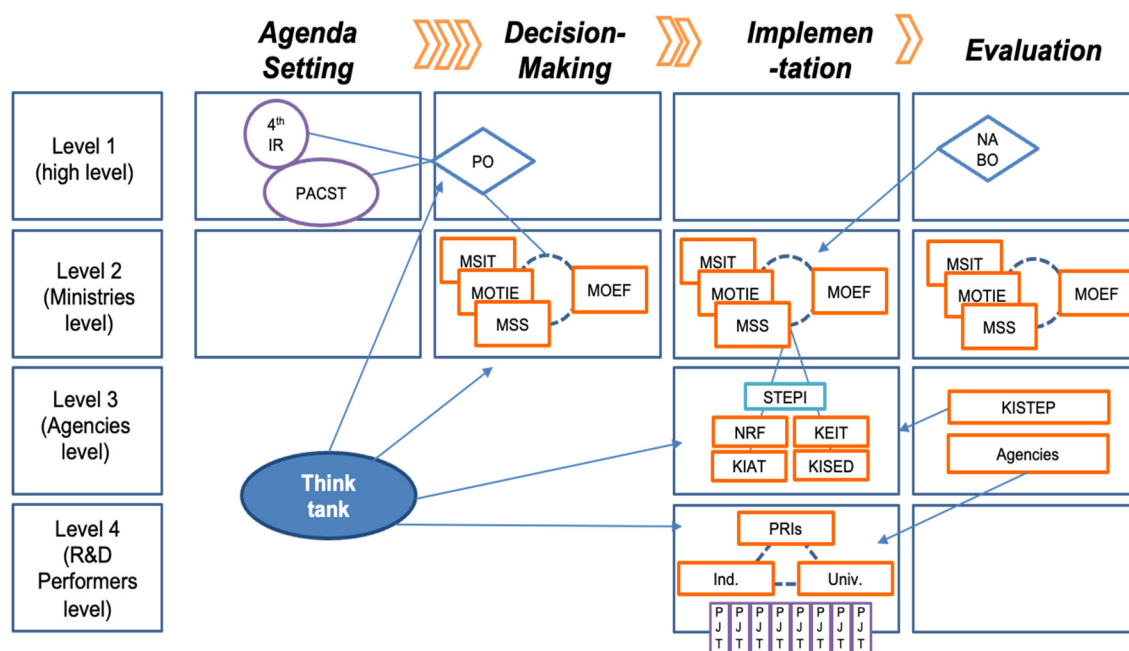
The implementation stage involves the intermediary agencies that connect the upper level ministries and lower R&D actors. Each ministry has its own intermediaries. For instance, NRF (National Research Foundation) allocates the R&D program of MSIT to university and individual scholars. While KIAT (Korea Institute for Advancement of Technology) and KEIT

1 Source: official homepage of PACST, 4th IR.

(Korean Evaluation Institute of Industrial Technology) mainly back-up the R&D programs from MOTIE, KISED (Korea Institute of Startup & Entrepreneurship Development) supports the Ministry of SMEs & Startups. STEPI sits on the fence between each ministry and supporting agencies. At Level 4, the organizations including university, industry and PRIs work on the R&D projects.

Each individual project at the Level 4 is generally evaluated by the supporting agencies that report the results to the Ministries. The joint R&D programs, that consist of multiple ministries, are evaluated by KISTEP (Korea Institute of S&T Evaluation & Planning), which was a part of STEPI. From the macro-view, NABO (National Assembly Budget Office) specialized in checking and monitoring the higher-level of budget programs by ministries.

[Figure 3-5] Mapping STI Policy Process & Think Tank



Source: Authors

The roles of think tanks in each stage of STI policy are presented in conformity with the actors and activities. At the agenda-setting stage, the think tank is required to have holistic policy coordination, help macro-system transition and adjust the goals and needs of main actors. At the subsequent stage, think tank identifies and compares the policy alternatives and then predicts policy outcomes in advance. The think tank usually does not get highly involved in the implementation stage but needs to align the conflicts between heterogeneous organizations and interest groups. Lastly, in the evaluation stage, the main role is to collect the evaluation results and analyze the implications for the feedback loop.

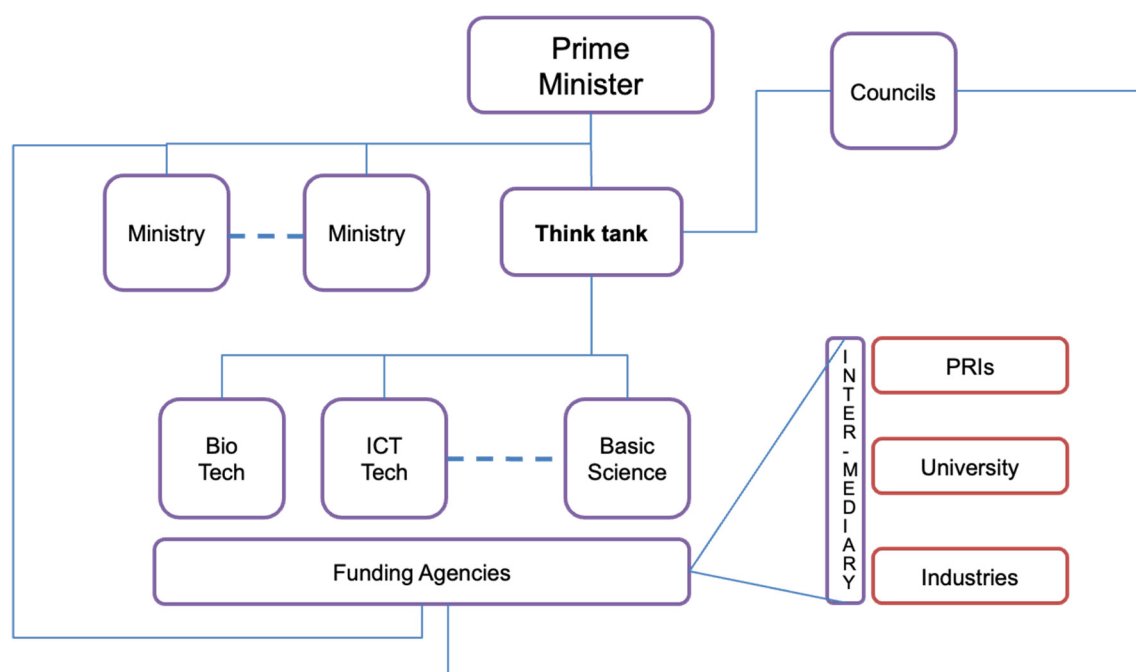
[Table 3-1] Mapping STI Policy Process & Required Role of Think Tanks

	Agenda aetting	Decision-making	Implementation	Evaluation
Main actors	- P.O. - National assembly - 4th IR - PACST	- National assembly - Ministries	- Ministries - Intermediary agencies - PRIs - Universities - Industries	- NABO - Ministries - Intermediary agencies - KISTEP
Activities	- Identify policy needs - Policy initiation - Vision setting	- Policy priority setting - Selecting policy alternatives - Budgeting	- Performing R&D - Provision of R&D outcome	- Program evaluation - PJT evaluation - Resource allocation
[Required] Functions of think tank	- Holistic policy coordination - Helping system transition - Screening & adjusting goals and needs	- Identifying & comparing alternatives - Predicting policy outcomes	- Aligning R&D efforts - Diagnosis of R&D process	- System evaluation & feedback

Source: Authors

Then, the illustration describes the ‘to-be’ positioning of utilizing its power to collect the relevant information and control conflicts. Ideally, the think tank needs to be put under the direct control of the PM in order to monitor the technology sectors by observing not only the agencies but also the implementation actors including PRIs, university and industries.

[Figure 3-6] STI Governance and Think Tank



Source: Authors

In summary, the main implication in this part is to derive the main role of think tanks in the process. Our discussion reaches an agreement that the main role is to coordinate STI policy by coordinating across technology sectors and aligning the policy actors from agencies to R&D actors. We also highlight that a think tank is required to have a more holistic approach in predicting, confronting, and planning the transition of policy systems.

5. Definition of Think Tank in Indonesian Context

5.1 Concept of Think Tanks

The UNDP Program defines think tanks as “... organizations engaged on a regular basis in research and advocacy on any matter related to public policy. However, the application of think tanks, including in Asian countries, may place less importance on independent research. They serve as the bridge between knowledge and power in modern democracies” (UNDP, 2003:6).

Similarly, the Cambridge dictionary defines think tank as a group of experts brought together, usually by a government, to develop ideas on a particular subject and make suggestions for action².

Think tanks are characterized as public policy research analysis and engagement organizations that generate policy-oriented research, analysis, and advice on domestic and international issues, thereby empowering policy-makers and the public to make informed decisions about public policy. These institutions often act as a bridge between the academic and policymaking communities and between states and civil society, serving in the public interest as an independent voice that translates applied and basic research into a language that is understandable, reliable, and accessible for policy-makers and the public. The key function of a think tank is ‘to connect policy ideas and knowledge with other researchers and institutions, and sometimes with people having different backgrounds or ideologies’ (NIRA, 2005).

The Anglo-American tradition regards think tanks as relatively autonomous organizations with a separate legal identity that engage in the analysis of policy issues independently of the government, political parties and pressure groups. Think tanks are commonly thought to provide independent, well-researched advice and advocacy on complex social issues.

² [https://dictionary.cambridge.org/dictionary/english/think-tank.](https://dictionary.cambridge.org/dictionary/english/think-tank/) / Accessed on 7 November 2021

(Stone, Diane. 2005). In general, think tanks produce accumulated ideas or concepts and inevitably disseminate them to the targeted spheres, especially policy-relevant fields (McGann, 2018).

Think tanks act as policy-knowledge intermediaries, research centers, and incubators of new ideas. As intermediaries, they bridge knowledge between academic experts, policy-makers, and civil society. As applied researchers, they turn multidisciplinary theories and empiricism into insights and recommendations packaged to inform and meet the needs of decision-makers. As innovators, they broaden the perspectives and possibilities of the policy world — for example, demonstrating the power of entrepreneurial solutions. Above all, think tanks provide information that is credible, relevant, and easy to understand. (CIPE, 2021)³

5.2 The Importance of Think Tanks

Think tank basically serves as a bridge between science and government policies. The complexity of the problem and its continuous development arise into challenges the government has to manage and overcome based on science knowledge. Think institute government tanks have the main task of formulating how science is transformed through policy research, so that recommendations for proven policies are generated.

As we've seen, the U.S. that has become the world's largest economy, is a successful country due to the functioning of the bridge between knowledge and policy in its government. Without negating other important factors, one of the factors is the existence of a short, medium, and long-term development strategy that are very well prepared, synergized, and scripted by think tanks that are then implemented consistently by the government.

In the context of public policy and development planning, the role of government think tanks is very important to support the birth of quality public policies and development plans that are realistic, measurable, and implementable. The development of the times demands a government that must be able to follow the progress of civilization. Today's government

³ <https://www.cipe.org/reports/how-to-guide-for-economic-think-tanks/the-role-of-think-tanks/> Accessed on 7 November 2021

must be able to provide greater benefits for its people through breakthroughs produced by science. One of them concerns government think tanks that play a major role in accelerating the development of a country.

Government think tanks basically function as a bridge between science and government policies (bridging research to policy, bridging knowledge to power). The think tank will formulate how this knowledge is transformed into policy research, so that recommendations and policy solutions are produced in the form of proven programs. So think tanks are needed to synergize all existing knowledge with government policies and programs.

5.3 Think Tank Classification

1. Think tanks based on independent civil society founded as a non-profit organization
2. Research institutions policies located in or affiliated with a university
3. Think tanks created by a government or sponsored by the state
4. A think tank-created company or affiliate of a business
5. More contemporary development specialties such as environmental think tanks (e.g. Institut Thai environment) and economic policy think tanks (e.g. Malaysian Institute of Economic Research), or a regionally focused think tank such as the Institute for South East Asian Studies (ISEAS) in Singapore. (Stone, Diane. 2005);
6. Some 'academic' style think tanks, focused on research, are geared towards university interests and building the base public knowledge
7. Political party think tanks (or candidates)

5.4 Think Thank Institutional Concept

Organizational arrangement and professional management determine the performance of

the organization. Robbins (1990) defines an organization as a consciously coordinated social entity, with a relatively identifiable boundary, that functions on a relatively continuous basis to achieve a common goal or set of goals. Ivanko (2013) sees organization as an activity, where organization is a conscious human activity to connect and coordinate production agents or techniques to combine processes for attaining goals with a purpose. Whereas, Mintzberg (1983) has the view that the organization is comprehensive in the perspective of the structure which he expressed as the structure of an organization being defined simply as the sum total of the ways in which its labor is divided into distinct tasks and then its coordination is achieved among these tasks. Mintzberg's view later became a role model in discussing the existence of think tanks in state-run organizations. In fact, a variety of Mintzberg organizational structures have been adopted for government organizations.

The transformation of government organizations is a necessity, whose aim is to improve the performance of public services through the creation of efficiency, effectiveness, flexibility, and adaptability according to the dynamics of society, the demands of public services, and the need for ideas for decision-making and policymaking, as well as encouraging the advancement of science and technology. The presence of government think tanks will become an accelerator of civilizational change in the digital era. The reason being that civilization is the broadest cultural entity defined by its main elements with a common purpose, such as language, history, religion, customs, institutions, and by human subjective self-identification. Moreover, civilizations have no clear boundaries and no exact beginning and end, while the composition and form of civilizations have changed over time.

Civilization is not only a meaningful and mortal entity but also long-lived, evolving and adapting, while being the most enduring of human associations. Meanwhile, according to M. A. Karim (2009), civilization is part of a culture that has a wide system of technology, building arts, fine arts, state-run systems, and science. He further emphasized through the general understanding used that civilization is part of culture that aims to facilitate and prosper human life. Along with changes in civilization in the digital era, adaptation efforts on the administration of government affairs require the existence and contribution of thoughts from think tanks.

Webster's New World Dictionary defines a think tank as (or factory) [slang]: a group or a center organized, as by a government or business, to do intensive research and problem-solving, esp. with the aid of computers and other sophisticated equipment (Neufoldt, 1991: 1391). In the perspective of a social scientist, Martsenyuk (2018:7), a think tank is defined as an organization that conducts research, policy analysis, or advocacy in one or more fields, and cooperates with national and local governments to develop, implement, and evaluate public policy. This understanding reflects the importance of the existence of think tanks, especially the role of think tanks in policy formulation and the public-policy cycle. Pautz (2011) defines think tanks comprehensively as non-governmental institutions that are intellectually, organizationally and financially autonomous from the government, political parties or organized interests; and formed with the aim of influencing policy. In line with that, Li, et.al. (2016) define think tanks as intellectually, organizationally, and financially autonomous institutions with the aim of influencing policy.

Internal think tanks can also be defined as local government institutions that have duties, functions, and roles, accompanied by responsibilities in R&D management to produce research-based planning and evidence-based policies through the use of digital technology tools or digital media channels. The goal is to create innovation, strengthen planning, find solutions, change civilization, make life easier, and improve the quality of policies (Herie Saksono. 2019)

R&D management is a managerial activity that includes program management/ research activities, development, assessment, implementation, engineering, operation, and policy evaluation. Civilization is defined as the highest human culture at this time, where all aspects of life rely on digital processes to accelerate the attainment of human welfare and sense of fulfillment (Herie Saksono. 2019).

Muhyidin, et.al. (2017) explain that government think tanks basically function as a bridge between science and government policies. The complexity of development problems, which over time become more severe, becomes a challenge for the government to be managed and overcome based on science. Government think tanks formulate how this knowledge is transformed through policy research to produce proven policy solutions.

What is the status of actual development of think tanks in Indonesia? Although not as fast as in developed countries, think tank groups in Indonesia have been much more active than before, and are increasingly contributing to public policy debates. Meanwhile, a workshop facilitated by Inspirit Innovation Circles and AusAID (October 2012) revealed the importance of think tanks in the context of public policy which need to be supported by the ability to: (1) present research results in critical situations that require quick decision-making, (2) find solutions to current problems, and (3) build credibility and effective communication. Another conclusion is that so far the strength of Indonesian think tanks lies in passionate human resources and think tanks' networks with the government, communities, NGOs, and universities. The next question is what is the map of institutions that can be categorized as Indonesian think tanks?

The inventory of institutions with the capacity as think tanks in Indonesia was actually carried out in 2012, the results of which were not satisfactory from being carried out spontaneously. It was identified that think tanks are still limited in the requirements to build institutions that implement studies and intellectuals, with issues that are discourse and cannot be transformed into policies. In Indonesia, several universities do have study centers but no inventory has been carried out whether they are included in the category of carrying out their functions and roles as think tanks. In fact, there are a good number of think tanks based on universities and corporations in Indonesia. However, the emerging innovations have yet to be translated into policy.

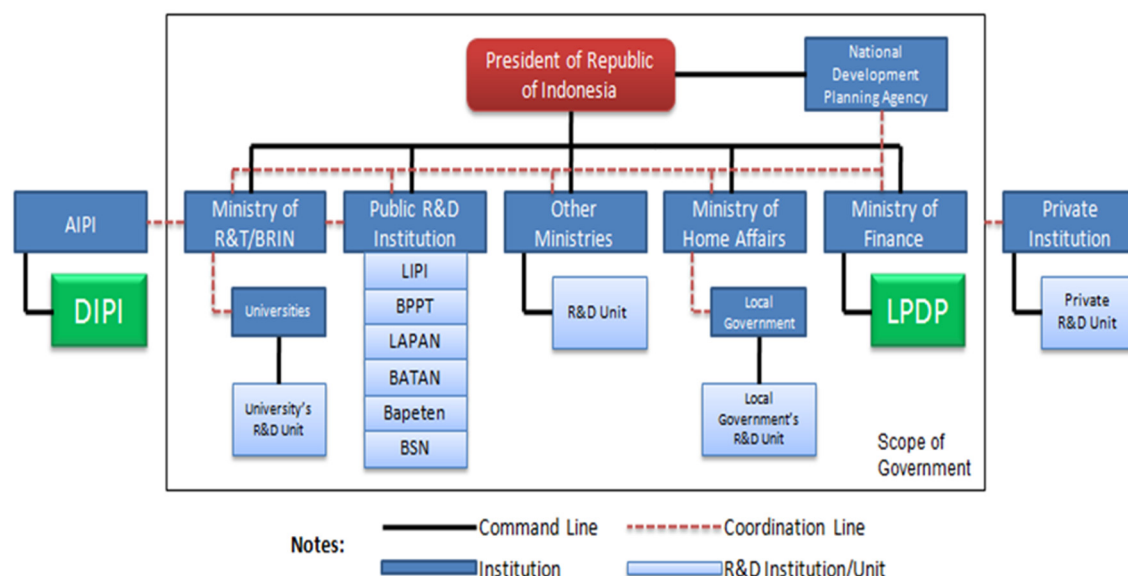
6. Overview of S&T Governance in Indonesia

6.1 The Indonesian S&T Governance Pre BRIN Formation

Indonesia is currently undergoing a very big change in the history of its S&T institutional governance. This is a mandate from the enactment of Law No. 11 of 2019 concerning the National System of Science and Technology, especially in Article 48. This article states the need for integration of R&D institutions in Indonesia which is embodied by the National Research & Innovation Agency (BRIN). This integration is intended to ensure the sustainability of one-way R&D activities in order to increase S&T capabilities in the national development. The agency was established by the government through the provisions of a presidential decree.

Before we enter the BRIN era, public R&D institutions in Indonesia were spread out in various institutional forms. The first form is public R&D institutions directly under the president, coordinated by the Ministry of Research & Technology (one of them is LIPI). The second form is R&D organizations under the ministries, state institutions, & universities, and in local governments. At that time, there were at least six R&D institutions directly under the coordination of the Ministry of Research & Technology, namely LIPI, BPPT, LAPAN, BATAN, Bapeten and BSN. Aside from that, almost every ministry/ state institution has an internal R&D unit.

[Figure 3-7] Governance of R&D Institution Before BRIN Era



Source: Hidayat et al (2020).

The results of the study conducted by Triyono et al. (2020) show that coordination of R&D programs only occurred among R&D institutions that are under the coordination of the Ministry of Research & Technology. However, it does not run optimally as the Ministry of Research & Technology doesn't have strong authority to create R&D institutions under its coordination running R&D programs in conformity with the national R&D programs. This is because the authority of R&D programs and funding thereof (DIPA-APBN^{4,5}) entirely resides in each institution. This results in programs overlap between institutions, which even compete with one another instead of collaborating. The poor coordination has also led to the scattering of R&D funds in various R&D institutions without a clear direction towards national R&D program objectives.

4 DIPA, *Daftar Isian Pelaksanaan Anggaran* (in Indonesia); Entry List of Budget Implementation (in English): the budget implementation document for each state-run institution in accordance with the various programs that have been proposed in the APBN

5 APBN, *Anggaran Pendapatan dan Belanja Negara* (in Indonesia); State Budget (in English): the state budget document for a period of one fiscal year

Actually, the mandate of 'integration' in Law No. 11 of 2019 concerning the National System of Science and Technology gave rise to several perceptions, which then produced two options for governance of R&D institutions. For the first option, the mandate of integration is defined as institutional integration. In this way, all (or maybe several) public R&D institutions (including LIPI) would be merged under the organizational umbrella of BRIN. The centralization of R&D institutions under the auspices of BRIN would facilitate the control of directed R&D activities in conformity with the national R&D program. This option is the most likely to be implemented because it comes from the political will of the elected president. Nevertheless, we realize that structural integration needs a long and comprehensive process to be done well. The integration of public R&D institutions from various sectors with a very broad field and scope is the main challenge that if not properly planned and designed, can become a threat to the continuity of R&D activities in Indonesia.

For the second option, the mandate of integration is not defined as institutional integration, but as only the integration of programs and funding thereof. This means that public R&D institutions remain in their current position, but the authority in the formulation of an R&D program and its budget are entirely taken over by BRIN. In other words, public R&D institutions (including LIPI) will only work to carry out what has been planned by BRIN. Public R&D institutions do not have R&D programs and its own funding, and the DIPA-APBN attached to the institution will only be used for the development of R&D resources; such as infrastructures, researchers and other human resource competencies, and networks. Some parties also refer to this option as 'strengthening the coordination'.

This option seems to be the best choice to be implemented in Indonesia because this integration is carried out without changing the existing institutional structure. However, this option has a major weakness in that it can't integrate R&D units under ministries (the proportion is more numerous than the amount of research institutions such as LIPI), due to the absence of a command and coordination line between BRIN and ministry's R&D units. However, R&D units will be working based on programs and duties given from their own ministries. The experience in Indonesia shows that without any clear lines of coordination, each institution tends to run on its own, so that collaborative processes are difficult to carry out. This will hamper the sustainability of one-way R&D activities in order to increase S&T

capabilities in the national development.

Then, the first option has become the government's choice, where the government finally decided to integrate all public R&D institutions by merging them into a uniform BRIN structure. This decision was realized by ratifying Presidential Decree No. 78 of 2021 concerning the National Research & Innovation Agency (BRIN). Thus, it can be said that BRIN is the only public R&D institution in Indonesia, apart from R&D units owned by universities.

6.2 Post BRIN Formation & Its Implication for STI Policy Think Tank in Indonesia

Before the BRIN era, Indonesia didn't have a clear and concrete think tank or policy institute, especially in the field of S&T. For example, is LIPI, where the institution has a work unit called the Center for Policy Research & Management of Science, Technology, & Innovation (P2KMI). However, P2KMI actually plays a major role in research in the fields of S&T and innovation policy. This function positions P2KMI as a center for policy research, so that its main role is more focused on producing scientific publications rather than policy advocacy. Accordingly, the function of the think tank is only supplementary, not the main one, in the implementation of P2KMI's role in disseminating the results of its policy research. This can be seen from a lack of clarity regarding the specific clients who will be the users of the analysis results. In turn, this has an impact on the minimal use of research results by policy-makers.

Policy analysis and policy research activities are different from one another, although they are cross-related and identical at the same time. Policy analysis is the application of existing knowledge to solve a public problem (Dunn, 2003). Moreover, William (1971) defines policy analysis as a process or activity of synthesizing information, including research results, to produce public policy recommendations. The main activity of policy analysis is the systematic collection of information and drawing of logical conclusions from that information. Thus, it can be said that policy analysis does not belong to the category of research but must be carried out scientifically; it must be systematic and logical, and use

information and knowledge, including the results of scientific research.

Policy research becomes one of the sources of information that can be used by policy analysts. This means that policy analysis is a further processing of various information, including research results, so that it is ready to be used in decision-making and public policy design. Therefore, according to Simatupang (2003), policy analysis is basically a vehicle for advocating various data, information, scientific evidence, and policy research for the policymaking process. It is this policy analyst who is actually needed by a think tank unit because it has the academic and political competencies to connect various types of evidence as well as the results of policy research with policy-makers. Meanwhile, P2KMI, as previously explained, is not fully equipped with these competencies.

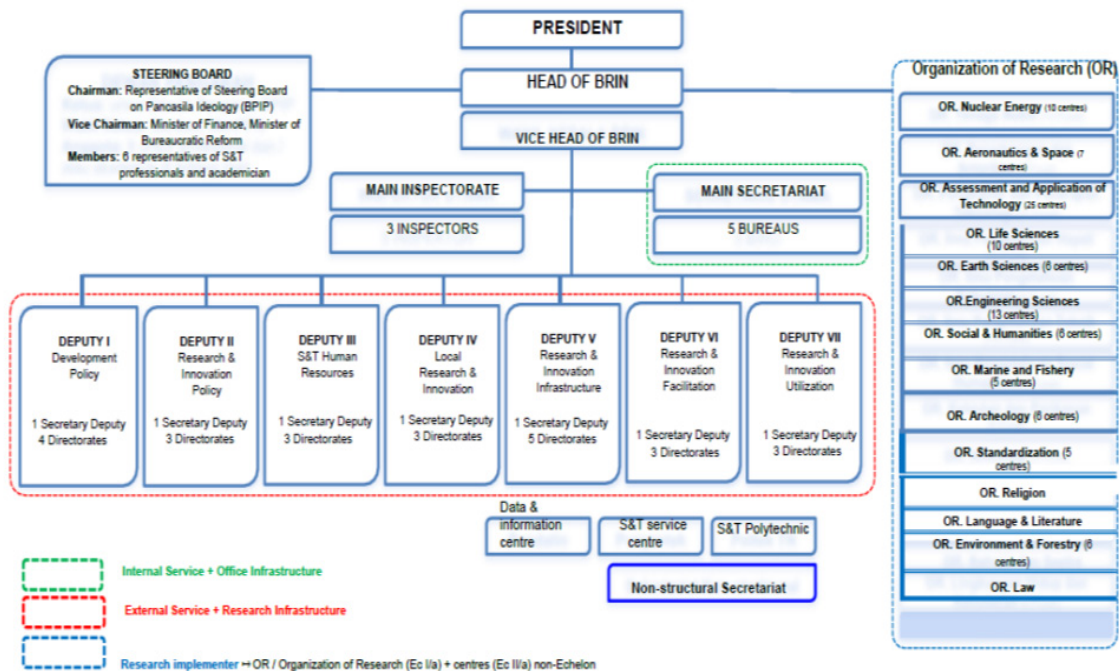
The Indonesian government seeks to make S&T the basis for making various development policies to encourage economic growth and improve industrial competitiveness. In an institutional context, this takes place through the establishment of BRIN which integrates all public R&D institutions to facilitate better coordination in the management of S&T resources in achieving national R&D programs. In this way, BRIN is expected to become the government's main think tank in various national development sectors to encourage evidence and scientific support-based policymaking.

As an institution that unites all government R&D activities, BRIN's scope covers the whole breadth of R&D from upstream to downstream. It starts with exploration of science in the form of basic research, continuing on to applied research, then moves on to development in the form of inventions until research results and inventions flow downstream to users to push innovation. All the basic and applied research activities take place inside the research organization (OR). The development and inventions flowing downstream to users become the main task attached to the function of the BRIN deputy and is part of the external services provided by BRIN.

In this case, a think tank becomes a part of downstreaming policy research results to policy-makers, so that all these functions will be facilitated under the BRIN deputy. But interestingly, even though all research activities (both basic and applied) are carried out in

the OR, but specifically for policy research, including STI policy research, the activities will be carried out by the BRIN deputy. It aims to bring the role of policy research closer to policy analysis to facilitate the transfer of knowledge in an effort to conduct policy advocacy.

[Figure 3-8] Draft of Organization Structure of BRIN



Source: BRIN (2021)

Based on BRIN Regulation No. 1 of 2021 concerning the organizational structure and work procedures of BRIN, this agency has seven deputies and 14 ORs, (while it is still possible to continue to grow in tandem with the transition process that is still ongoing today). From the seven deputies that have been formed, two of them have functions that are very closely related to BRIN's role as a national think tank.

The first is the deputy for development policy. The deputy for development policy has the task of carrying out the implementation of R&D as well as inventions and innovations in the framework of formulating policy recommendations and directions & principles of the design of national development planning as a basis in all areas of life. This deputy has duties as the think tank of the National Development Planning Agency (Bappenas) in formulating national

development plans and policies. In fact, to encourage collaboration from BRIN to fully facilitate Bappenas in preparing the entire science-based policymaking process, various directorate nomenclatures and their functions under this deputy are adjusted to the nomenclature of work units of Bappenas. This was done to facilitate coordination with Bappenas and aimed at promoting the role of BRIN as a think tank in national development policies.

The second is the deputy for research & innovation policy. The deputy for research & innovation policy has the task of carrying out the implementation of R&D as well as inventions and innovations in the framework of the formulating policies in the field of research and innovation, including a master plan for the advancement of science and technology, and roadmaps for research and innovation. This deputy has duties for the BRIN think tank in formulating national STI plans and policies.

[Table 3-2] Type of Researches Based on Institutional Arrangement: Case of BRIN-Indonesia

Basic research	Applied research	Policy research	STI policy research
Economic researches	Finance & Tax researches	Food policy research	Research & development (R&D) policy research
Social-cultural researches	Agriculture & Food researches	Transportation policy research	Science-techno-park (STP) policy research
Earth science researches	Fishery & Marine researches	Education policy research	Technology-based Industrial policy research
Engineering researches	Health researches	Immigration policy research	Innovation ecosystem & policy research
Life science researches	Security & Law researches	Tourism policy research	R&D finance policy research
Other science basic researches	Other science applied researches	Other policy researches	Other STI policy researches

Source: Author's Analysis

BRIN divides four types of research supporting performance of BRIN with a national scope. The first is basic or core researches that are aimed at developing derivative research fields such as economic, socio-culture, earth science, engineering, life science researches, and other science basic researches.

The second one is applied or derivative researches that are aimed at closing basic research to real-sector conditions. For the most part, it is aimed to address currently existing issues in particular sectors, such as finance & tax, agriculture & food, fishery & marine, health, security & law researches, and other science applied researches.

The third one is policy research which is one of the applied research types that focuses more on addressing public issues and impacting problem solving in specific or sectoral public issues. In this research, stakeholders or governmental actors are the target of these research fields because it leads to specific guidance in governing many actors of both government and non-government organizations. The examples of this research are food policy, transportation policy, education policy, immigration policy, tourism policy researches, and other policy researches. The last is science-technology-innovation (STI) policy research which is the specific policy research focusing on STI issues such as technology catch-up, knowledge-based economy (KBE), innovation system, innovation ecosystem, technologies-based industry (TBI), etc. Those examples are research & development (R&D) policy, science-techno-park (STP) policy, technology-based industrial policy, innovation ecosystem & policy, R&D finance policy researches, and other sti policy researches.

All research types are required to create a national innovation environment, BRIN accordingly will close from basic, applied, and policy researches, including STI policy researches. Science for national and science-based policies is the main goal of BRIN in creating national innovation environment supporting national economy development of Indonesia. Therefore, basic and applied researches are integrated into policy researches and STI policy researchers. Basic and applied researches are the domain of research centres at the organization of research (OR), while policy researches are the domain of Deputy I – Development Policy and STI policy researches are the domain of Deputy II – Innovation & Research Deputy. Thus, ORs as science and technology (S&T) suppliers and the deputy as science and technology (S&T) users are connected to make a robust national innovation environment in Indonesia.

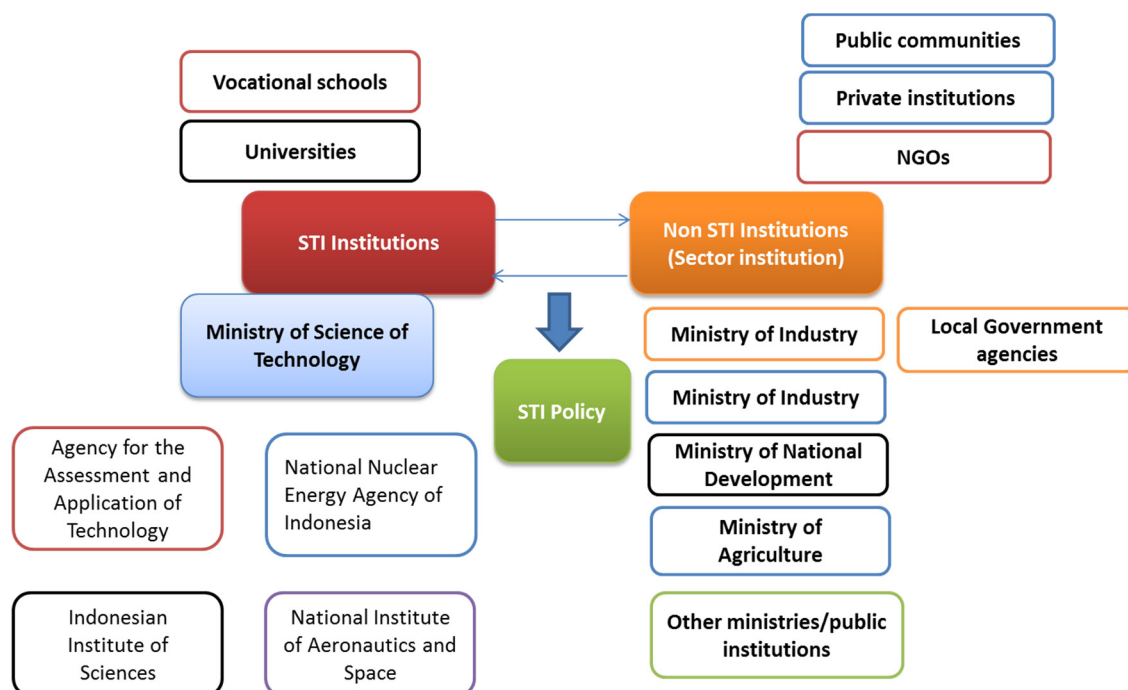
7. Ideal STI Policy Think Tank & Practice in Indonesia

7.1 Ideal STI Policy Think Tank Relationship

The ideal STI policy think tank relationship is not dominated by STI institutions, but rather by non-STI institutions. Specific to Indonesia, before integration within BRIN, STI institutions are vocational schools, universities, and related education institutions. Again, public research and development (R&D) institutes are the Indonesian Institute of Sciences (LIPI), Agency for the Assessment & Application of Technology (BPPT), the National Nuclear Energy Agency of Indonesia (BATAN), and National Institute of Aeronautics & Space (LAPAN). The four R&D institutes are under the coordination of the Ministry of Research & Technology (Kemenristek) (refer to Figure 3-9 on the left side).

Non-STI institutions are sectoral institutions focusing on specific/ particular sectors in public domains. They are private institutions such as enterprises, bank; non-government organizations (NGOs) such as foundations, associations, public communities; and state-run agencies. The last are state-run agencies which are central and local government agencies. Central governments include the Ministry of Industry, Ministry of National Development, Ministry of Agriculture, and other ministries/ state-run agencies at the central government level (refer to Figure 3-9 on the right side).

[Figure 3-9] Ideal STI Policy Think Tank Relationship before Integration within BRIN



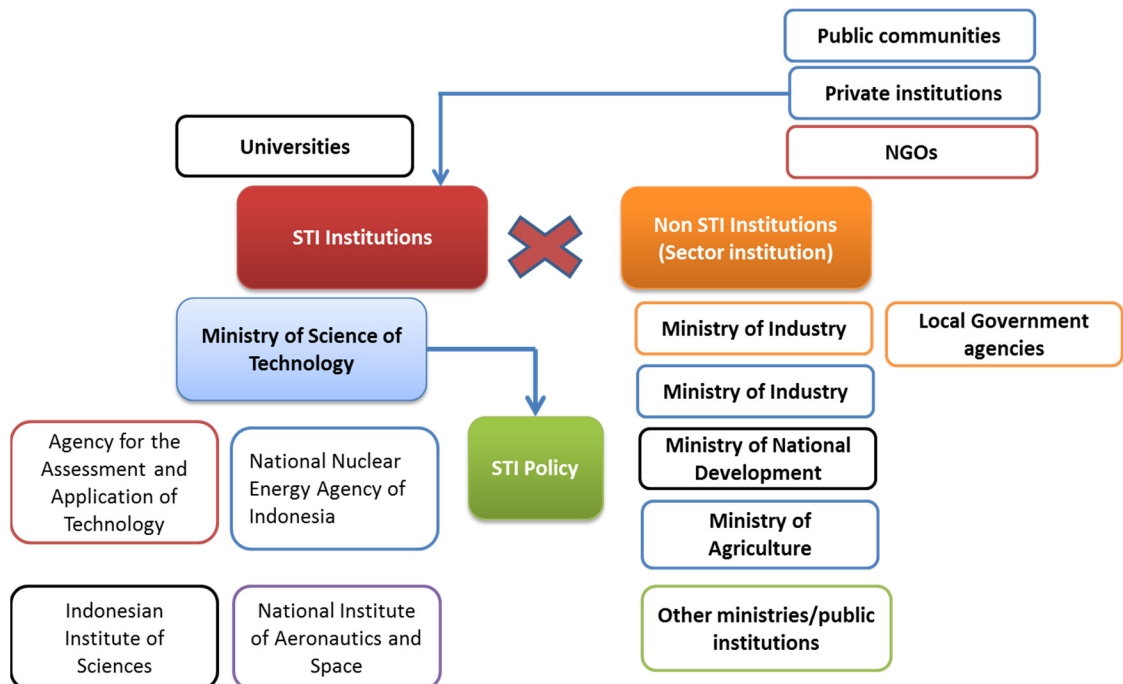
Source: Author's analysis referring to Dodgson (2000) & Edler et al. (2016)

Science-technology-innovation (STI) policy is not merely policy-addressing R&D activities, but also policy focusing on development of a national economy system in each country. Therefore, a linkage between R&D and non-R&D institutes is required (Dodgson, 2000). STI policy are a meeting point between the supply side of knowledge (universities, R&D institutes, training units) and demand side of knowledge (communities, firms, other users) (Edler et al., 2016). Role of STI policy is to close any gap or to connect between STI institutions as S&T suppliers (left side) and STI institutions as S&T users (right side). STI policy does not exist without a specific sector that needs to be addressed. It means that STI policy is only a means to address public specific issues based on science-based problem solving such as a knowledge-based economy (KBE), technologies-based industry (TBI), innovation system, etc. However, STI policy is not single entity but rather involves many aspects and actors influencing the STI policymaking process.

7.2 The Practice of STI Policy Tank Relationship in Indonesia

Practice of an STI policy think tank relationship in Indonesia is not an ideal type as suggested in Figure 3-9 above. The studies of Lakitan (2013), Asmara et al. (2019), and Triyono et al. (2020) show that weak and silo coordination among the Indonesian STI and non-STI actors had been an age-old unresolved problem until this study was finished at the end of November 2021. After the new order regime collapsed in 1997 and the reformation regime emerged in 1998 to the present, there is no an integrated mechanism to bring up R&D results yielded by researchers from public and private R&D institutes and universities. To be known, in the new order regime, the central government through the Agency for the Assessment & Application of Technology (BPPT) & Ministry of Research & Technology (KRT) has strong power to mobilize public R&D research institutes and state-owned strategic enterprises (BUMNIS) in a single mechanism to commercialize R&D results.

[Figure 3-10] The Practice of STI Policy Think Tank Relationship before Integration within BRIN



Source: Author's analysis referring to Dodgson (2000) & Edler et al (2016)

STI institutions are only universities and public research and development (R&D) institutes such as the Indonesian Institute of Sciences (LIPI), Agency for the Assessment & Application of Technology (BPPT), the National Nuclear Energy Agency of Indonesia (BATAN), and National Institute of Aeronautics & Space (LAPAN). They are coordinated by the Ministry of Research & Technology (Kemenristek) (refer to Figure 3-10 on the left side). For this case, the Ministry of Research & Technology is the single actor to mobilize all public S&T resources including LIPI, BPPT, BATAN, and LAPAN. The role of the Ministry of Research & Technology is very limited, not reaching fully private R&D institutes. Universities are not main authority of the Ministry of Research & Technology, while intervention of the Ministry of Research & Technology is limited on R&D incentives without mobilizing their resources.

Input of STI policy at the Ministry of Research & Technology is also from public communities, non-government organizations, and enterprises. Those actors deliver their problems and propose suggestions to the Ministry of Research & Technology related to the many issues of STI. Unfortunately, there is no connection between STI institutions and non-STI institutions to respond to many STI issues from these actors, and the Ministry of Research & Technology will maximize internally existing resources to respond to STI issues suggested by them. Indeed, coordination mechanism from the Ministry of Research & Technology to public R&D institutes (LIPI, BPPT, BATAN, and LAPAN) is often quite difficult.

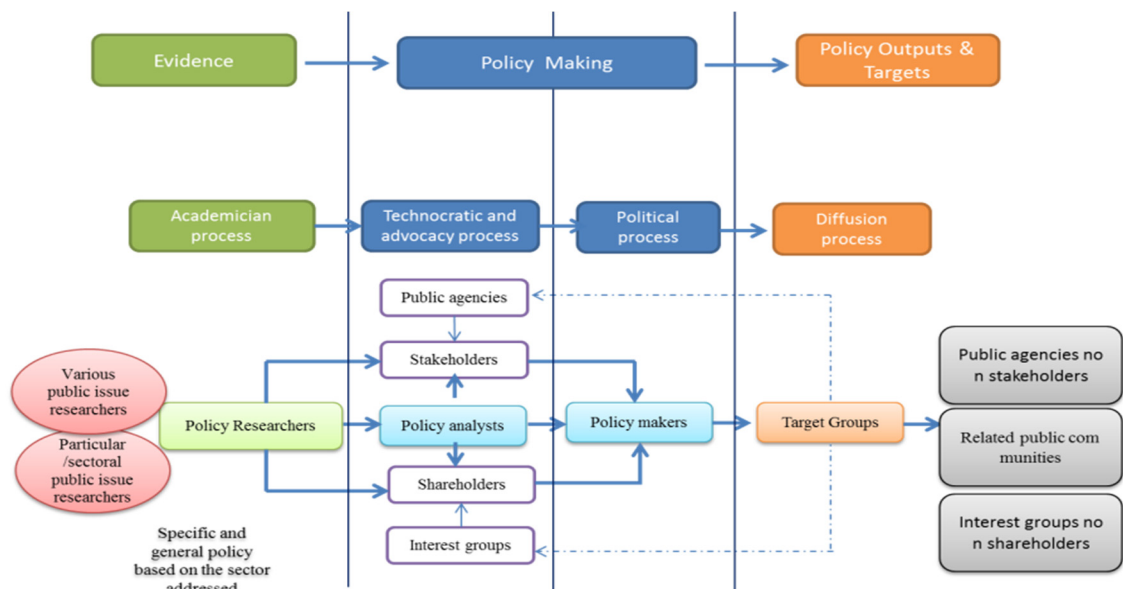
Non-STI institutions as knowledge users such as the Ministry of Industry, Ministry of National Development, Ministry of Agriculture, and other ministries/ state-run agencies at central and local government levels (refer to Figure 3-10 on the right side) are not much influenced from the Ministry of Research & Technology in making policy at each particular sector. Each ministry makes policy based on considering a specific or sectoral issue derived from internal suggestion /input at units of each ministry concerned. Even, each ministry has an internal research and development (R&D) agency (Balitbang) to support internal programs as well as propose directly the policymaking process and provide policy recommendation to internal stakeholders (up to minister level), strongly supported by substantial human resources, facilities, and funding. Therefore, it is strong reason why other ministries and agencies at national and regional levels do activities which is separated from STI institutions.

8. Ideal STI Policy Think Tank: Concept & Early Design

8.1 Ideal Evidence-Based Policymaking in Indonesia

Ideal Evidence-Based Policymaking (EBPM) is a policymaking model by which decisions of stakeholders are strongly based evidence through R&D results or existing studies related to particular issues of policies. Role of academic experts at public and private universities and R&D institutes is very pivotal to produce knowledge feasibly applied to policy recommendation. As we know, bridging R&D results (knowledge) to policy-makers is big challenge for each country around the world. Thus, many scholars pay attention to study about what are the shortcomings and how to solve them. EBPM was firstly introduced in the United Kingdom (UK) in 1997 where Prime Minister Tony Blair said that policymaking should be based on research findings to minimize various interests from many actors involved (Parson, 2006).

[Figure 3-11] Ideal Flowchart of Evidence –Based Policymaking



Source: Authors with modification from Dye (2002)

System theory of policy proposed by Dye (2002) divides the three interconnected elements of input, process, and output. Under that theory, input is everything that is used as raw materials to be considered in the next step of the political process. Simply put, process is a stage by which a policy discussed and determined by involving various actors with their respective motives, ranging from government and non government agencies including firms, association, communities, social organizations. In this step, political actions are very dominant. Finally, the last step is policy outputs and targets. Every policy has outputs and group targets resulting from the political process. Output is the expected results in the short-term period (1-5 years), whose impact can be received directly. In addition, outcome is positively expected and non-expected results in the middle to long-term period (more than 5 years), whose impact cannot be received directly. For the most part, outcome is part of the result of other policies

In this study, input is substituted to be evidence, the political process substituted to be policymaking, and output substituted to be policy outputs and targets. We break down the three steps above to be more specific elements particularly based on stages and main actors concerned. The stages include 1) the academic process to produce evidence (input), 2) technocratic and advocacy processes to make policy including political actions (process), and 3) the diffusion process to deliver and implement policy results to beneficiaries (policy outputs and targets).

In the first stage, academic action is conducted by policy researchers and academicians related to policy studies. They accommodate various public and particular/ sectoral issues resulting in addressing a particular/ specific policy. Considering other general policies are also input of this stage. Providing 'ingredients' is main result of the first stage. In the second stage, technocratic and advocacy process are played by policy analysts. They accommodate the public interests of stakeholders (state-run agencies) and private interest of shareholders (private agencies). Thus, the political process is the arena of policy-makers. This stage is very important to convert scientific actions (evidence) into political actions (policy). Therefore, the role of actors in this stage is to 'cook' the 'ingredients' as well as 'dish them out' to policy-makers for political purposes. In the third stage, the diffusion process is carried out by target groups consisting of non-stakeholders from state-run agencies, related public communities,

and non-shareholders from the internal group. Their roles are not only as players of this policymaking process, but also as direct and non-direct beneficiaries of that output policy.

This study categorizes human resources as players in the evidence-based policymaking (EBPM) process to the four categories of policy researchers, policy analysts, policy-makers, and target groups. Based on our in-depth interviews, we use five state-run agencies both at central and local government levels which are close to policymaking activities. However, not all institutions have completed human resources to support practice of EBPM in Indonesia (Table 3-3).

[Table 3-3] Characteristics of State-Run Institutions on Evidence-Based Policymaking

Institution(s)	Policy researcher(s)	Policy analyst(s)	Policy-maker(s)	Target group(s)
Ministry of Finance	Embedded	Embedded (2014)	Very specific and clear (finance policy-makers of central and local government)	Public communities on finance
Ministry of Planning & Development	Not applicable	Not applicable	Very specific and clear (planning policy-makers of central and local government)	Public communities on planning program
Ministry of Industry	Not applicable	Embedded since 2020	Very specific and clear (industrial policy-makers of central and local government)	Public communities on industrial affairs
Public Administration Agency	Not applicable	The first policy analyst in Indonesia (education centre for Indonesian analyst policy)	Clear, specific, and limited on internal institution	Public administration communities
Local government R&D agency of East Java Province	Embedded since 1999	Embedded since 2021	Not specific and not clear both at internal and external institutions	Not specific and not clear both at internal and external institutions

Source: Research Results

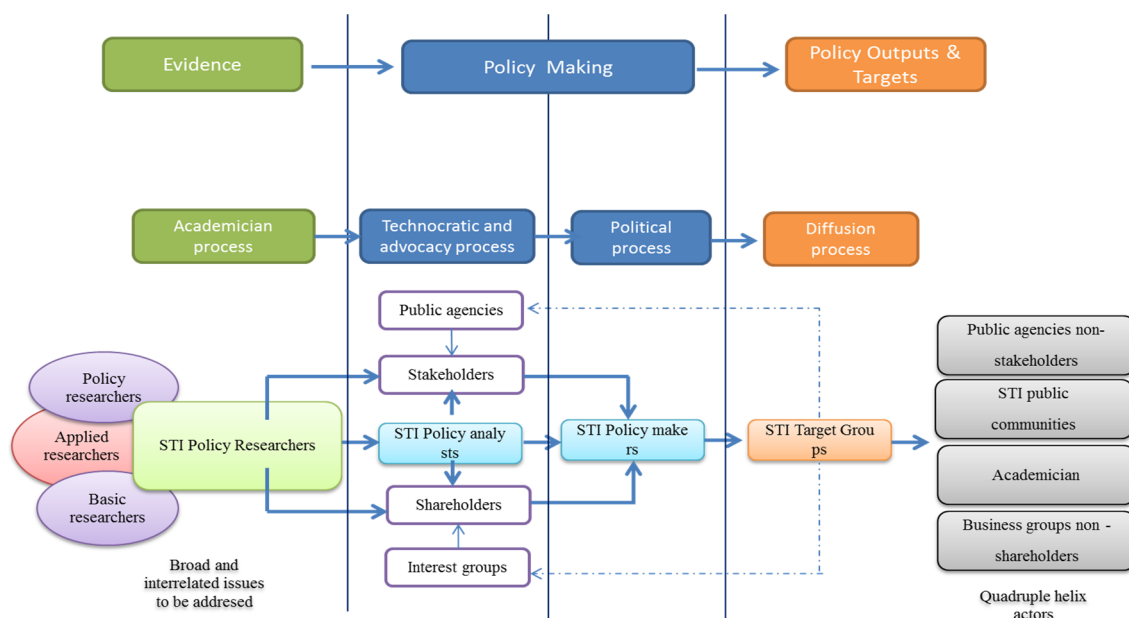
We exemplify five state-run agencies in Indonesia based on the availability of human

resources in doing and supporting EBPM, namely 1) Ministry of Finance, 2) Ministry of Planning & Development, 3) Ministry of Industry, 4) Public Administration Agency, and 5) local government R&D unit of East Java Province. Based on our analysis, the Ministry of Finance is the top state-run agency to produce EBPM. This ministry has all specific and skilled human resources with clear policy-makers and target groups to receive impact of this policy. Ironically, local government R&D agency of East Java Province, the largest R&D agency at local government in Indonesia as well as representative of R&D agency at local government in Indonesia rank the lowest position to make EBPM based on availability human resources. Indeed, this agency focuses on R&D activities addressing many public issues at local government level.

8.2 Ideal Evidence-Based STI Policymaking in Indonesia

As discussed above, the ideal evidence-based policymaking (EBPM) process is not separated from five categories of human resources playing their role to support EBPM in Indonesia. It is general EBPM model that can be applied to specific/ particular policies in Indonesia or maybe in other countries. This study uses that model to describe sequentially of EBPM on science-technology-innovation (STI) policy in Indonesia. Stages, role, and human resources are fully adopted from that model, and STI policy becomes main issue in this context.

[Figure 3-12] Elements of Evidence-Based STI Policymaking in Indonesia



Source: Authors with modification from Dye (2002)

In the first stage, academic action is taken by policy researchers and academician related to policy studies. They accommodate non-STI issues resulting in addressing a specific STI policy. Considering other general policies are also the input of this stage. Providing 'ingredients about STI' is main result of the first stage. In the second stage, technocratic and advocacy processes are carried out by STI policy analysts. They accommodate the public interests of stakeholders (state-run agencies) and private interest of shareholders (private agencies) who aim to develop STI policy. Thus, the political process is the arena of policy-makers, not STI actors. This stage is very important to convert scientific actions (evidence) into political actions (policy) in formulating STI policy based on evidence.

Kuhlmann et al (2010) said that innovation policy is a political process at the end process, so it is cannot separaetd from political actions. Therefore, the role of actors in this stage is to 'cook' the 'ingredients' of STI and its related issues as well as 'dish them out' to STI and non-STI policy-makers for political purposes. In the third stage, the diffusion process is carried out by STI target groups consisting of STI and non-STI government agencies (categorized as non-stakeholders), STI communities, and business, and academic experts. Their roles are not only as players of this policymaking process but also as direct and non-direct beneficiaries of that output policy. In STI policymaking, the presence and network of academic experts, business, government, and communities (ABGC) are the key reason why STI policy can be formulated, implemented, and evaluated. According to Carayannis et al (2018), actors of ABGC or quadruple helix are the main elements to initiate and support innovation practice. Their existence is not based on the self interests of each actor, but rather on accomodating each interest to achieve common goal.

For Indonesia, we use five STI institutions that have the capacity and authority to conduct R&D activities or make STI policy from the formulation, implementation, and evaluation phases. They are four R&D institutions such as the Indonesian Institute of Sciences (LIPI), Agency for the Assessment & Application of Technology (BPPT), National Institute of Aeronautics & Space (LAPAN), National Nuclear Energy Agency of Indonesia (BATAN). Additionally, policy analysts can increase the accuracy of policy decisions, and accurate policy decisions will have greater impact on policy target groups. They are STI institutions at a central government level before all of them are integrated under the umbrella of the National Research & Innovation Agency (BRIN) (Table 3-4).

[Table 3-4] Characteristics of Public R&D Institutions Referring to Evidence-Based STI Policymaking Flowchart

Institution	STI policy researcher(s)	STI policy analyst(s)	STI policy-maker(s)	STI target group(s)
Indonesian Institute of Sciences (LIPI)	Embedded (Education centre for researchers)	Newly embedded (2018)	Not specific, not clear, embedded only at internal policy-makers	General not specific, all public communities/ targets
Agency for the Assessment & Application of Technology (BPPT)	Not applicable	Firstly embedded at S&T Institution since 2014 , formally in 2018	Not specific, not clear, embedded only at internal policy-makers	Particular communities concerning to technology engineering and design, STP communities
National Institute of Aeronautics & Space (LAPAN)	Not applicable	Embedded since 2018	Aeronautics-related policy-makers	Specific on aeronautics communities
National Nuclear Energy Agency of Indonesia (BATAN)	Not applicable	Embedded since 2020	Nuclear-related policy-makers	Specific on nuclear communities
Ministry of Research & Technology (Kemenristek)	Not applicable	Embedeed since 2020	Clear but limited only at internal policy-makers and few local government R&D units	Specific on STI communities

Source: Research results

Based on the Table 3-4 above, the human resources are categorized as STI policy researchers, STI policy analysts, STI policy-makers, and STI target groups. Each STI institution has its drawback that must be complemented by other STI or non-STI institutions. This study has analyzed each institution based on the human resources suitable for the mission of the institution. BAPAN and LAPAN are very suitable to policymaking actors because they have

specific and limited actors. Again, both public STI institutions are each a single authority in doing nuclear energy development for BATAN and aeronautics-space development for LAPAN. LIPI is in tune with academic experts/ researchers because LIPI has STI policy researchers embedded in the internal institution.

At the same time, LIPI policy-makers are aimed at general institutions. BPPT has STI policy analyst firstly among public STI institutions, but this institution does not have specific target groups such as BATAN and LAPAN, and does not have STI policy researchers such as LIPI. The Ministry of Research & Technology (Kemenristek) is an institution that has single authority to coordinate all public STI institutions at national and local government levels in Indonesia with specific and focused policy-makers, namely the Minister of Science & Technology, but the resources such as human resources, funding, facilities are very limited.

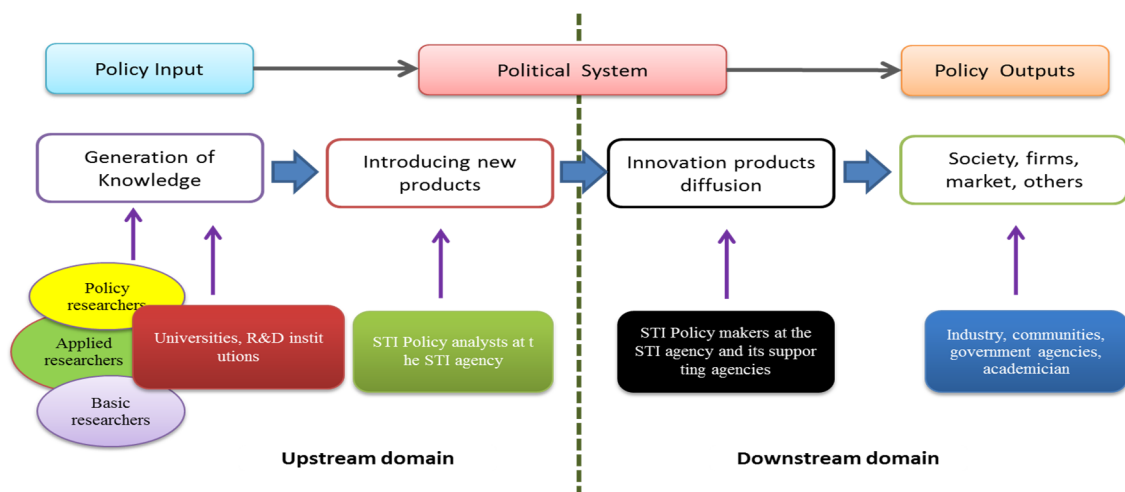
All STI institutions are new to having STI policy analysts because this profession was massively launched from 2018 to 2019. The presence of STI policy analysts as key partners of STI policy researchers is very pivotal to yield evidence-based policymaking. Additionally, the Research & Technology Minister (Kemenristek) has one STI policymaker. All STI institutions need to collaborate together to complement STI human resources and support STI policymaking activities. However, they are now integrated under the new BRIN organization to coordinate STI policy more efficiently in making STI policies.

9. Proposed Designs for Indonesian STI Policy Think Tank

Ideal model of evidence-based policymaking (EBPM) as previously discussed consists of the three parts of evidence, policymaking, and policy outputs & targets. Based on that model, each stage can refer to the role of government in executing innovation policy. Edler et al. (2016) provide the three functions of innovation policy, namely generating knowledge, introducing new products, and innovation products diffusion. Each function can be played at each stage, where evidence is the function of generating knowledge, policymaking is the function of introducing new products, and policy outputs & targets is that of innovation product diffusion. Based on these functions of STI policy at each stage, this study proposes three designs of Indonesian STI policy think tank, namely, 1) STI policy as upstream and downstream policies, 2) STI policy as micro, meso, macro-policies, and 3) STI policy as an integrated process. Each design will be elaborated as follows.

9.1 STI Policy Think Tank as Upstream and Downstream Policies

[Figure 3-13] STI Policy Think Tank as Upstream & Downstream Policies



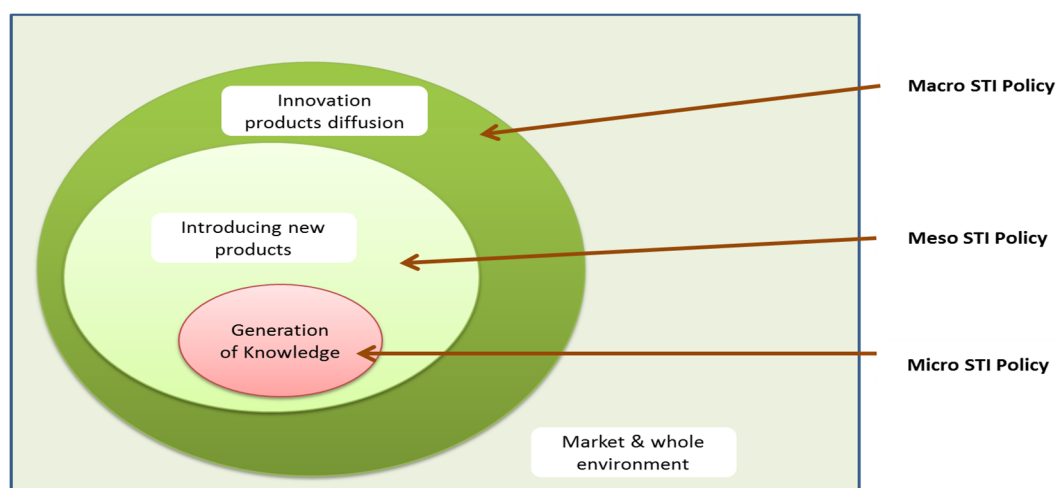
Source: Interpreted and developed from Dye (2002) and Edler et al. (2016)

The model of upstream and downstream policies is adopted from how an innovation policy is made and delivered to target groups by means of the commercialization process. It refers to the process model developed by Dye (2002) in which policymaking consists of the three steps of input, a political system, and output. Therefore, each stage refers to Edler et al (2016) based on the concepts that input is generating knowledge, political process is introducing new products to stakeholders, shareholders, and related actors, and output is innovation products diffusion to policy targets. The policy targets include businesses (non-shareholders), government agencies (non-stakeholders), academic experts, and communities' actors.

Generating knowledge corresponds to academia such as universities and R&D institutions by which they have the activities of basic & applied researches and public policy/ STI policy research. Introducing new products corresponds to public policy/ STI policy analyst and policy-makers (stakeholders, shareholders, other actors) to produce STI policymaking. And innovation products diffusion corresponds to official staffs at public STI institutions after STI policies are legitimated. However, those activities are sequential from the early stage to end stage. However, there is feedback on all the steps aimed at evaluating and giving insights at the first or second stages.

9.2 STI Policy Think Tank as Micro, Meso, Macro-Policy

[Figure 3-14] STI Policy Think Tank as Micro, Meso, Macro-Policy



Source: Interpreted & developed from Edler et al. (2016) and Vlado & Chatzidakis (2020)

The model of micro, meso, macro-policies refers to Vlados & Chatzinikolaou (2020) studying about strengthening entrepreneurship through an integrated competitiveness policy. This study is closely related to science-technology-innovation (STI) policy because actors of government, business, academia, and communities are key players in addressing competitiveness issues. They define that macro, meso, and micro-policy is concerned with:

“The macroeconomic policy usually affects macroeconomic sizes by using forecasts on the short-to-medium term projected evolution of the national economy. In the traditional macroeconomic context, the extent and diffusion of economic growth is a priority, and in this way, in the absence of the factors contributing to economic growth, the overall economy can move backward in developmental terms when it does not emphasize properly on the overall organic ingredients (firms, sectors, spaces, institutions, and natural equilibrium) of the socio-economic system” (page, 4)

“Meso-policy is selective and focuses on specific areas of socio-economic interest. Contemporary meso-competitiveness policies (as opposed to general macro-level policies) appear to aim at innovative firms or sectors and specific regions” (page, 5)

“These developments in the literature imply that new micro-level policies aim at strengthening the entrepreneurship capacities of the specific socio-economic subsystems, with the primary tools being education, training, funding, research, and all other factors that can boost the competitiveness of the specific actor. Micro policies focus primarily on the general fostering of entrepreneurship, the promotion of science and technology within firms, and the enhancement of human, intellectual, and social capital in the targeted socio-economic space.” (page 5)

“The micro-meso-macro competitiveness policy is more of an overall system than a fragmented policy approach. This overall approach can be a —system of competitiveness that takes into account all socio-economic development/

underdevelopment characteristics.” (page 6)

Definition of micro, meso, macro-policies shows that STI policy intervention can be conducted at three schemes by which each scheme is mutually related to the other schemes. It is clear that the role of an STI policy think tank relies on what is specific intervention that government agencies will undertake to make STI policymaking. According to Edler et al (2016), the three STI policy functions of generating knowledge, introducing new products, and innovation products diffusion can be placed at each of the schemes at either micro, meso, or macro-policy. Generating knowledge comes under micro-policy, introducing new products is under meso-policy, and innovation products diffusion under macro-policy. Each one has a respective policy function at its scope of policy. To simplify this understanding, we provide the following in Table 3-5 below.

[Table 3-5] STI Policy in Frame of Micro, Meso, and Macro-Policy

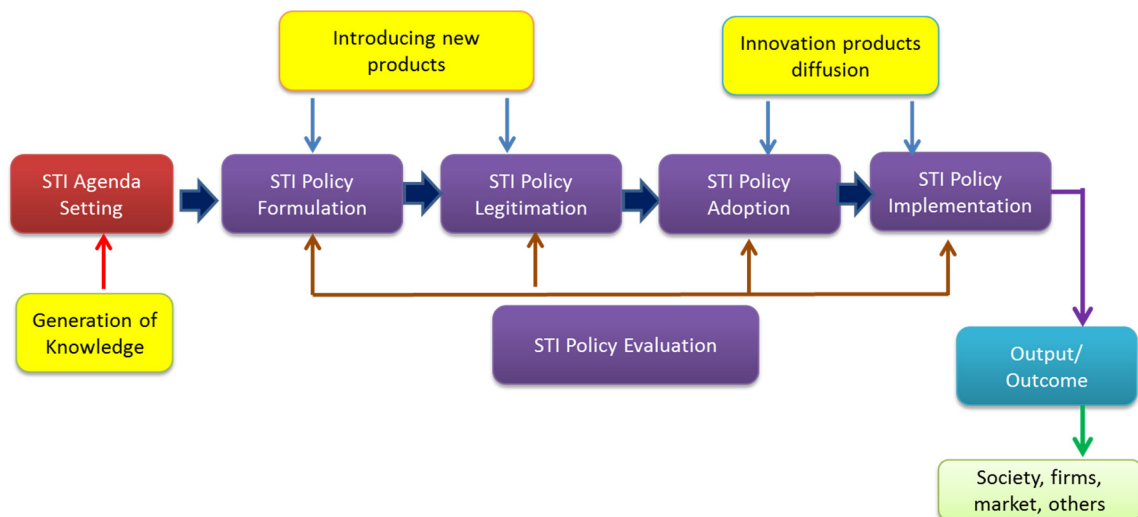
Scope of policy (Vlados & Chatzinikolaou, 2020)	Policy focus	STI policy function (Edler et al., 2016)	Policy focus
Micro	Strengthening firms, universities, R&D institutes, educational & training institutions	Generating knowledge	Reinforcing universities, R&D institutes, educational & training institutions to produce knowledge
Meso	Sectoral socio-economic fields such as creating innovation cluster at a region, fostering a particular technology development	Introducing new products	Growing new technologies/ products in a region/country, proposing new ideas/ actions to many actors related to propose new products/ technologies
Macro	Economic growth, technology diffusion, national institutions	Innovation products diffusion	Disseminating new products & technologies to target groups/ users/ markets

Source: Author's with referring to Edler et al., (2016) and Vlados & Chatzinikolaou (2020)

The Table 3-5 above has specifically categorized the roles of government to intervene in the STI field. Scope of policy proposed by Vladoš & Chatzinikolaou (2020) and STI policy function proposed by Edler et al., (2016) have the same policy focus in concept and reality. It means that both make no difference in making STI policy. This study proposes that evidence-based STI policymaking can be understood by means of micro, meso, and macro-policies. In this regard, we say that micro-policy is the scope by which evidence of policy conducted by academic experts (universities, R&D institutes, training & educational centres) is produced. Meso-policy is the scope by which policy analysts, policy-makers (shareholders, stakeholders, and other related actors) introduce, formulate, negotiate, and stipulate STI policy to be formally known by public or target groups. The last one is macro-policy, which is the scope by which formal STI policy is disseminated or communicated to target groups or policy users or public communities at general. In such last activities, macro-policy can be an 'open place' by which public communities, target groups, or policy targets can provide suggestions, insight, or other comments to the input for policy-makers and policy analysts in a meso scope as well as for academic experts in a micro-scope.

9.3 STI Policy Think Tank as An Integrated Process

[Figure 3-15] STI Policy Think Tank as an Integrated Process



Source: Interpreted and developed from Nugroho (2011), Andersson (2011) and Edler et al. (2016)

Policy Model of circle process of this study is adopted from Nugroho (2011) and Andersson (2011). According to them, policy is a sequential process in which each stage cycles from one step to another step. There is no end stage of a policy; instead, it only terminates in the temporary period after a public problem has been addressed. Surely, each policy has an output/ outcome aimed at specific target groups such as communities, academia, firms, government agencies, non-government organization(s), among others. Those stages start from agenda setting, policy formulation, policy legitimation, policy adoption, policy implementation, output/outcome, and policy evaluation at each stage. The interesting stage is the agenda-setting one. In this first stage, all public issues are accommodated and confronted to get more attention from policy-makers (stakeholders & shareholders). This stage is where the main issue of a policy formulation is determined. While subsequent stages are mostly triggered by the main issue at agenda setting, this process is very critical in making public policy.

In the case of STI policy referring to Edler et al (2016), STI policy functions are generating knowledge, introducing new products, and innovation products diffusion. In this context, generating knowledge involves agenda setting, then introducing new products involves STI policy formulation and STI policy legitimation, and innovation products diffusion involves STI policy adoption and STI policy implementation. At agenda setting, STI is not a single issue, but one of many and various issues to be confronted and chosen to be the main issue at the next stage of the policy-circle process. Output and outcome of policy are public communities, users, firms, or general public which are called target groups. They are beneficiaries that will receive the direct and indirect impacts of STI policy. Thus, making STI policy is not only based on policy-makers' interests, but also considering target groups' needs and source of trustworthy information knowledge which is derived from the activities of generating knowledge from academia (universities, R&D institutes, policy analysis units, & other knowledge sources).



CHAPTER 4

Building Monitoring & Evaluation Systems for National Research & Development Programme in Indonesia

Youngjoo Ko

President, Daejeon Institute of Science & Technology for Enterprise and People (DISTEP)

Kusnandar

Researcher, BRIN

Karlina Sari

Researcher, BRIN

Ria Hardiyati

Researcher, BRIN

Sofia Yuniar Sani

Researcher, BRIN

Galuh Syahbana Indraprahasta

Researcher, BRIN

Chapter 4. Building Monitoring & Evaluation Systems for National Research & Development Programme in Indonesia

1. Introduction

In 2016, Indonesia launched the country's new vision on research and development (R&D). In this regard, the National Research Master Plan (*Rencana Induk Riset Nasional* or RIRN) 2017-2045 has been promulgated and expected to provide stronger guidelines underlying the country's R&D activities. A new national R&D agenda was seen to be necessary given the overarching performance of previous R&D framework (that lasted between 2006 and 2017 under the National Research Agenda or ARN umbrella) that was argued to be not optimal, mainly due to a less clear direction and lack of inter-institutional collaboration. In the same vein, it was also expected that with the RIRN document's presence, the functioning of R&D activities in supporting the country's economic development could be much improved. As can be clearly reflected through its vision, RIRN was envisioned to support the vision 'Competitive & Research-based Sovereign Indonesia in 2045'. The term 'competitive Indonesia 2045' implies that research should be translated into inventions and innovations that will ultimately have an impact on increasing the nation's competitiveness. Meanwhile, the term 'research-based sovereign' implies that Indonesia should gain control and master advanced science & high technology, so that the country will be able to compete globally. This vision is then poured into several targets as indicated by Table 4-1.

[Table 4-1] Five Annual Targets as Stated in the RIRN

Indicators	2015	2020	2025	2030	2035	2040	2045
Researcher ⁶	1,071	1,600	3,200	4,800	6,400	8,000	8,600
GBAORD/ GDP (%)	0.15 ⁷	0.21	0.42	0.63	0.84	1.05	1.26
GERD/ GDP (%)	0.20	0.84	1.68	2.52	3.36	4.20	5.04
Ratio of researcher candidate ⁸	5.6	20	40	60	80	90	100
Researcher productivity	0.02 ⁹	0.04	0.08	0.10	0.14	0.18	0.22
Multifactor productivity (MFP) (%)	16.7	20	30	40	50	60	70

Source: MORTHE (2016)

As an umbrella and long-term document, RIRN is further broken down into national research priority status (*Prioritas Riset Nasional* or PRN) that is basically a five-year R&D plan on an annual basis. The current period of PRN (2020-2024) is deemed to be the first step to establishing the (new) R&D ecosystem in the country. It can be argued that while PRN is the operational plan of RIRN that comprises more concrete and detailed directions for R&D activities, its overall system (i.e. structure and mechanism) is still developing. Given this context, it is important to have an evaluation mechanism for an R&D program in Indonesia under the PRN umbrella to function as a feedback loop mechanism in correcting and improving PRN in general and R&D program in particular.

Since its first establishment in 2020, there has been no specific discussion and mutual agreement among the participating decision-makers on how the evaluation mechanism will be developed and put into the overall PRN system accordingly. Based on Korea's Science & Technology Policy Institute (STePI)-LIPI study report in 2020, it is indicated that the Government of Indonesia does not have any clear evaluation mechanism for an R&D program yet. The mechanism tends to vary among different institutions. A more interesting point is that in spite of this variety, the evaluation has strongly focused on the administrative

6 Person(s)/ million population

7 Data in 2014, including R&D employee' salary, total allocation

8 Ratio of master & doctoral students to bachelor students

9 Data in 2014, SCImago citation index

aspect vis-à-vis aspects related to the substance and development implication of an R&D program. A robust substantial-related assessment has been thus largely absent accordingly.

Meanwhile, the experience of South Korea, for instance, has shown that the evaluation system for an R&D program is an evolutionary process. That being said, the mechanism is a 'living system' in that it is open to change following the need as well as the dynamic nature of R&D activities and their relation to the country's changing development priorities. Having started to develop its R&D evaluation system from around 1997, the Government of South Korea continues to improve this system to date. Additionally, South Korea has proved itself as a leading economic country with a strong R&D foundation. It is thus interesting to use South Korea's experience as teachable case studies. Although there exist some differences between South Korea and Indonesia, we believe there are also some points that can be learned from South Korea in the context of how to design and develop an appropriate evaluation system that supports a well-functioning R&D agenda.

Having this background, this study therefore aims to fill this empirical gap and is thus intended to contribute to design an evaluation mechanism for PRN 2020-2024. Given the broad nature of PRN, the current study limits its scope of analysis to the two following aspects: (1) proposal selection of research projects under the PRN scheme and (2) monitoring and evaluation of research projects under the PRN scheme. In each of these aspects, we mainly focus on digging into (1) the structure thereof, (2) actors involved therein, and (3) indicators for proposal selection and monitoring & evaluation of research projects. In this study, we also highlight the aspect of the proposal selection process because the quality of a research project depends heavily on (a) how it relates to specific development priorities, (b) whether it has the potential to provide economic, social, and/or environmental benefits, and (c) whether it should promote collaboration.

2. Research Evaluation: Basic Concept & Principles

This section discusses the principles of research evaluation system, based on the literature review, as the basis for this study. The literature review consists of the definition of monitoring and evaluating (M&E), elements in M&E, steps in designing an M&E for a program, and principles of research evaluation systems.

There is arguably limited formal knowledge and few methodologies for monitoring and evaluating (M&E) program implementation, and relatively limited recognition in the practitioner-oriented arena (Neumann, Robson, and Sloan, 2018).

2.1 Monitoring and Evaluation

Monitoring is a regular collection of data/ information (daily, monthly, annually) related to the progress of programs (Kusek & Rist, 2004; Nirmala & Kumar, 2017; Neumann, Robson, and Sloan, 2018). Monitoring aims to understand the progress of programs, get early indications whether the programs are on the track, and provide stakeholders with data/ information for evaluation (Kusek & Rist, 2004; Nirmala & Kumar, 2017; Neumann, Robson, and Sloan, 2018).

Evaluation is an assessment of (ongoing or completed) programs based on obtained data/ information (Kusek & Rist 2004; Nirmala & Kumar, 2017; Neumann, Robson, and Sloan 2018). The evaluation aims to assess the relevance & fulfilment of the objective, the results (outcome, impact), and efficiency of the programs, to identify the reasons for the success and failure of programs (teachable lessons) and determine the sustainability of programs (terminated & continued) (Kusek & Rist, 2004; Nirmala & Kumar, 2017; Neumann, Robson, and Sloan, 2018).

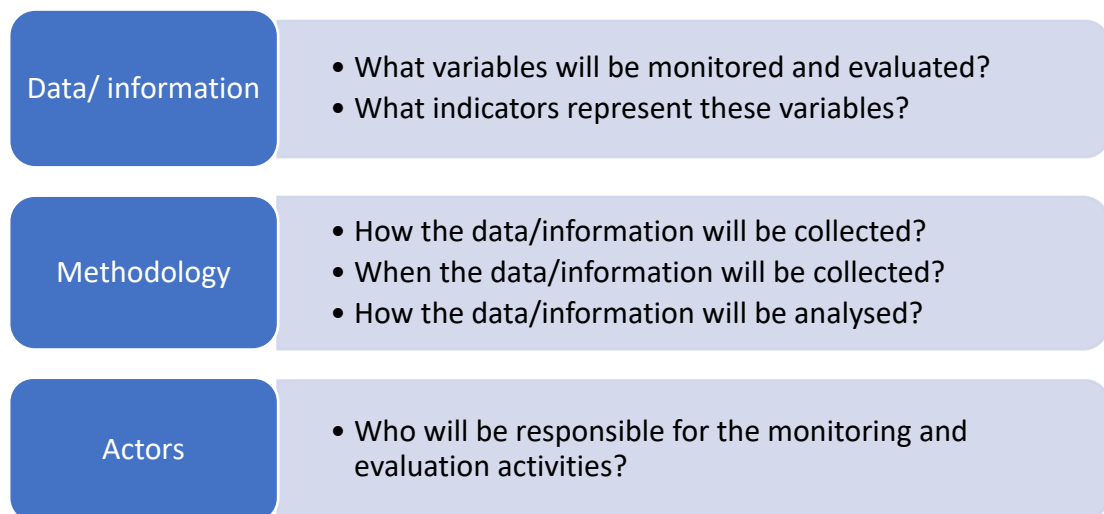
Even though monitoring and evaluation are different activities, they are complementary to each other and cannot be separated (Subrahmanyam 2017; Kusek & Rist, 2004).

In this study, M&E is defined as a regular collection and interpretation of data/ information related to research activities to understand its progress and gain teachable lessons from these activities (Reed et al. 2021; Edwards & Meagher, 2020).

2.1.1 Elements in monitoring and evaluation

There are three elements in M&E: 1) data and information; 2) methodology; and 3) actors (Waylen et al., 2019; Segone, 2010; Subrahmanyam, 2017), as shown in Figure 4-1.

[Figure 4-1] Elements in Monitoring and Evaluation

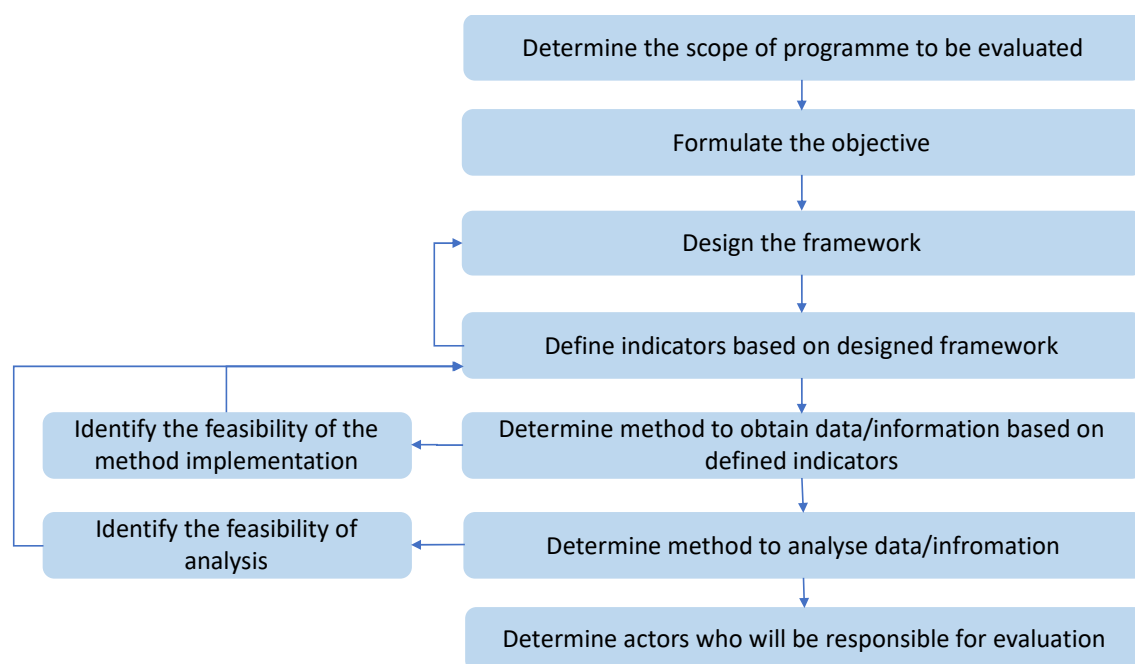


Source: Adapted from Subrahmanyam 2017; Waylen et al. 2019; Segone 2010

2.1.2 Steps in designing an M&E for a Program

General steps in designing M&E for a program are presented in Figure 4-2. It is synthesized based on some studies (Wholey, Hatry, and Newcomer, 2010; Kareemulla, 2017; Subrahmanyam, 2017; Waylen et al., 2019; Segone, 2010).

[Figure 4-2] Steps in Designing Monitoring and Evaluation



Source: Adapted from Wholey, Hatry, and Newcomer (2010); Subrahmanyam (2017); Waylen et al. (2019); Segone 2010; Kareemulla (2017)

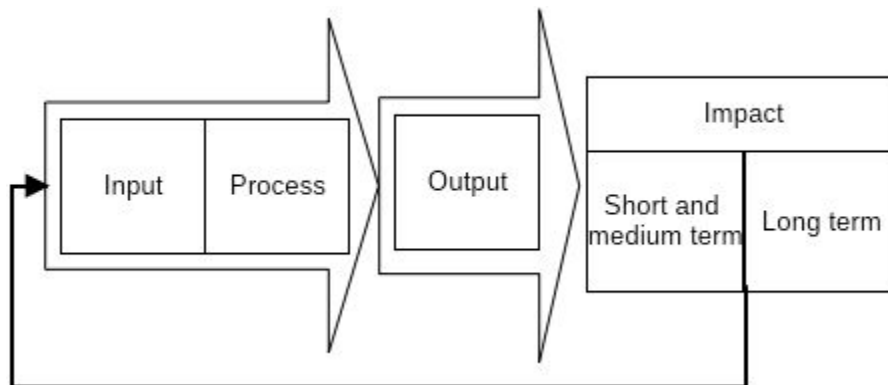
2.2 Research Evaluation Systems

Based on its objective, research evaluation can be distinguished into formative or summative (Reed et al., 2021) ones. Formative focuses on the achievement of research and its impacts, while summative focuses on ongoing monitoring and learning to improve the research activities (Galas et al., 2018; Reed et al., 2021). This study will cover these two kinds of monitoring and evaluation.

Building upon previous work on research monitoring and evaluation (Donovan & Hanney, 2011; King, 1987; DBarnett & Gregorowski, 2013; Galas et al., 2018), a framework for research monitoring and evaluation is designed as shown in Figure 4-3. The framework is designed based on the logic model of the payback proposed by (Donovan & Hanney, 2011) complemented by other previous works on research monitoring and evaluation (Galas et al. 2018; DBarnett & Gregorowski, 2013; King, 1987).

The framework follows the nature of the research process comprised of input, process, output, and impact variables (Donovan & Hanney, 2011; Galas et al., 2018).

[Figure 4-3] Framework for a Research Evaluation System



Source: Adapted from Donovan & Hanney (2011); King (1987); DBarnett & Gregorowski (2013); Galas et al. (2018)

Research evaluation system in which this study focuses on is the proposal selection and research monitoring & evaluation system that are discussed below.

2.2.1 Proposal Selection

The study of Warta and Morenon (2020) on the Science Europe Initiative for Mutual Learning has proposed aspects regarding the proposal selection : 1) Soundness of methodology, 2) feasibility of research, 3) resource allocation aligned with objectives, 4) applicants' expertise and complementary expertise of researchers, 5) dissemination plan, 6) novelty of the research question, 7) potential economic & social Impact, 8) potential transfer/ commercialization, 9) potential contribution to public policies, and ethical considerations.

In most cases, the review system used is a single external peer review (actor) combined with a panel review.

2.2.2 Monitoring and Evaluation of Research Projects

To design and carry out the M&E of a research project, the data, method, and actors must be defined. Based on the framework for research monitoring and evaluation (Figure 4-2), indicators for research and monitoring are identified for every variable. Examples of indicators of research monitoring and evaluation for variables of input and process are provided in Table 4-2.

[Table 4-2] Indicators of Research Monitoring & Evaluation for Variables of Input & Process

Input	Process
• Human resources • Financial resources • Time • Material • Infrastructure	• Plan • Activities • Communication and coordination • Learning process of the research team • Dissemination of (early) results

Source: Adapted from King (1987); Galas et al. (2018)

Concerning output, it can be distinguished based on its research contribution. The research contribution also connects to the kind of anticipated impact. The contribution of research can be distinguished into (Strahan, Keating, & Handmer 2020; European Commission 2018):

- a) **Scientific:** The research contributes to scientific knowledge that benefits researchers (other researchers & the team). It is not limited to scientific publications but can also be open databases of research results.
- b) **Economy:** The research produces innovative artifact(s) (e.g. products, processes, methods) that can be adopted/ developed/ mass-produced by industries.
- c) **Society:** The research produces something that can benefit society, either directly or indirectly. It encompasses practical knowledge for society, intervention methods, and policy recommendations for the government.

Concerning impact, research outputs cannot directly affect science, economy, and

society. Advanced processes that take time are needed. In addition, it is also affected by the target user's behavior to the research output. Therefore, intermediary domains to enable research impacts to manifest should be considered. They consist of target users' awareness, decision-making changes, and adoption (Reed et al., 2021). Meanwhile, concerning the scope, research output can have impacts on different scales on an individual, team, institution, regional, national, and international basis (Reed et al., 2021).

Reed et al. (2021) distinguish five methods of research monitoring and evaluation: statistical methods, system analysis methods, oral and arts-based methods (ethnography), indicator-based approach, and evidence synthesis approach. This study focuses on the indicator-based approach, in which the defined indicators can be used as guidelines to assess (through measurement or questions) the research (inputs, process, outputs, impacts) (Reed et al. 2021; Galas et al., 2018). The data and information collected can be quantitative or qualitative (Reed et al., 2021; Galas et al., 2018).

Research monitoring and evaluation can be performed by both, internal and external actors (Galas et al., 2018). Internal actors are researchers who are also involved in the research activities. Meanwhile, external actors can be funding institutions or independent reviewers from academia, industries, government, and communities.

3. Methodology

This section discusses the method implemented in building a research evaluation system for PRN, encompassing proposal selection and research monitoring & evaluation. It consists of understanding the current situation followed by designing the evaluation system. In understanding the current situation, we collected information from the actors involved in PRN (i.e. researchers, project management officer of PRN, research funding manager). Then, in building research evaluation, we gathered and considered as much as possible ideas from the actors involved through multiple techniques explained below. In fact, discussions with representatives of the actors involved were facilitated. Therefore, to a certain extent, the participatory approach has been implemented in this study.

As mentioned in Section 1, in building the research evaluation system for PRN, we also take teachable lessons from the research evaluation system in South Korea as an example of a successful country in developing the R&D system. The understanding of South Korea and PRN systems was done in parallel.

3.1 Understanding the Research Evaluation System in South Korea

Understanding the research evaluation system in South Korea was mostly done through communication and discussions between the Indonesian (BRIN) team with an appointed expert from Korea. The communication and discussions were performed through email and monthly online meetings from June-October 2021.

In addition, there was a two-day training session on September 2021 conducted by STEPI Korea cooperated with BRIN. In this training, there was a presentation and discussion about the research evaluation system in South Korea.

3.2 Understanding the Current Situation of PRN Research Evaluation System

In order to have a better perspective about the proposal selection process and mechanism evaluation for PRN 2020-2024, multiple techniques for gathering data and information were conducted as follows below:

3.2.1 In Depth Interview with researchers

In August 2021, multiple individual interviews were conducted with five researchers who each represent a different research focus of PRN 2020-2024. Those interviews were conducted through virtual meetings. Every researcher shared his/her perspective about the implementation of PRN 2020-2024.

3.2.2 Interview with Project Management Office (PMO) of PRN from MoRT/BRIN

PMO of PRN were appointed through agreement with the research funding agency (LPDP) to manage the daily basis of PRN activities. PMO of PRN plays significant roles encompassing the proposal selection and evaluation processes of PRN activities. Therefore, it is important in this study to have an interview with **the office** in order to obtain information about PRN management, and related subject matters.

Interview with the PMO of PRN was conducted on August 2021 through virtual meetings.

3.2.3 Interview with research funding manager (LPDP)

The Indonesia Endowment Fund for Education or “Lembaga Pengelola Dana Pendidikan” (LPDP) takes a role as one of the main funding agencies for PRN 2020-2024 through the agreement between LPDP and MoRT/BRIN signed in 2018.

The interviewed with LPDP officers were performed on the 4th week of August 2021. From this interview, the information about the funding mechanism for PRN 2020-2024 was obtained.

3.2.4 Survey I Through Online Questionnaire

The team of this study conducted the survey by sending the online survey (see Appendix 3) to 305 researchers involved in PRN 2020-2024 (the data was obtained from MoRT/ BRIN). The survey was also sent to related stakeholders including the PMO of PRN and LPDP. The survey has the aim to obtain the perspectives of actors involved on the evaluation system of PRN. In addition, it also gathered the information about challenges encountered in the research evaluation system of PRN 2020-2024.

The online survey was sent to the respondents (through emails) on August 2021. The survey was conducted online for about two weeks. The respondents included 90 researchers who gave the response to the questionnaire.

3.3 Designing the Research Evaluation System of PRN.

In designing the research evaluation system for PRN, multiple FGDs with representatives of the actors involved and a second survey through an online survey were conducted in this study. These activities are discussed below.

3.3.1 Focus Group Discussion (FGD)

There were 2 FGDs in this study as follows: 1) the first FGD discussing the proposal selection on the 2nd week of September; and 2) a second FGD discussing the research monitoring and evaluation on the 3rd week of September. Both FGDs focus on three aspects, namely 1) the structure & mechanism, 2) actors, and 3) indicators. The results of interviews and online surveys are the basis for the FGD participants to design the proposed research evaluation system for PRN.

Regarding FGD participants, the team of this study selected 5 participants representing the actors involved in PRN 2020-2024. FGD participants consisted of 3 researchers from Engineering, Life Sciences, and Social Science, 1 PMO of PRN, 1 LPDP officer. In addition to these 5 key individuals, the team also invited 3 new BRIN officers from the deputy of

research & innovation policy.

3.3.2 Survey II Through Online Questionnaire

The results of FGDs, then, are synthesized by the team of this study to design the proposed system for research evaluation of PRN. Afterward, a second survey through an online survey (see appendix 4) was conducted with the aim of: 1) communicating the proposed research evaluation system; and 2) obtaining feedback from the respondents regarding the proposed system. This second survey was also conducted by sending the survey to all 305 researchers involved in PRN 2020-2024 and related actors. The results of this survey are important to improve the proposed system of proposal selection and research monitoring & evaluation of PRN 2020-2024.

4. Results

4.1 Research Evaluation System in South Korean

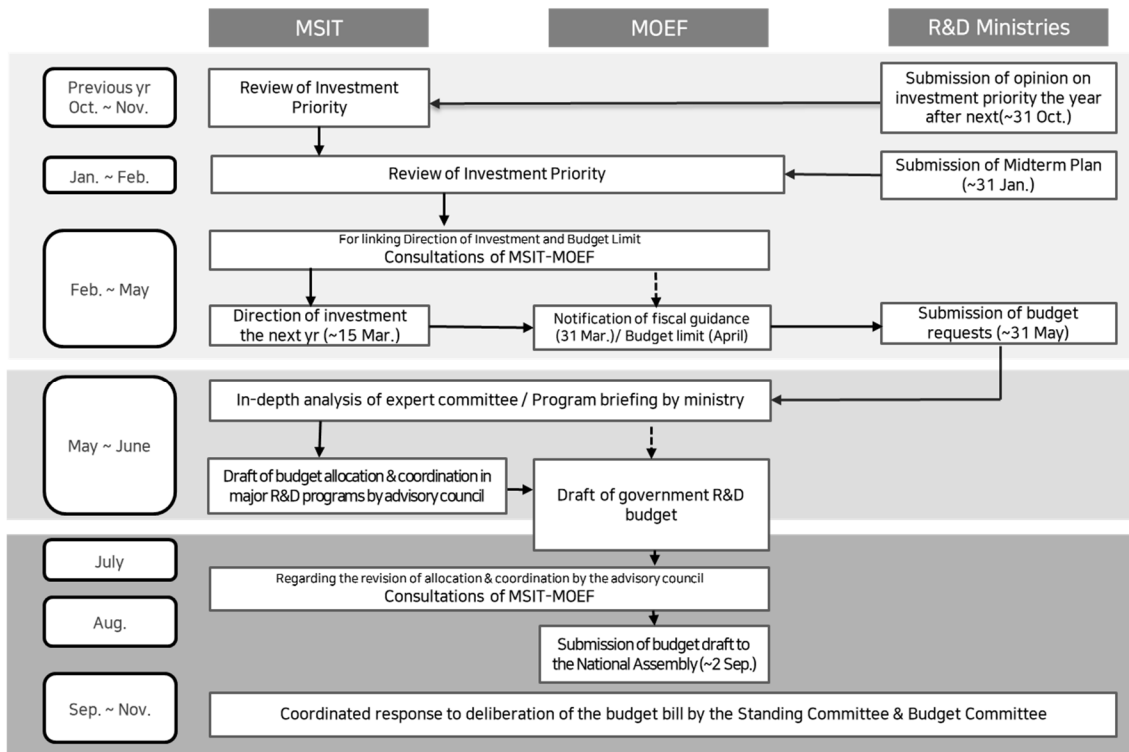
Korea's performance evaluation system was started in 2006 with the Law on Performance Evaluation & Management of National R&D Program enacted in December 2005. The focus of the law was to evaluate performance achievement based on the results of R&D activities, and evaluation process was managed by the Ministry of S&T. The law was revised in 2008 changing not only focus into enhancing efficiency of R&D but also managing authority into the Ministry of Strategy & Finance. A plan for improving government performance evaluation was launched in 2013 shifting focus to quality of performance, social, and economic impacts. The Ministry of S&T started to manage the evaluation process again.

According to the Law, the ex-ante process of national R&D programs begins with proposing a departmental 5-year midterm plan to the monetary authority (Ministry of Economy & Finance, MOEF) and the competent science and technology ministry (Ministry of Science & ICT, MSIT). The Ministry of S&T reviews the investment priority among departmental programs and discusses the setting of investment guidelines and budget limit with the monetary authority. Once the departmental limit of the R&D budget is set by the monetary authority, the Ministry of S&T establishes next year's investment guidelines (i.e. direction, criteria, etc.) and notifies each ministry. The applicants work on budget requests reflecting the new programs by referring to these guidelines and expenditure limit, and submit them to the Ministry of S&T as shown in Figure 4-4.

The budget for departmental R&D programs is compiled through a feasibility evaluation process. In particular, it is mandatory for large-scale programs with a total expenditure of over KRW 50 billion (USD 45 million) and a subsidy of over KRW 30 billion (USD 27 million) to conduct a preliminary feasibility test. Items for the test and other general considerations in the planning process will be addressed in more detail later on. The draft budget of government R&D is established after the departmental allocation and coordination process.

The monetary authority compiles a budget in practice based on the draft, followed by deliberation and confirmation of the budget bill by the National Assembly.

[Figure 4-4] Process of National R&D Budget Allocation & Coordination



Source: Ministry of Science & ICT, Korea (2020)

The 2001 Regulations on the Management, etc. of National R&D Projects (Presidential Decree No. 17429) did not regulate any ex-ante evaluation at the program level, and subsequently, the detailed provisions regarding planning were added. Since the program is established by the government without a competitive process between various actors, it seems that the planning process is greatly emphasized, without the separate ex-ante evaluations (E.H. Lee et al, 2020).

The main structure of Korea's program evaluation consists of self-, meta-, and in-depth evaluations. Self-evaluation of each program is conducted by ad-hoc committees organized by ministries who formulated and implemented these programs. Self-evaluation has a 3-

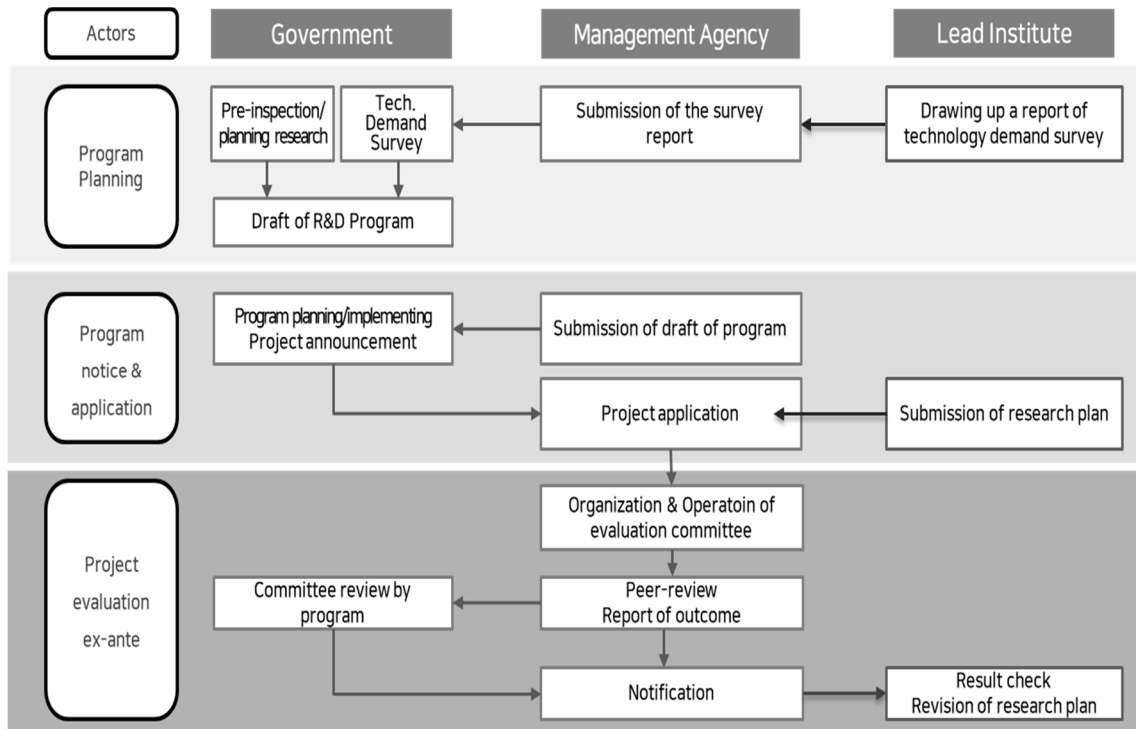
year evaluation cycle during the program lifetime and 'yes or no' questions of indicators on a weighted basis. The product of self-evaluation is grade classified into five categories determined by scores such as 'excellent-good-moderate-unsatisfactory-poor'. Next year's budget is increased for 'excellent' and 'good' programs, or otherwise cut for 'unsatisfactory' and 'poor' programs. Two consecutive 'unsatisfactory' or 'poor' grades proceed with conducting in-depth evaluation while programs of three consecutive 'unsatisfactory' or poor grades should be terminated. The results of self-evaluation should be submitted to MSIT which is responsible for meta-evaluation about self-evaluation results. KISTEP supports the meta-evaluation with external ad-hoc committees. Meta-evaluation results are submitted to the National Science & Technology Council (NSTC) with the performance grades. The results of meta-evaluation are used to allocate and coordinate next year's government R&D budget as well.

In-depth evaluation has some criteria for selecting long-term R&D program(s) with large budgets, R&D program(s) that are suspected to overlap with each other, R&D program(s) that are jointly implemented by several gov't ministries, R&D program(s) that are at issue in nation and R&D program(s) that MSIT judges are necessary to evaluate

In summary, the total results of all kinds of evaluation are utilized for improving the program running system & research environment and coordinating budget allocation, as well as for consulting program planning and some aspects policy development.

At the project-level, in general, the request for proposal (RFP) of each research project is established in the process of planning the higher program. Research management agencies make project announcements using an RFP, and researchers from university-industry-government apply for it with their research proposal. The ex-ante evaluation proceeds with the received proposals, leading to the final result being confirmed and notified in Figure 4-5.

[Figure 4-5] Process of Selecting National R&D Projects



Source: Ministry of Science & ICT, Korea (2020)

The selection process at the project level is very different from that of a program. Whereas the selection of a project is generally organized from the perspective of excellence, the research program is mainly evaluated by its priority and feasibility among departmental programs under the budget limit. Even if it is recognized to be superior, it may be difficult to conduct the program due to the budgetary shortfall.

The Regulation of National R&D projects stipulates the criteria of the ex-ante evaluation for a project as follows.

- Creativity and faithfulness of a research and development plan; Connectivity with the objectives of a national research and development project;
- Levels of research environments, such as human resources, facilities, and equipment for research;

- c) Overlapping with any research & development tasks which has been promoted or is being promoted as national research and development;
- d) Adequacy of the security levels;
- e) Adequacy of installation of research facilities and equipment;
- f) Possibility of linkage and cooperation in or out of the Republic of Korea in performing the research & development tasks;
- g) Effects resulting from the outcomes of research and development;
- h) Usefulness of the outcomes of research and development, such as technology transfers, commercialization, and follow-up research;
- i) Levels of research capacity and research code of ethics of a person in charge of research;
- j) Whether any publicly announced technology or intellectual property rights exist.

Korea's evaluation system both in program and project level can give some insights to Indonesia in terms of evaluation structure, participating actors, specific indicators and utilization of the evaluation results.

4.2 Current Situation of Research Evaluation System of PRN

This section discusses the current situation of the research evaluation system of PRN 2020-2024. Based on the scope of this study, the research evaluation system discussed here encompasses the proposal selection and research monitoring & evaluation system. In this section, firstly, the PRN is explained to give information about funding, focus and structure of PRN that are connected with the proposal selection and research monitoring & evaluation system discussed afterward.

4.2.1 PRN: Funding, Focus, and Structure

Prioritas Riset Nasional (PRN) or National Research Priority is a five-year operational R&D program based on the National Research Master Plan (RIRN). The first period of PRN is currently running from 2020 to 2024. PRN focuses on achieving output in the form of innovative product development. It aims to integrate research activities conducted by various research institutes and universities through the establishment of research consortiums based on certain research focuses and product development targets.

In the case of funding, there are two types of PRN 2020-2024 funding:

1. Institutional funding

It is a top-down government funding disbursed to research institutes and universities. Research programs are planned and conducted by each research-conducting institution. This type of funding is exclusive for internal researchers of the institutions only.

2. Project funding

This fund is provided by research fund managers. Research programs are determined by the fund managers and the programmes of the call for proposal are open for all research institutes and universities in Indonesia. However, for PRN, there is a special case in which the Indonesia Endowment Fund for Education (LPDP) plays a role in providing funds, while the program is fully managed by MoRT/ BRIN.

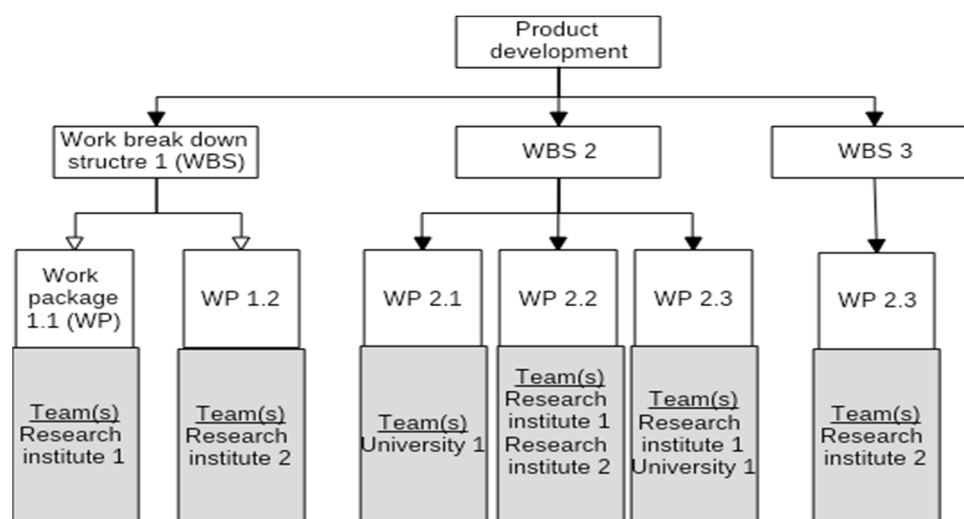
Based on RIRN's focus, there were discussions between BRIN, research institutes, universities, and industry (mostly enterprises owned by government) to determine the product development targets for PRN 2020-2024. Table 4-3 provides the product development targets of PRN 2020-2024 based on its focus.

[Table 4-3] PRN 2020-2024 Focus & Examples of Product Development Target

Research focus	Examples of product development target
Agriculture	High quality seeds
Energy	Bioenergy, nuclear power plant
Health	Vaccines, medicines
Transportation	Electric vehicle, high speed train
Engineering	Food packaging, earthquake resistant building
National defense & security	Rocket, radar
Maritime	Harbour tug dual fuel, processed seafood
Social	The change of society in digital era
Multidisciplinary	Satellite, smart water management systems

To achieve the targeted product developments, the research activities for PRN follows the hierarchical structure based on the elements of targeted products as illustrated in Figure 4-6. Every product development consists of several work breakdown structures (WBS), and then every WBS is divided into several work packages (WP) or research teams that consist of researchers from one or more research institutes/ universities. Every product development is managed by the program manager, and every WBS and WP has a dedicated coordinator.

[Figure 4-6] PRN Structure



Source: Constructed by Authors

The funding, focus, and structure of PRN determine the mechanism of proposal selection and research monitoring & evaluation of PRN 2020-2024, which will be explained in the next section.

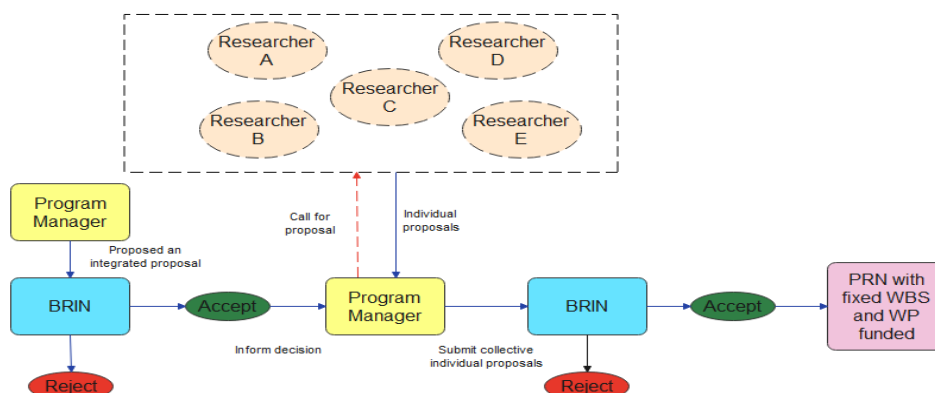
4.2.2 Proposal Selection

This section discusses the current situation of the proposal selection system of PRN 2020-2024. The discussion is divided into: 1) structure and mechanism; 2) actors involved; and 3) indicators.

■ Structure and mechanism

The results of in-depth interviews with respondents show that there are two different schemes of proposal selection in PRN 2020-2024. In the *first scheme* (Figure 4-7), the program manager (PM) writes an integrated proposal to determine the scope of the research project, including a 4-year roadmap. Then the PM submits the integrated proposal to MoRT/ BRIN. When the proposal is selected to be funded, the PM invites researchers in the related field to propose individual researchers. Afterward, the PM selects the individual proposals, then submits them to MORT/ BRIN. MoRT/ BRIN gives the final decision on which proposals will be accepted or rejected.

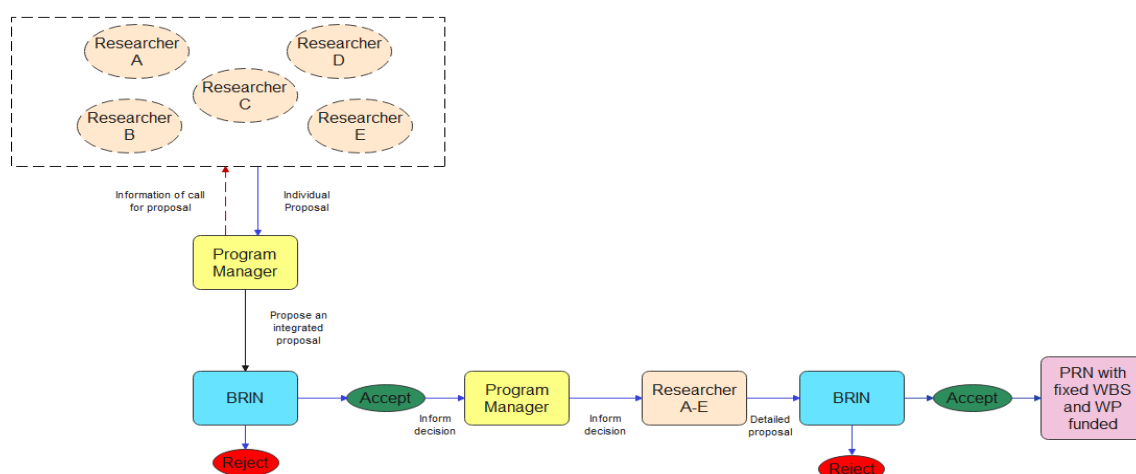
[Figure 4-7] First Scheme of PRN Proposal Selection



Source: Constructed by Authors

In the *second scheme* (Figure 4-8), the PM invites researchers from various institutions to send a proposal under a specified topic, then the PM selects proposals that fit determined criteria. From those selected proposals, the PM formulates an integrated the proposals into integrated proposal, then sent it to the MoRT/ BRIN. If the integrated proposal is accepted, the PM will inform the involved researchers to send their individual proposals to BRIN. BRIN then selects which proposal will be funded.

[Figure 4-8] Second Scheme of PRN Proposal Selection



Source: Constructed by Authors

Currently, BRIN only receives single-year research proposal. Therefore, researchers who conduct multi-year studies must submit another follow-up proposal for each following year. The same proposal selection mechanism applies for the sequel research projects.

■ Actors involved

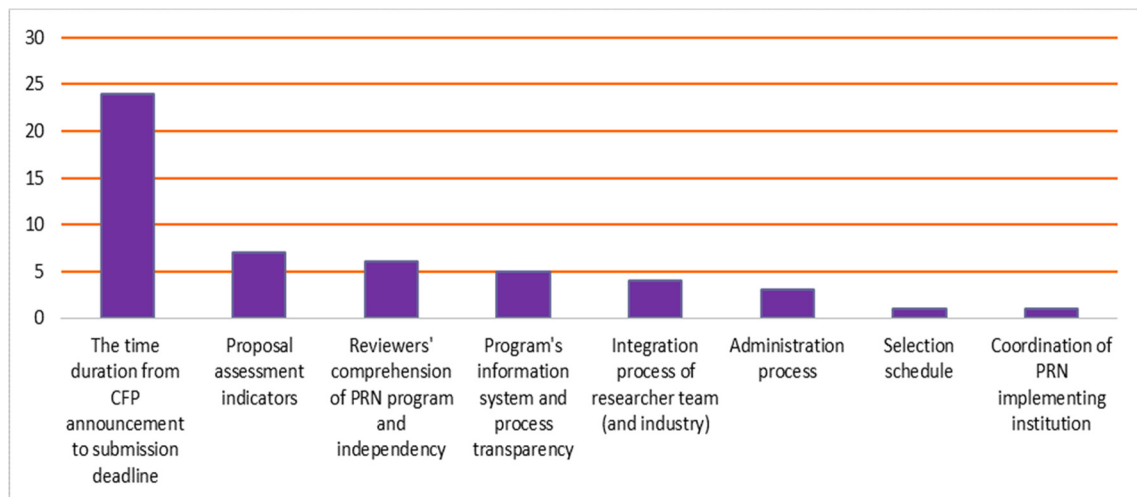
Based on in-depth interview and FGD results, BRIN as PRN project management officer plays a main role in the process of proposal selection. There is information that experts specialized in the PRN focuses are also involved in reviewing proposals. However, the criteria used by BRIN in assigning the experts is not clear.

■ Indicators

In selecting integrated proposal, BRIN and the appointed reviewer team assess the proposal presented by PM. However, there is no clear information about indicators in selecting the proposal. Meanwhile, individual proposals are also assessed by BRIN (and PM, in some cases). There is also no clear information on the criteria of funded individual proposals.

The survey result shows that there are several problems identified in PRN proposal selection (Figure 4-9). Almost 50 percent of respondents complain that the duration between call-for-proposal announcement and proposal submission deadline is too short, less than a week. Other problems are attributed to unclear and non-transparent proposal assessment indicators set by BRIN which are (13.7%), reviewers' capacity and independency (11.7%), difficulty in accessing the information system and lack of process transparency (9.8%), difficult integration process of research team (and industry) (7.8%), complicated and unclear administration process (5.9%). All these problems are attributed to the management of PRN implementation.

[Figure 4-9] Identified Problems in PRN Proposal Selection



Source: Related survey

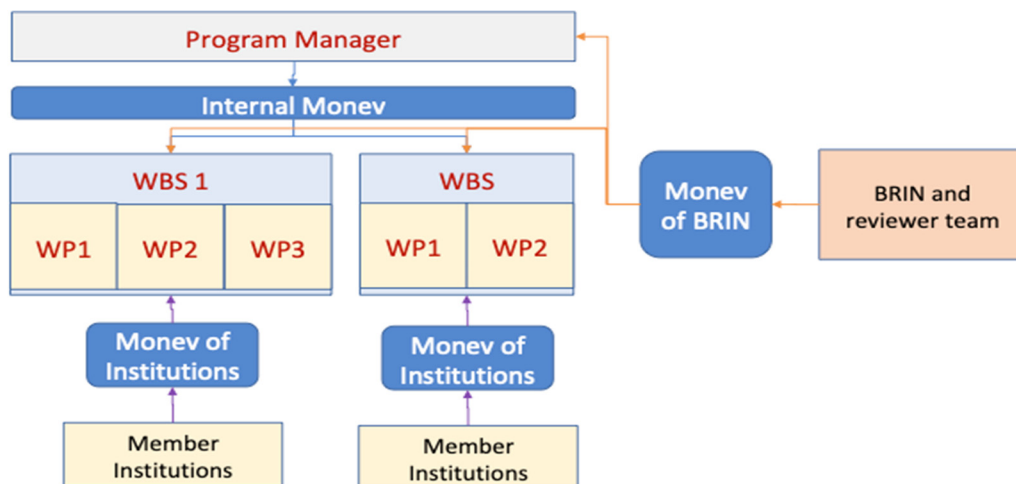
4.2.3 Monitoring and Evaluation

This section discusses the current situation of the research monitoring and evaluation system of PRN 2020-2024. In the proposal selection discussion, it is also divided into: 1) structure and mechanism; 2) actors involved; and 3) indicators.

■ Structure and Mechanism

During the running of PRN projects, monitoring and evaluation (M&E) is conducted to assure the achievement of every product development target. The three stages of M&E are identified in PRN 2020-2024 as illustrated in Figure 4-10.

[Figure 4-10] PRN's Current Mechanism of Internal & External M&E



Source: Constructed by Authors

The three stages of M&E are identified as follows:

1. Internal monitoring and evaluation by program manager

The program manager (PM) along with his/her team monitor and evaluate both substantial and administrative aspects, with an emphasis on the former, of each work packages (WP) and work breakdown structure (WBS). The team consists of a program manager, research coordinator, and administrative coordinator. As for the

reviewer, in several cases, an internal M&E also involves related external actors, such as academic experts or regulators. In general, an internal M&E aims to prepare the M&E conducted by MoRT, although sometimes it can formulate the improvement of WP's project plan. Therefore, an internal M&E usually takes place a few weeks before an external M&E. Typically, an M&E is executed by conducting presentations from WBS/ WP coordinators, followed by discussions with the PM team (and external reviewer). However, there is no standard mechanism in an internal M&E, and so each PM has his/her own M&E mechanisms and schedules.

2. External monitoring and evaluation by BRIN

An external M&E is conducted by BRIN for the PM and WBS/ WP. The reviewer teams for this M&E consist of BRIN officers and experts appointed by BRIN. Usually the M&E is performed twice a year. The first M&E takes place in the middle of project period as a requirement for the second phase of fund disbursement, while a second M&E is conducted at the end of the project year. The scope of this M&E consists of administrative matters and research progress. Concerning the research project, it focuses on targeted products and potential adopters. Not unlike an internal M&E, this external M&E also requires every research coordinator (WP/ WBS) to present their research progress, accompanied by the program manager. The presentation follows the template provided by BRIN that consists of the output targets (innovative products), a roadmap to achieve the target, and the budget. The presentation is followed by a Q&A session conducted between reviewers and presenters. This M&E expects results of project plan revision/ improvement or even changes in the structure of WP or WBS.

3. Monitoring by the institution.

This M&E is conducted in particular to the WPs funded by the institutions. The mechanism follows the standard of M&E mechanisms from every institution, including its frequency and schedule. In general, the scope of M&E encompasses administrative matters (i.e. budget) and research progress. Usually, the reviewers

come from the internal institutions (appointed formally by the institutions) and external actors (chosen based on his/her expertise).

■ Actors involved

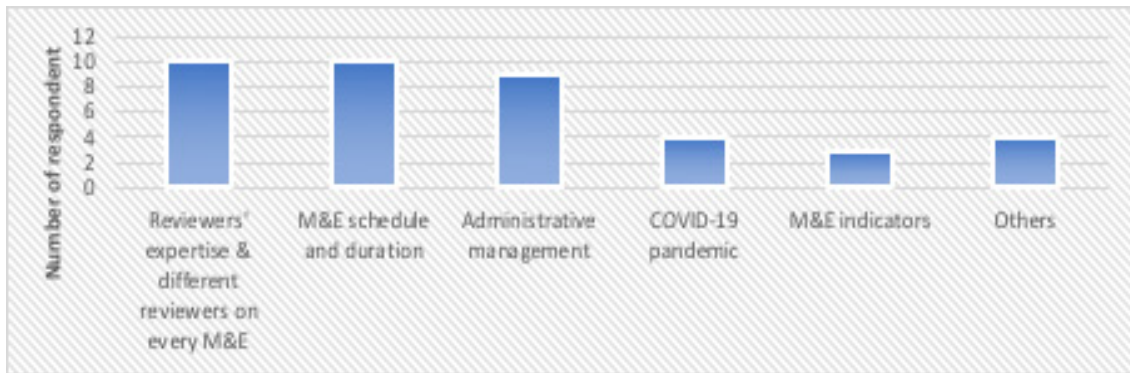
The program manager plays a key role for internal M&Es, while BRIN and academic experts are responsible for external M&Es. In addition, the lead research institutes are also important actors for M&Es at an institutional level.

■ Indicators

For an M&E in an administrative aspect, budget realization is the main indicator to assess whether the research is well-run. However, there is no clear information about indicators used to assess research progress.

Similar to proposal selection, the problems about M&E of PRN mostly lie on the reviewers and M&E management itself (Figure 4-11). Around 25 percent of the respondents said that the reviewers' expertise often does not comply with the research field and they are always assessed by different reviewers on every stage of an M&E. Another 25 percent complain about the M&E schedule that is spontaneous and too frequent, and the duration is too short to receive valuable feedback from reviewers. The complicated administrative management is also a popular answer concerning M&E problems (22.5%). Respondents (7.5%) also state unclear M&E indicators as an issue that needs to be resolved. Any external constraint such as COVID-19 pandemic also factors as an issue since M&E needs to be conducted online.

[Figure 4-11] Identified Problems in PRN M&E



Source: Related survey

4.3 Proposed Research Evaluation System for PRN

This section discusses the proposed research evaluation system for PRN encompassing the proposal selection and research monitoring & evaluation. The results of the online survey and the FGDs (discussed in Section 3) are the foundations in designing this proposed evaluation system.

4.3.1 Proposed Proposal Selection System for PRN

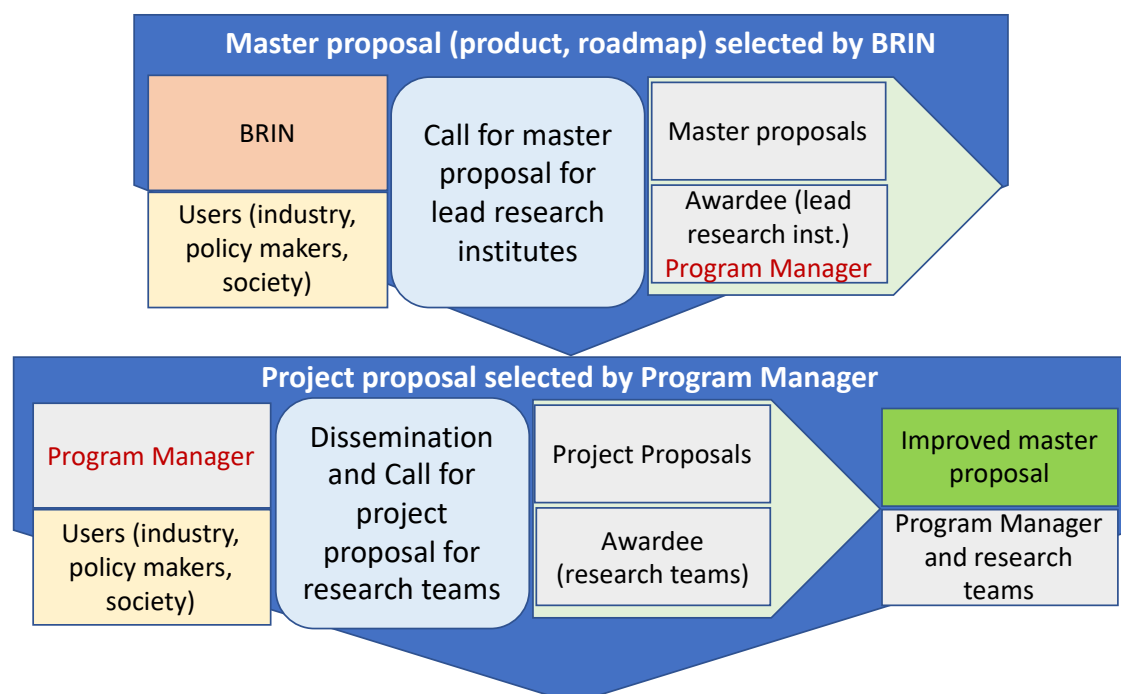
The proposed proposal selection system comprises the elements of: 1) the structure and mechanism; 2) actors involved; and 3) indicators. The corresponding three elements are explained below.

■ Structure and mechanism for proposal selection.

Regarding the structure and mechanism for proposal selection of PRN, two levels of selections are proposed as follows: 1) the selection of master proposals (integrated proposals for the current system); and 2) selection of project proposals (individual proposals for the current system). The master proposal assessment focuses on the targeted product development and roadmap to achieve the targets. Meanwhile, the assessment of project

proposal focuses on the contribution of the individual project to achieve the targets defined in the master proposal, including the detailed research activities. The proposed structure and mechanism for proposal selection are presented in Figure 4-12.

[Figure 4-12] Proposed Structure & Mechanism for PRN Proposal Selection



Source: Authors

The selection of the master proposals is the responsibility of BRIN. In this process, BRIN cooperates with potential users (i.e. industries, policy-makers, civic group officials) to design programs of product development targets for PRN. These product development targets are designed based on the needs of the potential users. Then, BRIN opens a call for proposals on research institutes in Indonesia (i.e. research centres (PR) under BRIN, departments under universities). After the process of proposal selection, the list of granted proposals along with the list of research institutes as awardees is obtained. Every awardee (the research institute) then establishes a *program manager team*, which consists of a leading researcher (main contributors of master proposals) and the management of each research

institute including its administrative officers.

Meanwhile, the selection of project proposals is the responsibility of the program managers (PM) to whom the team's master proposals have been granted. In this process, the PMs also cooperate with potential users (i.e. industries, policy-makers, civic group officials), who are related to the team's product development targets, to make a call for project proposals on research teams. The awardees of this call for proposals then become the research teams of the PMs. Then, these research teams are organized into WP and WBS. Afterward, the PMs along with their research teams improve their master proposals (based on the selected project proposals) as the basis for the agreement (including budget, targets) between the PMs and BRIN.

Based on the structure of PRN (consisting of the PM, WBS, WP), the role of each PMs is pivotal to coordinate and integrate the research activities to achieve the team's product development targets. However, in the current situation, they have less authority to perform these roles. Therefore, giving the PMs full authority in the selection of project proposals is seen to address this situation. However, it is not sufficient, and so the PMs should also be given fund management authority.

■ Proposed actors involved in proposal selection.

In the process of proposal selections, the reviewer teams must be established, for both master and project proposals. The reviewer teams must: 1) have the relative capacity in their fields, 2) be independent, and 3) represent the related stakeholders.

For master proposal selection, the reviewer teams (that this study proposes) consist of:

- 1) *Regulators*, e.g. BRIN, Ministry of National Development Planning (Bappenas), National Agency of Drug & Food Control (BPOM).
- 2) *Users*, e.g. industries, policy-makers (ministries, local governments), civic group officials (NGOs, arts and cultural figures).
- 3) Academic experts.

Meanwhile, for project proposal selection, the reviewer teams (that this study proposes) consist of:

- 1) *The program managers*
- 2) *Academic experts*
- 3) *Users, e.g. industries, policy-makers (ministries, local governments), civic group officials (NGOs, arts and cultural figures).*

■ Proposed indicators for proposal selection.

In proposal selections, the reviewer teams must have the guidelines on assessing the proposals with the indicators of proposal selection. The proposed indicators for the selection of both master and project proposals encompass the four variables of 1) research design, 2) input; 3) output; and 4) impact. Every variable has indicators as provided in Table 4-4.

[Table 4-4] Proposed Indicators for Master & Project Proposal Selections of PRN

Variable	Indicator
Research design	• Problem definition • Objective • Method
Input	• Research infrastructure • Human resources
Output	• Technology Readiness Level • Product/Technology
Impact	• Based on defined output

Source: Authors

Based on these indicators, the proposal can be assessed quantitatively or qualitatively. To this end, additional studies are required to design the detailed measurement method for every defined indicator.

4.3.2 Proposed Monitoring and Evaluation System for PRN

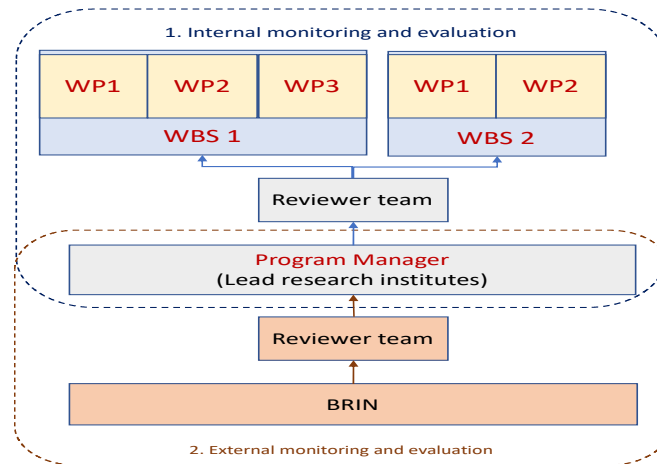
As the proposal selection system, the proposed system for PRN monitoring and evaluation also encompass the elements of: 1) the structure and mechanism; 2) actors involved; and 3) indicators. These three corresponding elements are explained below.

■ Proposed structure and mechanism for monitoring & evaluation.

In the case of the structure and mechanism for monitoring & evaluation, two levels of monitoring and evaluation are proposed: 1) internal monitoring and evaluation by program manager (PM); and 2) external monitoring and evaluation by BRIN. The proposed structure and mechanism for monitoring & evaluation of PRN are presented in Figure 4-13. The internal monitoring and evaluation have the aim to monitor and evaluate the research activities at the level of WP and WBS. This internal monitoring and evaluation are the responsibility of the PMs.

Meanwhile, external monitoring and evaluation have the aim to monitor and evaluate the research program at the level of the PMs. This external monitoring and evaluation are the responsibility of BRIN. Therefore, in this proposed system, BRIN does not perform monitoring and evaluation directly to the level of WP and WBS anymore (as in the current system), as these have been done by the PMs.

[Figure 4-13] Proposed Structure & Mechanism for Monitoring & Evaluation for PRN Research



Source: Authors

■ Proposed actors involved in the monitoring and evaluation.

For internal monitoring and evaluation, in this proposed system, every program manager forms a reviewer team. The members of the reviewer team that this study proposed consist of:

- 1) *Program manager team*
- 2) *Academic experts*
- 3) *Users, e.g. industries, policy-makers (ministries, local governments), civic group officials (NGOs, arts and cultural figures)*
- 4) *Administration office of lead research institutes (esp. for budget)*

Meanwhile, in external monitoring and evaluation, BRIN also forms the reviewer teams. The reviewer teams can be different for every focus and product development target depending on its research field and sector. However, the stakeholder representatives in the reviewer teams that this study proposed are similar consisting of:

- 1) *BRIN officers (including administration)*
- 2) *Academic experts*
- 3) *Users, e.g. industries, policy-makers (ministries, local governments), civic group officials (NGOs, arts and cultural figures)*

■ **Proposed indicators for the monitoring and evaluation.**

As in the proposal selections, for monitoring and evaluation, the reviewer teams must have the guidelines **consisted of** so-called indicators of monitoring and evaluation. The proposed indicators for the monitoring and evaluation encompass two variables of 1) output; and 2) process. Every variable has indicators as provided in Table 4-5.

[Table 4-5] Proposed Indicators for PRN Monitoring and Evaluation

Variable	Indicator
Output	• Research performance indicators (Progress of targeted outputs, Intellectual property, publications)
Process	• Budget: realisation • The involvement of users (industries, policy makers, society) • The progress of technological readiness level

Source: Authors

Based on these indicators, in addition to evaluating the PRN research projects/ programs, the reviewer teams give program managers/ research teams feedback to improve their research.

5. Conclusion

This study aims to build an evaluation system for R&D programs in Indonesia, in particular under the PRN 2020-2024 umbrella. In this study, the South Korea's experience has been taken as teachable case studies. Some lessons learned from the case of South Korea then are adapted to the Indonesian context in more detail as inputs to the structure and management of PRN 2020-2024. In building the evaluation system of PRN in this study, we have tried to involve as much as possible the participating actors (i.e. researchers, project management officer of PRN, research funding manager) through multiple means such as in-depth interviews, online surveys, and multiple FGDs.

Aligned with the finding of the STEPI-LIPI study report in 2020, this study has found that there is still no clear evaluation system for PRN, notably for: 1) the structure and mechanism; 2) actors involved; and 3) the indicators. This situation has clearly generated challenges, hindering the optimal implementation of PRN 2020-2024, such as less coordination & integration of projects, less-organized monitoring & evaluation, or excessive actors involved in monitoring and evaluation.

The evaluation system proposed in this study is thus expected to resolve the challenges mentioned above. Accordingly, two levels of structure in proposal selection and research monitoring & evaluation can give program managers (PM) more authority to coordinate and integrate their research projects. In addition, such a structure would also make the division of responsibility between BRIN and the PMs clearer, notably for proposal selection and research monitoring & evaluation. The proposed actors involved have represented important stakeholders that we believe can provide PMs and research teams feedback on their research programs. Meanwhile, the proposed indicators are expected to make the process of proposal selection and research monitoring & evaluation activities more organized.

In this study, we have also worked to address the issue of inter-institutional collaboration in R&D programs/ projects. Accordingly, coordination and integration are expected to be

improved by giving more responsibility and autonomy to PMs. These are essential for PRN programs to produce innovative products. It should be noted that PMs cannot fully perform their roles if they do not wield the authority (to a certain extent) in managing research funding as it pertains to the current situation. Therefore, political support and change in the R&D management system are required to alter the current situation.

While the proposed design resulted in this study clearly indicates a decentralized management system, the recent reorganization of research institutions in Indonesia through the establishment of the National Research & Innovation Agency (BRIN) has raised a new issue worthy of discussion. Although more studies are required to gauge the implication of BRIN to our proposed system, there has been a tendency that BRIN has strengthened its centralized and fragmented funding management, a system that we do not recommend as it may create a new and added bureaucratic process into the existing R&D management system.

6. Follow-up Work

In the process of this study, in addition to the issue of the research evaluation system of PRN, several issues in PRN implementation have also been identified. However, due to funding and time constraints, these identified issues cannot be fully answered in this study. Therefore, follow-up work is needed to complement the findings of this study. The issues that emerged in this study on which further work is needed are briefly discussed below.

6.1 Study on PRN Design

In the current situation, PRN 2020-2024 was designed to implement all the R&D programs starting in 2020, and it was expected to produce innovative products by 2024 (Table 4-6). It should be noted that between 2020 and 2024, no new program has been planned, nor after 2024 as well.

[Table 4-6] Design of PRN 2020-2024

2020	2021	2022	2023	2024	2025	2026	...
Call for proposal							
49 product development targets				49 innovative products	?	?	?

Source: Authors

However, innovation is a continuous process. Therefore, continuous R&D programs (refer to Table 4-7) are highly important to produce innovative products that are expected to contribute to the country's economic growth. To achieve this, additional study on redesigning PRN is essential.

[Table 4-7] Examples of Designing Ongoing Research Program

2020	2021	2022	2023	2024	2025	2026	...
Call for proposal 1	Call for proposal 2	Call for proposal 3	Call for proposal 4	Call for proposal 5	Call for proposal 6	Call for proposal 7	Call for proposal n
Programmes 1	Programmes 2	Programmes 3	Programmes 4	Programmes 5	Programmes 6	Programmes 7	Programmes n
				Products/ Technology 1	Products/ Technology 2	Products/ Technology 3	Products/ Technology n

Source: Authors

■ Study on co-funding management for PRN

PRN 2020-2024 applies a co-funding scheme, in which funding for PRN research projects can thereby come from LPDP through MoRT/ BRIN, member institutions, and other funding sources. From the perspective of research funding management, this scheme can overcome the problem of limited research funding assigned for PRN. However, based on the findings in this study, this scheme has led to the challenge of coordinating research activities. As a result, additional study on co-funding management for PRN is important.

■ Study on commercialization of research output of PRN programs

As previously mentioned, the main objective of PRN 2020-2024 is to produce innovative products that are expected to contribute to the country's economic growth. However, until now, most of the R&D programs still largely focus on research activities. To the best of our knowledge, there is still no clear plan to commercialize products resulting from these research programs. As a result, further research on commercialization of research output of PRN programs is needed accordingly.

6.2 Teachable case studies from other countries (i.e. South Korea) to get industries more involved in national research programs

Based on the findings of this study, one of the main challenges in PRN 2020-2024 implementation is to involve industry actors. Even though several previous research projects have involved industry actors, the main challenge still exists. Therefore, teachable case studies and lessons from successful experiences of countries (i.e. South Korea) on how to engage industries more actively under national research programs become essential to carry out.



CHAPTER 5

Laws & Regulations of Science & Technology

Seong-soo Kim

Professor, Hankuk University of Foreign Studies (HUFS)

Changyul Lee

Researcher, Hankuk University of Foreign Studies (HUFS)

Jung-Hun Kim

Researcher, IBS Consulting Company

Dadang Ramdhan

Researcher, BRIN

Maulana Akbar

Researcher, BRIN

Anggini Dinaseviani

Researcher, BRIN

Sri Supadmi

Policy analyst, BRIN

Chapter 5. Laws & Regulations of Science & Technology

1. Introduction

This chapter analyzes the current regulations and laws on science and technology in Korea and Indonesia, and thereby provides suggestions for certain regulatory frameworks for Indonesia. This chapter covers the following subjects.

The second section deals with the category of regulatory framework that involves science and technology, and also covers the issues of content, scope, classification, and framework of regulations in Korea.

The third section discusses the current regulatory framework for science and technology in Korea, and introduces the developments of R&D regulations that are based on the ten-year retrospective between the 1950s and 2010s. In particular, this research distinguishes between the laws for improvement of science and technology and laws for regulating science and technology. It also introduces the major regulations in certain time periods and discusses the adoptions of regulations and specific issues in the amendment procedures.

The fourth section analyzes the current regulatory framework for science and technology in Indonesia, while examining in more detail the developments of R&D regulations that are based on the ten-year retrospective between the 1950s and 2010s. In particular, this section also discusses main Issues regarding science and technology in Indonesia according to Law No. 11-year 2019 concerning the national system of science and technology.

The fifth section then examines the main issues in the process of regulatory developments in the area of science and technology in Korea. More importantly, it focuses on certain regulatory issues in the change of science and technology regulations in Korea, and analyzes

specific problems.

The final section provides a summary and suggestions for the regulatory framework for science and technology development in Indonesia.

2. Typology of Laws of Science & Technology

2.1 Contents & Scope of the Laws & Regulations for Science and Technology

The classification standard can be decided based on the inquiry how to define science and technology. The content and scope can be distinguished based on the three categories, such as narrow, broad, and extended categories (Yoon et al., 2013: 131-132). The narrow concept merely covers the area of regulations that directly involves the improvement of science and technology and enhancement of industrial use. In other words, the narrow definition of science and technology laws covers the regulatory measures for the improvement and enhancement of science and technology. The wide concept of science and technology laws covers the narrow definitions of science and technology laws & of adverse effects as well as the regulations of performance management of science and technology. The extended definition includes all the regulations that deal with science and technology. In general, the wide concept of science and technology can be considered as science and technology laws that can be discussed here. Therefore, it defines the scope of science and technology laws based on improving and regulating science and technology on a state level.

The categorization of the wide scope of science and technology laws in the scope of improving and regulating this sector is a result of the shift of social recognition on science and technology. Since the importance of science and technology is notable in our modern society, we often emphasize its role as both an economic growth driver and solution provider for diverse social and future problems. In other words, science and technology can contribute not only to the economic development of the state-run apparatus but also to solutions for social and future problems in our society, such as climate change, aging society, energy consumption, quality of life, etc. Science and technology is one of the key policy instruments for society and the general public, and involves major policies such as economy, education, diplomacy, etc. Therefore, it indicates that the laws of science and technology should be understood as a wide concept of regulations for science and technology.

In particular, the development of science and technology on a state level is often influenced not only by initiating science and technology improvement but also raising certain industries that are directly and indirectly associated with specific science and technology. Therefore, in order to understand the framework and content of science and technology laws, it is necessary to discuss the relevant issues based on the scope of wide definition that was discussed above. Accordingly, the subject that is covered in this chapter should be the laws for the improvement of science and technology, enhancement of R&D, and supports and aids for the industries which are closely related to science and technology.

2.2 Category of Laws of Science & Technology

When we decide to interpret the scope of science and technology laws in a wider sense, it is essential to define the method and analysis standard for categorizing science and technology laws. In effect, there are several different approaches to the categorization of science and technology laws which are founded on various academic works. This research first provides the literature reviews on Korean laws of science and technology. Afterward, we will discuss the classification system of science and technology laws.

The Korean Ministry of Government Legislation provides a classification of four science and technology sectors, such as administrative structure & general provisions, improvement of science and technology, nuclear, and astronomical weather research¹⁰. This area used to be the work of the science and technology ministry in Korea. This classification appears to be too narrow because it does not reflect the laws related to science and technology policy which belong to the jurisdiction of other state-run ministries. For discussing a wider categorization method, there are numerous academic works as follows. Lee et al. (2006) argue that it is necessary to categorize the laws of science and technology based on a state-run ministry, such as general provision, administrative structure, R&D and improvement, research coordination & international cooperation, research organization, workforce training, education & organizational culture, research performance management & expansion, environment & safety regulation, nuclear, astronomical weather research, and

10 Korean Law Information Center <https://www.law.go.kr/LSW/lsAstSc.do#AJAX> (accessed on 10 May 2021)

other areas of law. Yang et al. (2012 and 2017) also categorizes science and technology laws as follows: R&D regulation, performance-evaluation regulation, technology protection regulation, technology utilization regulation, general infrastructure regulation, research institute regulation, workforce training regulation, technology standard regulation, organization regulation, science and technology safety regulation, and lastly science and technology ethics regulation. Yoon et al. (2013) categorizes the laws of science and technology based on legislative purposes, such as general regulatory provision and sector-specific general policy, organization regulation for science and technology, improvement of science and technology regulations, and law of regulating science and technology. A part of science and technology improvement includes the law of STI and the development and enhancement of industry, the law of performance management of R&D and enhancement of technology utilization, and the law of organizing the science and technology foundation, organizational culture, and training for developing specialists in science and technology.

This research is founded on the work of Yang et al. (2012 & 2017) and provides an analysis standard for science and technology regulations. The research provides a standard of analyzing science and technology laws as follows: the objectives of law, policy goals, technology sector, scope of regulations, competent ministry, and enactment period. In particular, the policy goals are divided into the eleven categories that were suggested earlier. This classification has some noteworthy advantages compared to other academic works. It contains the objectives of the laws of science and technology which were already discussed in other works. Meanwhile, it allows an effective analysis on a regulatory framework of science and technology which is based on various points of view.

We can provide the standard of classification in detail, which is explained as follows.

Firstly, the standard following the objective of laws can help categorize (i) the law of science and technology improvement and (ii) law of regulating science and technology.

Secondly, the policy goal standard can be used to classify (i) the law of R&D, (ii) law of performance management, (iii) law of technology protection, (iv) law of technology utilization, (v) law of infrastructure, (vi) law of developing research institutes, (vii) law for

developing science and technology workforces, (viii) law of technology standard, (ix) law of organization, (x) law of science and technology safety, and (xi) law of science and technology ethics.

Thirdly, the standard of regulatory scope covers (i) the general provision of policy on state-run science and technology, (ii) general provision for the competent science and technology ministry policy, and (iii) law for implementation of science and technology policy. The standard of scope of regulation is helpful to understand the overall framework of relevant Korean laws of science and technology, and this will be further discussed below.

Fourthly, the competent ministry becomes the standard of the classification method for categorizing the competent authorities of science and technology laws, and one of them is the Ministry of Science & ICT (MSICT). In Korea, there are several state-run ministries that are responsible for the sectors of science and technology, education, industry, public health, defense, etc. These ministries have certain jurisdictions based on laws involving science and technology. Interestingly, each ministry's title and scope can be changed where there is a reorganization scheme.

Fifthly, the technology standard consists of (i) the common technology, (ii) IT, (iii) BT, (iv) NT, (v) ET, (vi) ST, (vii) CT, and (viii) nuclear. Lastly, the standard of enactment period is based on the ten-year retrospective method. This classification method can be used to analyze the current legislation and features of regulatory changes arising from relative environmental changes.

Among the classification methods, this paper reviews and examines the legislative objectives and period of legislation that are based on the focus of the improvement and regulation of science and technology in Korea. Afterward, it further analyzes the distinctive features in the process of adoption and amendment to the relevant acts.

[Table 5-1] Method & Standards for Analyses of Science & Technology Laws

No.	Category	Classification criteria	
1	Legislative purpose	① S&T Laws for Promotion	② S&T Laws for Regulation
2	Policy purpose	① Technology development ③ Technology protection ⑤ Infrastructure ⑦ Workforce training ⑨ Administrative organization ⑪ S&T ethics	② Performance evaluation ④ Technology utilization ⑥ Fostering R&D institution ⑧ Technical standard ⑩ S&T safety
3	Fields of technology	① Common technology ③ BT (Bio technology) ⑤ ET (Environment technology) ⑦ CT (Culture technology)	② IT (Information technology) ④ NT (Nano technology) ⑥ ST (Space technology) ⑧ Nuclear energy
4	Levels of Laws	① Rules governing National Science & Technology Policy ② Rules governing S&T policy of individual ministries ③ S&T policy rules implemented by individual ministries	
5	Ministerial jurisdiction	① NSTC(National Science & Technology Commission)	
		② M. of Education ④ M. of Industry	③ M. of S & T ⑤ M. of Health and Welfare
6	Enactment year	⑥ M. of Environment ⑦ M. of Land, Infrastructure & Transport ⑧ M. of National Defense ⑨ M. of Agriculture, Food, & Rural Affairs ⑩ Nuclear Safety & Security Commission, etc.	
		① 1950s ② 1960s ③ 1970s ④ 1980s ⑤ 1990s ⑥ 2000s ⑦ Post-2010s	

Source: Revised from Yang et al. (2012). p.162

2.3 Three Dimensions of the Laws of Science & Technology

According to Yang et al. (2012), the Korean regulations of science and technology can be divided into the following three levels of (i) the general regulations at the cross-ministerial level, (ii) general regulations of science and technology policy agenda at a ministerial level,

and (iii) general regulations of the implementation of science and technology policy at a ministerial level.

To begin, the general regulations at a cross-ministerial level indicate the laws that govern the science and technology policy and state-run R&D plan from the total periodic perspectives. Therefore, it refers to the laws that can adjust the cross-ministerial issues and have the implementation functions. This sector includes the Framework Act on Science & Technology, which provides legal measures on the general organization for the state-led science and technology policy, the principles of enhancing the state-run R&D sector, and parts of the direction of the state-led science and technology policy.

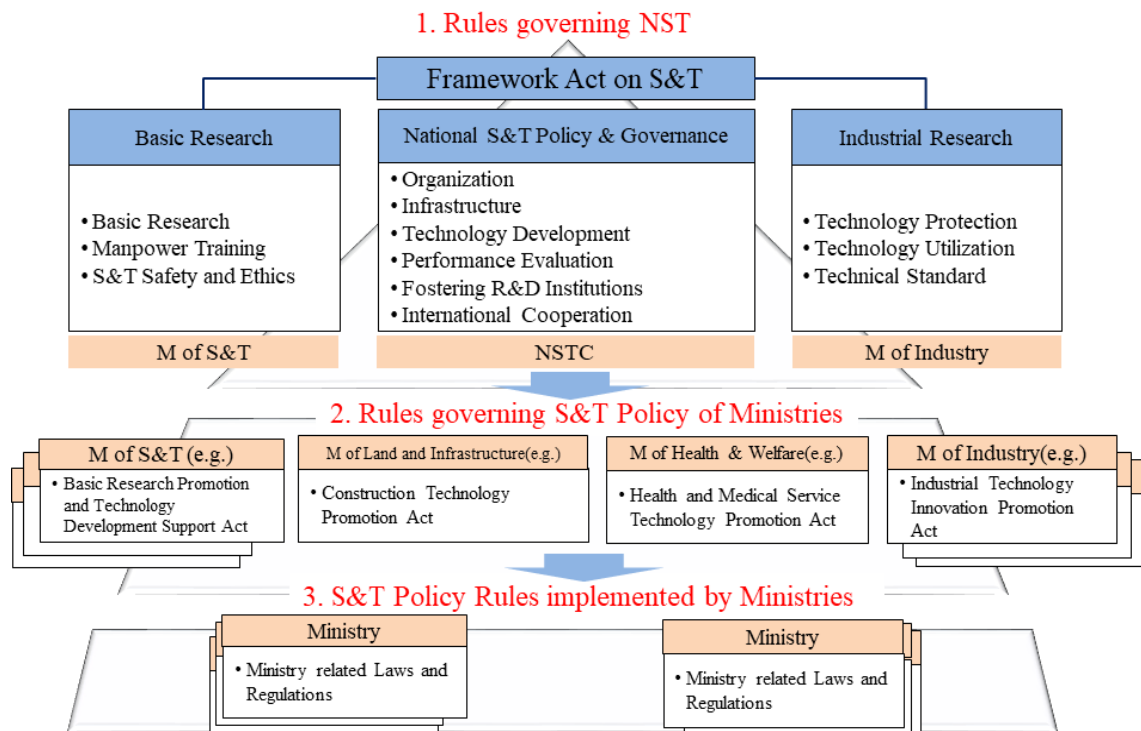
The National Science and Technology Commission (NSTC) is responsible for the decision-making on state-level science and technology issues. At the time of writing in 2021, it was named the Presidential Advisory Council on Science & Technology (PACST) in Korea. PACST has the task and duty of performing overall policy governance of the state-led strategic plan of science and technology. Lastly, the general regulations at a cross-ministerial level can be divided into several parts, such as the general regulation of science and technology policy, general regulation of industry policy, and general regulation of fundamental research policy. In particular, the general regulation of science and technology policy is comprised of the law of organization (including planning), law of infrastructure, law of R&D, law of performance management, law of developing research institutes, and law of international cooperation. The general regulation of industry policy embraces the law of technology protection, law of technology standard, and law of technology utilization. This approach is based on the view that the sector of science and technology should be understood as an industry. The general regulation of fundamental research policy comprises the law of fundamental research, law of workforce training, law of science and technology safety, and law of ethics. This category indicates the law of managing science and technology, considering the fundamental research and connection with universities from an educational perspective.

The general regulations of science and technology policy agenda at the ministerial level indicates the laws of adjustment and enforcement functions at a ministerial level in the following areas of science and technology policy and state-run R&D sector. Each ministry

has jurisdiction over a specific science and technology policy. Of course, there are differences between ministries based on the competencies and laws that permit certain jurisdictions. For example, the MSICT has discretion in managing and adjusting the policies on the improvement of fundamental research under the Basic Research Promotion & Technology Development Support Act. Moreover, the Ministry of Health and Welfare has jurisdiction over the R&D of public health and medical technology under the Health & Medical Technology Promotion Act.

Finally, the general regulations of the implementation of science and technology policy at the ministerial level inform the laws of policy implementation of science and technology that exclude the two above. Each ministry performs the implementation of the law involving science and technology when there is a clear statement of jurisdiction.

[Figure 5-1] Three Dimensions of Laws of Science & Technology



Source: Revised from Yang et al. (2012). p.165

3. Laws of Science & Technology in Korea

Since the Republic of Korea was established in 1948, we have seen continuous developments of laws of administrative frameworks of science and technology and those of improving and regulating science and technology in the face of the state's demand and societal changes. This paper provides the two categories of the relevant laws, such as the law of improving science and technology and law of regulating science and technology. In particular, this reviews the major legislations and revisions, which help better understand the changes and developments of the laws of science and technology, and thereby examines the features of legal developments.¹¹

3.1 The 1950s

Since the Republic of Korea government was established in 1948, the 1950s period was important for being the time of the foundation of state-led development. At that time, there was no effective and systemic foundation for legislation because it was the stage of providing a legislative framework. As a result, the Korean legal system of science and technology was implemented not based on the specific state's goal but rather on the purpose of solving certain problems. One of the first meaningful legislations was the adoptions of the Trademark Act in 1949. All the while, there was an important enactment for improving science and technology, which was the adoption of the Atomic Energy Act of 1958. This law was about regulating a certain sector of technology, but it was the first legislation for the purpose of improving science and technology in Korea. This law covers the areas of nuclear power involving R&D, utilization and management, etc., and thereby enhances academic advancement and triggers industrial development.

¹¹ This section reorganized the contents according to the purpose of the study based on STEPI (2017: 25-51).

3.2 The 1960s

A meaningful discussion on the Korean regulations of science and technology was made in the 1960s. During this time, the Korean government took a necessary approach to establish a systemic foundation of basic legislation for the state's science and technology policy. For instance, Korea instated the laws of technology standard and establishment of science and technology institutes. Moreover, the government provided the laws for the systemic development and improvement of science and technology in the fields of machines, shipbuilding, electronics, etc. For starters, in the early 1960s, Korea established certain standards for R&D and systems for ensuring property rights in the industrial sectors. For example, Korea enacted the Measures Act of 1961 to provide standards involving various measuring and gauging standards. In addition, Korea enacted the Industrial Standardization Act for enhancing the quality of mining and manufacturing products and improving productive efficiency. Moreover, at this time, the Utility Model Act and Design Act were adopted for the guarantee of industrial property rights.

In 1996, Korea enacted the Korea Institute of Science & Technology Promotion Act, and thereby established an institute specialized in science and technology which could initiate effective R&D promotion. In 1967, the Science & Technology Promotion Act was provided, which became one of the important and founding legal measures for improving science and technology in Korea. This Act became the founding legal measures for providing an overall policy on science and technology and its implementation, while Korea initiated legislation for improving industrial development in various founding sectors.

Moreover, in the late 1960s, various regulations were adopted in order to trigger the state's economic development in certain sector industries. For instance, Korea provided new regulations, such as the Machinery Industry Promotion Act, Shipbuilding Industry Promotion Act of 1967, and the Electronic Industry Promotion Act of 1969. Through providing these Acts, Korea could actively introduce R&D plans and train specialists of technologies.

3.3 The 1970s

In the 1970s, Korea started considering that science and technology was one of the most important tools for the development of state-led economy and industries, and tried to enhance this sector, and so Korea adopted legislation for promoting technology development. In particular, Korea continued to adopt numerous legislations for supporting the sectors of refinery, steelmaking, airline industries, etc. Moreover, Korea launched the state-funded research institutes for certain sector industries for assisting these fields. From legal perspectives, Korea stipulated various standards and procedures for activities involving technological services. Diverse technology-related qualification systems became a part of governance by instating the relevant Acts.

Firstly, in 1972, the government enacted the Technology Development Promotion Act. Moreover, policy for fostering sector-specific industries extended to other fields of industries. For instance, Korea enacted the Petrochemical Industry Promotion Act and Iron & Steel Industries Promotion Act of 1970, thereby improving the sectors of petrochemical and iron & steel industries. Additionally, Korea enacted the Aviation Industry Promotion Act for developing the aviation industry by supporting R&D efforts in this area.

In addition, several Acts were provided to create the state-funded institutes for the research of science & technology and education. In 1970, the Act on the Agency or Defense Development was enacted for ensuring the potential developments of science and technology in the area of defense. In 1973, the Korean Atomic Energy Research Institute Act was passed, which made the government implement the overall R&D policies on atomic energy. In 1976, the Korea Science and Engineering Foundation Act was enacted to establish the Korea Science and Engineering Foundation for improving the capability of science & technology research, science education, and international coordination with other countries in the area of science and technology. Regarding the regulating measures, Korea enacted the Technical Services Support Act in 1973 which provided the process and standard of implementing various technical services. By means of instating the National Technical Qualifications Act, the Korean government could manage and regulate diverse technical qualification systems.

3.4 The 1980s

The 1980s is the time period of extending the state's R&D sectors. Therefore, there were a number of regulatory changes and developments for supporting R&D in Korea. In other words, the Korean government could rearrange and reorganize the state-funded research institutes and other technology-related institutes by providing certain legislations. In order to reinforce R&D support, Korea enacted the Research of Basic Sciences Promotion Act in 1989, which became the systemic measures for stimulating the basic science sectors. By providing this Act, the government could acquire competencies in developing outstanding science and technology workforces for creative R&D capacity in Korea.

Moreover, at this time, we have seen notable amendments to the Acts of the state-funded research institutes. To combine, expand, or reorganize these institutes in its push to rearrange, Korea had to adopt legislations for new institutes. For example, Korea enacted the Korea Institute of Nuclear Safety Act of 1989, which allowed the government to restructure the existing Nuclear Safety Center to the Korea Institute of Nuclear Safety that became the institution responsible for safety R&D and assistance related to the nuclear industry. The roles and tasks of the state-funded institutes became segmented in this period, and a number of institutes, which performed the role of regulating the sectors rather than supporting the sectors, emerged during this time.

3.5 The 1990s

The Korean government provided the policy agendas for improving information technology and establishing the state's technology innovation system in the 1990s. As a result, it was necessary to adopt and revise the relevant regulations. At this time, Korea provided the foundation for the general regulations for state-run science and technology in Korea. Also, there were amendments to the regulations that covered the support and management of research institutes. Additionally, we have seen continuous revisions to the sector specific regulations for improving science and technology. In particular, facing rapid developments of information technologies, Korea adopted numerous regulations of the information

industries. Meanwhile, the number of legislations of regulating the science and technology sectors increased during this time, as opposed to other periods in history. Therefore, at this time, Korea revised the existing regulations for the fast-moving areas of innovation sectors.

The 1990s is important in terms of providing a foundation for enacting the Framework Act on Science & Technology of 2001. In 1991, the Presidential Advisory Council for Science & Technology Act was enacted for the creation of the Presidential Advisory Council. In 1997, Korea provided the Special Act on Innovation in Science & Technology that ensured the improvement of governmental funds, efficient implementation of R&D resources, enhancement of enterprises' R&D activities, and the systemic framework for supporting science & technology innovation, including incentives for scientists and engineers. This Act contributed to effective managements for the state's R&D innovation.

In addition, Korea improved the support and management system for special research institutes in the process of the state's technological innovation framework. In 1993, the Daedeok Science Town Management Act was enacted, and through this legislation, the Korean government could establish the Daedeok Science Town which became the center for educational and research institutes. This project was founded in 1974 and completed in 1993. By instating the relevant legal provisions, the government worked to enhance the effective management of Deadeok research institutes, and thereby utilize the outcome of research, while improving the coordination between universities and businesses. In 1999, the Act on the Establishment, Operation & Fostering of Government-Funded Research Institute, etc. was adopted. Before this legislation, several ministers had jurisdictions over certain industrial sectors, but this Act provided the control tower of the Prime Minister for monitoring and controlling the state-funded institutes for R&D. In summary, this Act enabled the government to reinforce the research outcomes of the institutes and evaluate the managements of the institutes.

For the progress of the industrial sectors, several Acts for information technologies and knowledge-based economies were adopted. In 1995, the Framework Act on Informatization Promotion was enacted, which allowed the government to initiate the policies on the state's informatization. Korea also adopted the Framework Act on Electronic Commerce of 1999

which was the law for the development and standardization of technologies for electronic commerce and research on standards involving e-commerce in Korea and other countries. Moreover, the Act on Special Measures for the Promotion of Venture Businesses was enacted in 1997, an act which has aimed to provide a general framework of the supports for start-up or venture companies that specialized in new knowledge-based technologies.

In addition, considering the regulatory field, the Korean government tried to provide certain measures for specific industrial technology sectors, and provided the legal framework for environmental protection and relevant technologies. For starters, Korea revised the contents and titles of the regulatory provisions on industrial technology in order to fit into the environmental changes. The Korean government expanded the scope of its funding and support, thereby protecting enterprises' innovation technologies. For instance, Korea amended to the Technical Services Support Act in 1992, which was the entire revision of the Act, and this amendment allowed the extension of private sectors' autonomy and support policies on engineering technologies. Moreover, the Manufacturing Standardization Act was renamed to the Industrial Standardization Act by revising the title of the law. This eventually confirmed the scope of the industrial standards meant to actively reflect the sector of cutting-edge industry and innovation. Regarding the overall standardization, Korea also provided the Framework Act on National Standards in 1999, which established the standards in various technology sectors. Moreover, there was an amendment to the Unfair Competition Prevention Act, which was renamed as the Unfair Competition Prevention & Trade Secret Protection Act in 1998. This Act protects the knowledge and know-how technologies of individual enterprises. Lastly, Korea enacted the Water Quality Conservation Act in 1990 for protecting environment and the Natural Environment Conservation Act in 1991 for conserving environmental resources.

3.6 The 2000s

One of the most important legislations in the 2000s is the Framework Act on Science & Technology that includes the general regulation for Korean science and technology. The Korean government enforced this Act to reorganize the regulatory framework of science and technology sectors. Additionally, there were noteworthy legislative approaches to shift the state-led policy of science and technology from the focus of R&D to the agenda for implementing and utilizing the developed technologies. Accordingly, there were a series of legislations for the expansion of technologies and commercialization of relevant technologies. Moreover, Korea tried to adopt a new regulation and amend it to the existing regulations for enhancing R&D for science and technology. As the result, Korea adopted and revised the laws of air & space, nanotechnologies, conservation, maintenance, and development of agriculture, forestry, and fisheries resources, nuclear power & eco-friendly energy development, and other eco-friendly sectors. In addition, there were several problems and issues involving bioethics, laboratory safety, etc. in the research processes, which triggered the relevant regulations for solving these problems in the 2000s.

As mentioned earlier, one of the important developments in technology laws in Korea was the adoption of Framework Act on Science & Technology in 2001, which replaced the areas covered by the Science & Technology Promotion Act of 1967 and Special Act on Innovation in Science & Technology in 1997. This regulatory development was meaningful because former approaches to regulate sectors based on each ministry jurisdiction were changed to approaches of controlling the overall sectors for technology development policy as a whole. Under this new Act, the Korean government held jurisdiction over the issues of strengthening the function for budget allocation of state-run R&D innovation, improvement of local government's science and technology, enhancement of basic science research, etc. Overall, this Act intensified the framework of the state's technology innovation.

In particular, by restructuring the framework of science & technology innovation, the government initiated the push to rearrange the state-funded research institutes. In 2004, Korea instated the Act on the Establishment, Operation & Fosterage of Government-Held Research Institutions of Science & Technology that rearranged the control of the sector

under the Prime Minister being shifted to MSICT. Many expected that the council of state-run science & technology and MSICT would take the important role of business rationalization and development of the government-held research institutes. In addition to this, there was a regulatory development of a performance management system and utilization of these research institutes. In 2005, the Special Act on the Support of Daedeok Special Research & Development Zone, etc. was enacted, which has aimed at facilitating research development and innovation of universities, laboratories, and enterprises in specially designated regions.

Moreover, by rearranging the administrative framework of science and technology, Korea instated the laws for facilitating systemic management and utilizing the outcomes of state-run R&D innovation. In 2000, the Technology Transfer Promotion Act was enacted, which focuses on technology transfers, assessment of technologies, and distribution of technology information. There was a complete amendment to the Technology Transfer & Commercialization Promotion Act in 2006. The revised Act has emphasized not only technology transfers but also commercialization of applicable technology. Therefore, this Act affirmed the system of technology evaluation and implementation of technology, thereby providing a tool for technological finance that was related to technology evaluation. In 2005, the Act on the Performance Evaluation & Management of National Research & Development Projects, etc. was enacted, which permits the evaluation of R&D activity that is based on its performance and allows effective management and utilization of R&D results. In October 2006, the Act on Confidentiality & Protection of Industrial Technology Secrets was enacted. This Act's main purpose was to protect the major industrial technologies that have noteworthy impacts on the government by protecting the confidentiality of industrial secrets and important technology, thereby strengthening competitiveness of primary Korean industries.

One of the distinctive features of regulatory development in the 2000s is the noteworthy regulations of eco-friendly technology development, bioethics, prevention of harms from science and technology, and safety beyond the focus on R&D innovation. One of the examples of eco-friendly regulation is the Environmental Testing & Inspection Act of 2006. By instating this provision, the Korean government could establish policies and plans on

environmental testing and the relevant technologies. Moreover, we have seen several regulations of bioethics in this period. In 2001, the Transboundary Movement, etc. of Living Modified Organisms Act was enacted, which has aimed to ensure the safety of the development, production, import, export, and distribution of genetically modified organisms. In 2004, the Bioethics and Safety Act was enacted to provide an environment for the safe development and utilization of bio science and technologies for cures and preventions of human diseases. In 2009, the Act on the Acquisition, Management, & Utilization of Biological Research Resources was enacted to provide a foundation for initiating research, development and acquisition of biological resources and effective management of the resources. In the area of safety management, the Act on the Establishment of Safe Laboratory Environment was enacted in 2005. This Act provides the legal measures of ensuring the safety of labs of science and technology at universities and research institutes. It also stipulates the provision of financial compensation or damages for accidents at these labs.

3.7 The 2010s and Its Aftermath

In the 2010s, the Korean government initiated the regulatory reforms for future technologies, focusing on the sectors of internationalization, modernization, interdisciplinary systems, and a knowledge-based society. In other words, Korea provided certain legislations for the internationalization of science and technology, and revised several regulations to prepare for sustainable development, science & technological interdisciplinary studies, and intellectualization. Moreover, Korea reorganized the management system for basic research and R&D businesses, and there was an expansion of regulatory scope, which includes the laws of intellectual property rights and data protection.

Firstly, Korea enacted the Special Act on Establishment & Support of the International Science & Business Belt in 2011 in order to establish a framework for the internationalization of science and technology and facilitating modernization of technologies. This Act provided the foundation for the basic research environment at a global level and for interdisciplinary

business foundation, associated with basic research sectors.

In addition, there was a regulatory development that reflected the governmental efforts for supporting the interdisciplinary technology and industries and potential new technology development. In effect, Korea adopted the laws focusing on the interdisciplinary approach to industrial sectors, as well as the laws for facilitating each technology sector. In 2011, the Industrial Convergence Promotion Act was enacted, which has been geared at inter-industrial, technological, industrial, & inter-technological combinations, the facilitation of utilization-based R&D, and expansion of industrial convergence research outcomes and commercialization. One of the legislation cases of individual technology is the adoption of the Carbon Materials Convergence Technology Development & Promotion Support Act in 2016. This law aims to facilitate the developments and systemic supports for carbon materials convergence technologies. Moreover, one of the examples involving the future new technology sector is the legislation of the Hydrogen Economy Promotion & Hydrogen Safety Control Act in 2020. This Act was provided to recognize the importance of hydrogen as an important future industrial resource and safely manage & control hydrogen.

The Korean government has responded to the rapid development of technologies and provided the laws for supporting R&D innovation. In 2011, there were noteworthy complete amendments to two Acts, namely the Researches of Basic Sciences Promotion Act and Basic Research Promotion & Technology Development Support Act. Through these amendments, the Korean government could implement the policy for basic research supports and technological development facilitation. In 2020, the National Research & Development Innovation Act was enacted, which provides the systemic framework for the management of state-run R&D businesses. This Act contains the measures of state-run R&D business, including the call for business projects, evaluations, utilization of performance outcomes, etc., national R&D innovation environment, and sanctions involving research code of ethics in the field of state-run R&D businesses.

In the field of regulating sectors, various legal measures, such as the protection of intellectual property rights, bioethics, intellectualization, etc. In particular, the Framework Act on Intellectual Property was enacted in 2011. This Act was provided to establish the

framework for creation, protection, and implementation strategies involving the state's knowledge, property, and new knowledge property, with a provision including the measure for creating the research institute that performs R&D on the intellectual property-related system and policy. In the area of bioethics, the Act on Cord Blood Management & Research was enacted in 2010. This Act allows for the systemic management and research of cord blood derived on the stem cells for curing refractory diseases. Moreover, the Act on the Conservation & Use of Biological Diversity was enacted in 2012, and this law permits the policy establishment and implementation for R&D and innovation development strategy in the subject of biological diversity. As for the intellectualization sector, Korea instated the Act on the Development of Cloud Computing & Protection of Its Users in 2015. In order to facilitate the cloud computing sector, this Act provides for policy planning for R&D improvement in this field, while also aiming to improve the trusted security of cloud-computing services and users' privacy protection that includes the information release, damages, contracts, etc.

[Table 5-2] History of S&T Laws in Korea

Period of enactment	Legislative Purpose	
	List of enacted Acts	Regulations
1948-1950s	• Atomic Energy Act(1958)	• Trademark Act(1949)
1960s	• Korea Institute of Science & Technology Promotion Act (1966) • Science and Technology Promotion Act (1967) • A series of Acts for Fostering Individual Industries (e.g. Machinery, Shipbuilding, etc.)	• Measures, Standardization, Utility Model and Design Acts (enacted in 1961, respectively)
1970s	• Technical Development Promotion Act (1972) • A series of Acts for Fostering Individual Industries (Continued) • A series of Acts on the Establishment of Public Research	• National Technical Qualifications Act (1973) • Technical Services Support Act(1973)

Period of enactment	Legislative Purpose	
	List of enacted Acts	Regulations
	Institutions (e.g. Defense, Atomic Energy, Foundation for S&T, etc.)	
1980s	- Regulations on Daedeok Science Town Construction Promotion Committee (1980, Presidential Decree) - Basic Science Research Promotion Act(1989)	Korea Institute of Nuclear Safety Act(1989)
1990s	- Presidential Advisory Council for Science & Technology Act (1991) - Daedeok Science Town Management Act (1993) - Special Act on Innovation in Science & Technology (1997) - Act on the Establishment, Operation & Fostering of Government-Funded Research Institutes, etc. (1999) - Framework Act on Electronic Commerce (1999) - A Number of Acts in Response to the Informatization (e.g. Informatization Promotion, Promotion of Venture Business, etc.)	- Water Quality Conservation Act (1990) - Natural Environment Conservation Act (1991) - Unfair Competition Prevention & Trade Secret Protection Act (1998) (Protection of Company's Technology Information) - Whole Amendment of Technological Service Support Act(1992) and Industrial Standardization Act(1992)
2000s	- Framework Act on Science & Technology (2001) - Act on the Establishment, Operation & Fostering of Government-Funded Science & Technology Research Institutes, etc. (2004) - Special Act on the Support of Daedeok Special Research & Development Zone, etc. (2005)	- Technology Transfer Promotion Act (2000) - Transboundary Movement, etc. of Living Modified Organisms Act (2001) - Bioethics and Safety Act (2004) - Act on the Performance Evaluation & Management of National Research & Development Projects, etc. (2005) - Act on the Establishment of Safe Laboratory Environment (2005) - Act on Confidentiality & Protection of Industrial Technology Secrets (2006)

Period of enactment	Legislative Purpose	
	List of enacted Acts	Regulations
		• Environmental Testing & Inspection Act (2006) • Technology Transfer & Commercialization Promotion Act (2006) • Act on the Acquisition, Management, & Utilization of Biological Research Resources(2009)
2010s	• Special Act on Establishment & Support of International Science & Business Belt (2011) • Basic Research Promotion & Technology Development Support Act (2011) (Whole Amendment of Basic Science Research Promotion Act) • Industrial Convergence Promotion Act (2011) • Carbon Materials Convergence Technology Development & Promotion Support Act (2016) • Hydrogen Economy Promotion & Hydrogen Safety Control Act(2020)	• Act on Cord Blood Management & Research (2010) • Framework Act on Intellectual Property (2011) • Act on the Conservation & Use of Biological Diversity (2012) • Act on the Development of Cloud Computing & Protection of Its Users (2015) • National Research & Development Innovation Act(2020)

Source: Authors

4. Laws of Science & Technology in Indonesia

4.1 The Review of the Development of Science & Technology's Regulations & Laws in Indonesia

The three-dimensional concept expressed by Yang (2012) in Figure 5-1 has similarities with the policy system in Indonesia. As mentioned by Nugroho (2014), Indonesia has a continental policy system that divides the level of policy into three dimensions of macro, meso, and micro. Macro dimension, such as laws and state-led development plan, has a very general and not detailed point of view. Policies that follow up on these general regulations are in the realm of meso policies, which often involve regulations from the president, ministers, and government institutions. Meanwhile, operational policies that directly implement science and technology policies are on a micro-level. Accordingly, the comparison between South Korea and Indonesia in terms of S&T regulation and laws would be on the same dimension.

4.1.1 The 1950s

After Indonesia reached independence from Dutch colonialism in 1945, the government was preoccupied with various kinds of internal conflicts and wars. Consequently, the development of science and technology had not been a priority for the Indonesian government. There was almost no existence of S&T policies issued in the period 1945-1950. In the early 1950s, there were two trends in S&T policies applied as the initial foundation of science and technology in Indonesia.

The first of them was building human resources that were left behind. In the colonial era, the chaotic and discriminatory educational system caused indigenous communities to have less opportunity to receive formal education than other communities. The last survey in 1941 showed that only 10.8 percent of Indonesians could read and write (Central Bureau of

Statistics, 1947). Moreover, after independence, Indonesia only had 400 university graduates (Gross, 2014, p. 249). As a result, the first strategy of the government is to build the initial foundation by preparing policies in the education sector. Discriminatory educational values were changed with more inclusive education to educate the nation's life as stated in the 1945 Constitution (Article 31 Paragraph 3).

In 1950 there were two main policies in the regulation of the education sector: Law of Education No. 4-year 1950 concerning the regulation of governance and implementation of education and Emergency Law of Higher Education No. 7-year 1950 concerning the takeover of the legacy of the Dutch colonial universities, namely *Universiteit van Indonesia*¹², into the hands of the Indonesian government. During that period until 1959, there were various policies of the establishment of public higher education; Gajah Mada University (Government Regulation No. 23-year 1949), Airlangga University (Government Regulation No. 57-year 1954), Hasanuddin University (Government Regulation No. 23-year 1956), Andalas University (Government Regulation No. 24-year 1956), Teacher Education College (Government Regulation No. 23-year 1956), Padjadjaran University (Government Regulation No. 37-year 1957), and Institute of Technology Bandung (Government Regulation No. 6-year 1959).

Secondly, the Indonesian government generally developed two science and technology sectors of basic science and nuclear science according to Law No. 6-year 1969 concerning S&T Council establishment (MIPI)¹³ and Government Regulation No. 65-year 1958 concerning Atomic Energy Council & Atomic Institute. According to Gross, 2014 (261) MIPI was formed as a hub for the advancement of national science and technology. MIPI was created as an autonomous organization that provided input to the government on state-level issues, projects, and science & technology developmental activities. Meanwhile, the establishment of the Atomic Energy Council & Atomic Institute was a follow-up to the establishment of the State Committee for the Investigation of Radioactivity in 1954. One of

12 Hoger Onderwijs Ordonnantie 1946"; Staatsblad van Nederlands Indie 1947 No. 47), and "Universiteitsreglement 1946"(Staatsblad van Nederlands Indie 1947 No. 170)

13 MIPI stands for Majelis Ilmu Pengetahuan Indonesia

the main tasks of the Atomic Energy Council & Atomic Institute is to provide an infrastructure in nuclear power and serve as a place of talent education and research center of the nuclear power field.

4.1.2 The 1960s

In the 1960s, Indonesia experienced a major transition, characterized as from the Old Order to the New Order, in which there was a major strategy in science and technology policy between the two regimes. At the beginning of Sukarno's government in 1960, the position of universities was strengthened by the issuance of a new Law on Higher Education No. 22-year 1961 in which scientific fields were determined in four main faculties of religion, cultural studies, social sciences, and engineering science. Each field is assigned a research function. In this law, the government also supports the universities that are still deficient and not dispersed throughout Indonesian regions by providing subsidies to private universities.

Still in the same period, according to Provision of People's Consultative Assembly No. II-year 1960, for the first time, research was included in the national development plan in various fields. Its derivative regulations mandated national sectoral development to be based on research. This period was also marked by the development of S&T in the aviation field. The government, under the First Minister Ir. Juanda, formed the Astronautics Committee on May 31, 1962. Four months later, the Early Scientific and Military Rocket Project (PRIMA) was introduced by the collaboration between the Air Force and ITB. One of the success stories of such collaboration was when they succeeded in launching a rocket. After a series of activities, the government issued Presidential Decree No. 236-year 1963 governing the National Institute of Aeronautics & Space (LAPAN)¹⁴ which assigned this institute the main task of conducting research and development on aviation and space matters.

In the nuclear sector, Law No. 31-year 1964 concerning Basic Rules of Atomic Energy

14 LAPAN stands for Lembaga Penerbangan dan Antariksa Nasional

Institute mandated the use of atomic energy for research, education, health, agriculture, and industry. A year later in 1965, the first reactor was built in Bandung with a 250-kW reactor in operation. Along with this, the government established the National Atomic Energy Agency (BATAN) replacing the Atomic Energy Institute as the main institution to regulate and research nuclear power development.

During the shift to the New Order in 1966, a policy focus on the development of autonomous institutions was established. The Indonesian Institute of Sciences (LIPI)¹⁵ was formed based on Presidential Decree No. 128 -year 1967. LIPI superseded the function of DURENAS¹⁶ and MIPI¹⁷. As a result, LIPI's position is stronger than MIPI's in coordinating, integrating, and synchronizing all activities in the field of science and technology on a regional and national scale (Gross, 2014, p. 272). At that time, the holder of the highest science and technology authority and implementer was LIPI, but the executive function of implementing science and technology still resided the Ministry of Education & Culture.

In the late 1960s, the government constructed its five-year development plan (REPELITA), which encompassed all areas of development including science and technology (S&T) issues. In general, a direction of REPELITA I (1969-1974) was to improve industrial contribution. Therefore, in the late 1960s, the government issued the Law of Foreign Investment 1/67. The regulation opened investment opportunities in high-tech sectors such as telecommunications, aviation, and atomic power. In the case of S&T issues, the focus of science and technology development was on strengthening and establishing government R&D institutions, including improving the competencies and welfare of human resources and improving R&D facilities. The policies that responded to these issues were created in the 1970s.

15 LIPI stands for Lembaga Ilmu Pengetahuan Indonesia

16 DURENAS stands for Departemen Urusan Riset Nasional

17 Between the transformation from MIPI to LIPI, there were an intra-organizational change known as DURENAS and LEMRENAS

4.1.3 The 1970s

In the early half of the 1970s, there was a trend towards a more massive use of research in state-run bodies. In particular, the transfer of research affairs which was originally under the Ministry of Education and Culture to the new ministry that exclusively manages S&T development which is the Ministry of Research and Technology in 1974. In the same year, a research and development agency under departments was created in other ministries to strengthen research positions in supporting policy functions and ministry operations.

In this period, there were tendencies to a transition from the development of basic sciences to the use of industry and technology transfer. REPELITA II (1974-1979) which was still concentrating on research infrastructure development began to shift to developing science and technology in larger industries and prioritizing technology transfer in REPELITA III (1979-1984). In supporting priority areas in the REPELITA III, the Minister of Research Technology, B.J. Habibie, formed the Formulator of the National General Program in S&T (PEPUNAS IPTEK) which had a function as a partner group for the government consisting of experts from various fields of sciences to formulate priority research and technology programs. According to Oktavyanti et al (2013), science and technology activities in REPELITA III had been coordinated more intensively. Various expansions and improvements had been made in a variety of activities that support research development, including research facilities. To improve the quality of research, improvements had been made in the wider field of sciences, such as nuclear research, outer space flight, remote sensing, and regional mapping.

The policy agenda to accelerating industrialization has led to the creation of several research institutions and infrastructure. One of the research institutions is the Agency for the Assessment & Application of Technology (BPPT) through Presidential Decree No. 25-year 1978 which has a function to grow and develop mastery of technology and increase the utilization of technology. In addition, the concept of an integrated research center between LIPI-BATAN-LAPAN to synergize R&D institutions in Indonesia in supporting national development was built through the Research Center for Science and Technology (PUSPITEK) in 1976 through Presidential Decree No. 43-year 1976.

In short, the policies of the 1970s were a follow-up to the educational policies of the 1950s

and the development of R&D institutions in the 1960s. During this period, human resources and science and technology infrastructure were considered adequate to support important sectors, such as the national industrial sector, notably in preparing the initial foundation for applied research. Applied and engineering researches were considered profitable to support the increasing value of an investment through the foreign investment policy in order to prepare the government to catch up in science and technology.

4.1.4 The 1980s

Since the early 1980s, the effort to industrialize through various sets of policies had shown results. This could be seen from the contribution of the industrial sector, above all exports, which had been increasing and gradually sharing its role with the agricultural sector which dominated in the 1950s to 1970s. In this period until the New Order era ended in 1998, the average GDP growth of Indonesia had an average of 10.8 percent.

The highest regulator of the industry was Law No. 5-year 1984 governing industries that was made at the end of REPELITA IV. This law encouraged the development of industrial engineering by improving the quality of industrial products and setting industry standards through research. In 1989 a National Standardization Council (DSN) was created through Presidential Decree No. 7-year 1989 whose main task is to establish standardization policies, implement coordination, and foster cooperation in the field of national standardization.

To accelerate strategic industrialization through research and development during this period, B.J. Habibie as the Minister of Research & Technology determined nine industrial sectors supported by research funding for the state-run bodies, collaboration government, and companies, and top-down themes including 1) aerospace industry; 2) maritime and shipping industry; 3) land transportation equipment industry; 4) electronics and telecommunications industry; 5) energy industry 6) engineering industry; 7) agricultural tools and machinery industry; 8) defense and security industry, and 9) industries that grew due to the development of the eight sectors (Oktavyanti et al., 2013, p. 59). In implementing these policies, the Strategic Industry Development Agency (BPIS) was created based on

Presidential Decree No. 44-year 1989 whose function is to develop and manage 10 strategic state-owned enterprises. At that time, several strategic industries had been pioneered in Indonesia, such as the aircraft industry, PT. IPTN was established in 1985 and became the main actor in the strategic industry in the aerospace industry, PT PINDAD became a state-owned enterprise in 1983 as an actor in the defense and security industry, and PT PAL became a state-owned enterprise in 1980 and an actor in the maritime sector and other sectors.

The increasingly complex role of science and technology and growing need for science and technology policies had made the government strengthen the position of PEPUNAS to become the National Research Council (DRN) through Presidential Decree No. 1-year 1984 whose main task was to determine Indonesia's main national research program in 1987 which became the main reference for R&D and universities in the direction of research activities.

In short, the 1980s became an ideal momentum for Indonesia to develop a strategic industry through a strong research system. The role of the Ministry of Research & Technology was increasingly required in creating effective science and technology policies through strengthening DRN which was expected to get better policy input. Research financing schemes were directed at national needs, notably aimed at the nine priority sectors of the government and direct research production geared for national development.

4.1.5 The 1990s

During this period, there was a policy to strengthen the functions of the government agencies that had been formed. The role of national standardization carried out by DSN during this period was strengthened by the presence of the National Standardization Agency (BSN) through Presidential Decree No. 13-year 1997. In addition, Presidential Decree No. 10-year 1997 concerning the establishment of the Nuclear Energy Supervisory Agency (BAPETEN) was established whose task is to control nuclear power activities in Indonesia.

Leading scientists in Indonesia are also given a place to assess, study, and monitor the

development and utilization of science and technology in Indonesia through the Indonesian Academy of Sciences (AIPI) based on Law No. 8-year 1990. The AIPI is autonomous and not attached to any part of the government agency. The establishment of this body is one of the highest honors in the field of science and technology for Indonesian scientists who are elected as members.

In the field of Industry, in 1990, through a journal entitled 'Sophisticated Technologies: Taking Root in Developing Countries', B.J. Habibie who at that time served as the Minister of Research & Technology and Head of BPPT, explained Indonesia's strategy in technology transfer. Habibie (1990) describes four stages for Indonesia to implement technology transfers, such as 1) technology transfers through licensed products, 2) technology integration, 3) technology development, and 4) large-scale basic research. In supporting the technology transfer process, Habibie explained plans to duplicate PUSPIPTK as research centers in various locations in Indonesia such as Cibinong, Surabaya, Bogor, Medan, and Pasuruan. One of the biggest achievements at that time was the success of the Strategic Industry Program, particularly in aerospace sectors through IPTN by producing a prototype of the N250 aircraft. Several sets of domestic car policies through President Decree No. 2-year 1996 resulted in the first year of cooperation between Indonesia and Korea to produce domestic cars.

However, in the end, these policies and programs had to stop midway due to the monetary crisis experienced by Indonesia as well as ending science and technology planning through REPELITA in 1998. A series of science and technology policies related to industrialization that was developed from the 1970s to 1990s was one chapter of one of the general policies of the New Order under Suharto which gave a special place for science and technology to contribute to the country. During this long period, dozens of sciences & technology and R&D bodies were formed as well as a set of industrial regulations, technology transfer, and research focus for R&D and universities.

4.1.6 The 2000s

After 2000, or the so-called reformation period, was a new chapter for Indonesia after being led for more than three decades under the authoritarian New Order leadership. This period was Indonesia's decade with the highest number of leaderships where at least in ten years, there were four presidential changes. During this period, until early 2020, the science and technology planning document underwent various changes and the planning focus.

Planning on a national scale changed from REPELITA to the National Development Program (PROPENAS) 2000-2004, and medium-term development plan (RPJM) 1 (2005-2009). S&T policy planning is regulated in science & technology planning through two volumes of the National Policy and Strategy in S&T (Jakstranas Iptek) (2000-2004 and 2005-2009). In both Jakstranas, there are two relatively new policies which are the Development of Indonesian National Standards (SNI) to strengthen competitiveness and the program to strengthen the national innovation system (NIS). In addition, DRN also published the National Research Agenda (ARN) 2006-2009 as a guideline for national research programs.

During this period there was also an expansion of R&D agencies within government agencies. Previously, R&D consisted of autonomous R&D state-run bodies and R&D under the ministry, Jakstranas 2000-2004 also mandated the establishment of the Regional Research and Development Agency (Balitbangda) for each region as a guardian of government policies based on a mature science and technology foundation.

In this period, a several policies through laws and regulations were issued. Regulations that manage science and technology policies indirectly such as product protection and technology transfers in five laws, which are the protection of plant varieties (Law No. 29-year 2000), industrial designs (Law No. 31-year 2000), patents (Law No. 14-year 2001), trademarks (Law No. 15-year 2001), and technology transfers (Government Regulation No. 20-year 2005). The five regulations are interconnected in regulating the management of science and technology utilization from R&D to the market.

On the other hand, some policies enforce direct regulation including licensing to conduct research, specifically high-risk R&D (Government Regulation No. 41-year 2006), national

research system (Law No. 18-year 2002), improving engineering capability, innovation, and technology diffusion (Government Regulation No. 25-year 2007), reforming regulation on the National Research Council (Government Regulation No. 16-year 2005).

4.1.7 The 2010s and Afterwards

In this period, there were main references and regulations in science & technology policies. On a national development scale, there were two national mid-term development plans, RPJMN¹⁸ (2010-2014 and 2015-2019). In particular, the regulation of science and technology policies was in JAKSTRANAS in S&T 2010-2014. In 2017, due to ARN and JAKSTRANAS were considered not optimal as a reference for S&T development in Indonesia, ARN, and JAKSTRANAS were replaced by the National Research Master Plan 2017-2045 (RIRN). RIRN prioritizes research in the fields of food, energy, health, transportation, engineering products, defense and security, maritime affairs, social humanities, and other research fields. In the implementation of RIRN, a National Research Priority (PRN) is made for a period of five years 2020-2024 to produce research with integrity between research agencies, the government, and company partners.

In addition, Law No. 11-year 2019 concerning the national system of science and technology addresses issues such as the protection of research information, intellectual property rights, establishment of an endowment fund, implementation of R&D, and HR management. One of the mandates of this law is to establish BRIN which has the task of assisting the president in carrying out research, development, assessment, and implementation, notably inventions and innovations on a national scale that are integrated, as well as monitoring, controlling, and evaluating the implementation of BRIDA's duties and functions. This synergizes the functions of LIPI, LAPAN, and BATAN, as well as Balitbang under the ministry under the umbrella of a single BRIN organization.

Another regulation as an indirect regulation in the development of science and technology

¹⁸ RPJMN stands for Rencana Pembangunan Jangka Menengah Nasional

is Law No. 3-year 2014 governing industries which replaces Law No. 5-year 1984. In addition, Job Omnibus Law, Law No. 11-year 2020 which is a multi-sectors law, regulates various matters in science and technology policies, such as technology transfers, industry, and intellectual property protection.

In fiscal policy, the need for managing state budgets for education is managed by an endowment fund management mechanism by a public service agency. Therefore, the Education Fund Management Institute (LPDP) under the Ministry of Finance was formed as a source of funding for educational and research scholarship services which have reached 70.1 trillion Rupiah.

Other enactment promotion policies that apply during this period are Government Regulation No. 46 of 2012 governing observation and management of meteorological, climatological, and geophysical data, Government Regulation on Safety & Security of Nuclear Installations, policy set for the establishment of science technoparks, and renewal of Law No. 12-year 2012 governing education (12/2012).

To conclude, there are S&T policies in the 1950s to 2020s and afterward based on legislative purposes. Indonesia's S&T policies are still seeking a place to impose the S&T development to a higher level. There are some macro-policies during the whole period which are industry, foreign investment, education, transfer technology, and standardization that affect the development of S&T in Indonesia. The following table shows the history of S&T Laws in Indonesia.

[Table 5-3] History of S&T Laws in Indonesia

Year of enactment	Legislative Purpose	
	List of enactment promotion	Regulations
1945s-1950s	• Policy Series of Higher Education Policy in 1949-1959 • Law of S&T Council establishment (MIPI) No. 6-year 1969 • Government Regulation of Atomic Energy Council & Atomic Institute No. 65-year 1958	• Law of Education No. 4-year 1950 • Constitution of 1945 • Emergency Law of Higher Education No. 7-year 1950
1960s	• Establishment of the Department of National Research Affairs in 1962 • Presidential Decree of National Institute of Aeronautics & Space (LAPAN) No. 236-year 1963 • Astronautics Committee • The establishment of the National Nuclear Energy Agency of Indonesia (BATAN) • Presidential Decree of Indonesian Institute of Sciences (LIPI) 128/167	• Provision of People's Consultative Assembly II in 1960 • Law of Higher Education No. 22-year 1961 • Law of Basic Rules of Atomic Energy No. 31-year 1964 • Five-year development plan (REPELITA) I • Law of Foreign Investment 1/67
1970s	• The Establishment of Ministry Research and Technology (1974) • Research and Development Agency under Departments (1974) • The Presidential Decree of Research Center of S&T No. 43-year 1976 • The Presidential Decree of the Agency for the Assessment & Application of Technology (BPPT) No.25-year 1978 • Formulator of the National General Program in S&T (PEPUNAS IPTEK) (1978)	• Five-year development plan (REPELITA) II • Five-year development plan (REPELITA) III •
1980s	• Establishment of the National Standardization Council (DSN) in 1984 • Presidential Decree of Strategic Industry Development Agency (BPIS) No. 44-year 1989	• Five-year development plan (REPELITA) IV • Five-year development plan (REPELITA) V • Law of Industry No. 5-year 1984

Year of enactment	Legislative Purpose	
	List of enactment promotion	Regulations
	Establishment of Indonesia Aircraft Industry (IPTN) in 1985	Law of Patent No. 6-year 1989
1990s	- Law of Indonesian Science Academy (AIPI) No. 8-year 1990 - Transfer Technology Policy Series of B.J. Habibie - Dewan Riset Nasional - Establishment of National Research Council (DRN)	- Five-year development plan (REPELITA) VI - Presidential Decree Nuclear Energy Regulatory Agency No. 10-year 1997 - Presidential Decree of National Standardization Agency of Indonesia (BSN) No. 13-year 1997 - Presiden Decree of the Development of Domestic Car Industry No. 2-year 1996
2000s	- Law of National System of Research, Development, & Application of S&T No. 18-year 2002 - Government Regulation of Permission to Conduct Research No. 41-year 2006 - Government Regulation of Enhancing Engineering Capability, Innovation, & Technology Diffusion No. 35-year 2007 - National Innovation System - Government Regulation National Research Council (DRN) No. 16- year 2005 - Regional R&D (Balitbangda)	- National Development Program (PROPENAS) - Medium-term development plan (RPJM) 1 - White Paper of S&T - National Policy and Strategy in S&T (Jakstranas Iptek) 2000-2004 - National Policy and Strategy in S&T (Jakstranas Iptek) 2005-2009 - Law of Plant Variety Protection 29/2000 - Law of Trade Secret No. 30-year 2000 - Law of Industrial Design No. 31- year 2000 - Law of Patent No. 14- year 2021 - Law of Trademark No. 15-year 2001 - Government Regulation of Transfer Technology No. 20- year 2005 - National Research Agenda (ARN)
2010s & afterwards	Government Regulation of Implementation of Observation & Management of Meteorological,	- Medium-term development plan (RPJM) 2 - Medium-term development plan (RPJM) 3

Year of enactment	Legislative Purpose	
	List of enactment promotion	Regulations
	Climatological, and Geophysical Data No. 46-year 2012 • Policy Series in Science Technopark • Government Regulation of Nuclear Installation Safety & Security No. 54-year 2012 • Education Fund Management Institution (LPDP) • National Research Program (PRN) • National Research & Innovation Agency (BRIN)	• National Policy and Strategy in S&T (Jakstranas Iptek) 2010-2014 • Law of Industry No. 3-year 2014 • Law of Standardization and Conformity Assessment No. 20-year 2014 • Law of National S&T System No. 11-year 2019 • National Research Masterplan (RIRN) 2017-2045. • Omnibus Law No. 11-year 2020

Source: Authors

4.2 Main Issues Regarding Science & Technology in Indonesia According to Law No. 11-Year 2019 Governing the National System of Science & Technology

Law No. 18-year 2002 concerning the system of research development and application of science and technology in Indonesia has been replaced by Law No. 11-year 2019 governing the national system of science and technology. The enactment of the latter brings new hope for science, technology, and innovation (STI) development that supports national economic development. The latter considerably can bring some fundamental improvement towards the science, technology, and innovation (STI) development landscape in Indonesia (Survey Research STEPI-LIPI, 2020 – K Innovation ODA Program with Indonesia Chapter 5&6).

According to Law No. 11-year 2019, several issues have become the main concerns of the Indonesian government which can be mapped into six main structures as follows: Masterplan, implementation, policies & institutions, resources, networks, and guidelines & supervision.

4.2.1 Master Plan

Regarding technology planning and supervision on Law No. 11-year 2019, Indonesia has the

Master Plan for Advancing Science and Technology (RIPK)¹⁹. RIPK is becoming a guideline not only for conducting STI (Article 3 point 3) but also for the national long-term development plan (RPJPN)²⁰ and national mid-term development plan (RPJMN)²¹ (Article 3 point 2). According to Article 8 point 4, there are three types of RIPK classified by corresponding period based on long-term RIPI (25 years), mid-term RIPI (5 years), and annual RIPK. The Master Plan for Advancing Science and Technology is prepared by the central government in coordination with relevant stakeholders (Article 9 points 1 & 2). The availability of long-term STI strategic perspectives is important to maintain STI policy continuity that determines the effective allocation of STI resources (Aminullah, 2020). However, the government regulation to arrange the procedures or requirements to establish RIPK has not been enacted.

As previously mandated in Law No. 18-year 2002, Indonesia has its National Research Agenda (ARN)²² as a guideline for the research, development, and application of STI in Indonesia. ARN is used by related parties to manage their research project in Indonesia. The scope of the research area in every period of ARN varies according to the focus area mandated in the national mid-term development plan (RPJMN). According to Law No. 18-year 2002, National Research Council (DRN)²³ is the institution that should formulate the direction and priorities of STI development in Indonesia. During the ARN's formulation process, DRN forms a technical committee from several stakeholders such as government agencies, academic experts, R&D institutions (public & private), and industries which are monitored by the national parliament members (DPR). In Law No. 11-year 2019, the functions of DRN related to the research agenda formulation in Indonesia become less important, as shown from the abolition of DRN on Law No. 11-year 2019 (STEPI, 2020).

19 RIPK stands for Rencana Induk Pemajuan IPTEK

20 RPJPN stands for Rencana Pembangunan Jangka Panjang Nasional

21 RPJMN stands for Rencana Pembangunan Jangka Menengah Nasional

22 ARN stands for Agenda Riset Nasional

23 DRN stands for Dewan Riset Nasional

4.2.2 Implementation

According to Law No. 11-year 2019 Article 14 (1), the S&T activities consist of education, research, development, assessment, and application. These activities are coordinated by the central government and carried out by individuals, groups, business entities, government, private institutions, and universities (Article 13 point 2). In such S&T activities, training programs aim to increase the pool of technical human resources toward competitive and independent workforces (Article 16), and the research aims to strengthen basic and applied sciences, including social sciences, and create solutions to development problems (Article 19). To implement such activities, the central government needs to prepare human resources and improve the quality and suitability of science and technology and community service (Article 15).

In doing an assessment, the government emphasizes the importance of several related assessment activities of engineering, technology clearing, and technology audits, aiming to ensure the benefits of science and technology in solving development problems (Article 23 point 2). The purpose of engineering is to provide value, production processes, and better welfare of the community through testing, technology development, design, and operation by considering the technical, functional, business, socio-cultural, and aesthetic contexts (Article 24 point 1).

Another implementation form of Law No. 11-year 2019 is the application of S&T through implementing technology transfers, technology intermediation, diffusion of science and technology, and technology commercialization (Article 28). Technology transfers are prioritized for those who live in the territory of the Republic of Indonesia and for those who can utilize and control science and technology for the benefit of the community, nation, and government (Article 29 point 2). Technology transfers can be carried out through licensing, cooperation, and S&T services (Article 29 point 3).

Technological intermediation can be carried out by bridging inventions from technology-producing institutions to potential users and identifying the needs of potential users for the required technology (Article 31 point 1). Technology intermediation activities can be in the form of technology incubation, technology business meetings, partnerships, and promotion

of inventions (Article 31 point 2). Technological intermediation aims to ramp up the adoption of science and technology through increasing the capacity of science and technology, evaluating the readiness of technology users, and fostering increased absorption capacity of technology users (Article 32).

4.2.3 Policies and Institution

The role of S&T policy is particularly important in developing the National Innovation System (NIS) (Ragamustari et al., 2021). This section will discuss S&T policy and its institution according to Law No. 11-year 2021 regarding the national system of sciences and technology.

Law No. 11-year 2019 has a wider scope than those arranged in Law No. 18-year 2002. According to Law No. 11-year 2019 Article 42, research institute elements consist of research & development institutes, research & implementing agencies, universities, business entities, and supporting institutions, while there are no research and implementing agencies arranged in Law No. 8 of 2002. These additional elements in Law No. 11-year 2019 suggest that this law embraces more complex issues in the arrangement of science, technology, and innovation development than Law No. 18-year 2002. Moreover, Law No. 11-year 2019 emphasizes the measurement of research and technology indicators as an important factor in supporting data-driven policymaking²⁴. Such indicators include S&T data; research & development budget, and S&T workforces & output.

Law No. 11-year 2019 mandates the government to integrate all government research & development institutions into national innovation & research institutes (Article 48). The issue of such integration is also mentioned briefly in the Omnibus Law No. 11-year 2020 concerning job creation (UU Cipta Kerja). According to these laws, such a research institute has the main function to construct a national innovation system among all public R&D

²⁴Source: <https://theconversation.com/pentingnya-data-ipitek-di-indonesia-bagaimana-meningkatkan-kualitas-dan-kebaruannya-104000> retrieved 27 Juli 2021

agencies in Indonesia. The institute is called Badan Riset Inovasi Nasional (BRIN)²⁵ which is formally established through the enactment of Presidential Decree No. 33-year 2021 which was revised and changed to Presidential Decree No. 78-year 2021 concerning Badan Riset Inovasi Nasional (BRIN). Previously, Indonesia has not had any particular S&T and innovation policy nor a strong high-level coordinating body to plan and coordinate STI policies at a national level (Park & Kim, 2020). Many ministries, departments, and government agencies have planned and executed their R&D activities individually (Park & Kim, 2020).

The National Research & Innovation Agency (BRIN) was inaugurated on April 28, 2021 as a separate agency from the Ministry of Research & Technology²⁶ and under the direct control of the president. This policy has merged the Indonesian Institute of Sciences (LIPI), Central Agency for Assessment and Application (BPPT), Space and Aeronautics Agency (LAPAN), National Nuclear Energy Agency (BATAN), and other government research institutions or organizations into the National Research & Innovation Agency (BRIN). This unpopular policy has presented pros and cons between Indonesia's science technology and innovation (STI) community about how BRIN will boost innovation (Nature, 2021). BRIN plays a vital role in implementing a coordination mechanism and integrating sectors and institutions in the areas of policy implementation, program budget planning, and evaluation (Aminullah, 2020). If not for the establishment of BRIN, there would be setbacks for Indonesian sciences (Nature, 2021).

²⁵ In English, BRIN stands for National Research and Innovation Agency

²⁶ Currently, its function is merged into Ministry of Education and Culture and become Ministry of Education Culture Research and Technology

■ Repository and Dissemination of R&D Output (Articles 73 and 74)

According to Law No. 11-year 2019, science and technology institutions have the right to manage the results of their research, development, assessment, and application, and to establish an intellectual property management unit (Article 74 point 3). In terms of obligation, S&T institutions are required to disseminate information related to inventions and innovations (Article 73), including the dissemination of the results of intellectual properties (Article 74 point 2).

■ Ethics & Obligations related to Science and Technology (Articles 39 and 40)

Law No. 11-year 2019 arranges ethics (Article 39) and the obligation to submit and store primary data as the output of research, development, assessment, & application activities (Article 40). In detail, there is an ethics commission in charge of reviewing, determining ethical feasibility, and evaluating and supervising the implementation of the ethics code for research, development, assessment, and application following the field of sciences (Article 39 point 1 to 4). Such a commission has the authority to examine and impose sanctions (Article 39 point 5).

Law No. 11-year 2019 also arranges the obligation for funders and S&T human resources & institutions to submit and store primary data and output of research, development, assessment, & application activities (Article 40 points 1 & 2). Primary data is authentic raw data in various forms obtained from research, development, assessment, and application activities (Article 40 point 3). The output is an intellectual property resulting from research, development, study, and application activities (Article 40 point 4) which must be kept for a minimum of 20 years (Article 40 point 5). Management of mandatory data transfer and storage is carried out through a nationally integrated science and technology system (Article 40 point 8). The results of research, development, assessment and application activities must be used as a scientific basis in the formulation and determination of development policies (Article 41). Moreover, the results of the research must be published and disseminated by S&T human resources or institutions (Article 21).

4.2.4 Resources

The lack of research and development (R&D) funding becomes one of the main concerns of the national system of science and technology in Indonesia. The amount of gross expenditure on research and development (GERD) in Indonesia in 2018 was only 0.28 percent of total gross domestic product (GDP) or around IDR 41.43 trillion (World Bank, 2021). Such amount was still far below the world average which is 2.27 percent (World Bank)²⁷. The low expenditures on S&T R&D become a bottleneck in Indonesia's effort to develop S&T capabilities (Ragamustari et al., 2021). The lack of R&D funding considerably affects the output of public R&D performance. One of the indicators of low R&D performance is a lower number of patent applications registered by residents than by nonresidents. The patent application number registered by nonresidents was 4 times higher than that registered by a resident in 2007. Such an indicator shows that Indonesia's innovation output remains in sub-par performance (Park & Kim, 2020).

The primary source of the R&D budget in Indonesia is still from government spending, both central and local governments. The government contributes around 83.74 percent of the total R&D spending or around IDR 34.69 trillion. The government certainly need support from private and public sectors to strengthen the national STI system (Prasodjo 2020 on (STEPI, 2020)/ Survey Research STEPI-LIPI, 2020 – K Innovation ODA Program with Indonesia Chapter 5). The lack of participation from non-governmental sectors in R&D activities was probably due to the lack of regulations to support and give incentives to both industries, state-owned enterprises (SOE), or private R&D in order to collaborate and innovate through R&D activities.

²⁷ Worldbank.org, accessed on July, 2021

[Figure 5-2] Source of R&D Funding in Indonesia (in IDR trillion)



Source: MIRT/ BRIN (2019)

■ Human Resources (Articles 50, 51, 53, 57, & 58)

Law No. 11-year 2019 Article 49 states that science and technology resources should consist of (1) human resources, (2) funding, and (3) facilities and infrastructure. S&T human resources include researchers, engineers, lecturers, among others whose status include civil service officers, Indonesian National Armed Forces or Indonesian National Police, private workers or individuals managed by institutions, and employees who work at institutions stipulated by the law. To ensure the accountability of the S&T HR profession, a scientific professional organization was formed (Article 50 point 4). The retirement age limit for first and junior expert researchers/ engineers is 58 years old, intermediate experts 65 years old, and primary experts 70 years old (Article 53 point 1). For researchers and engineers who have entered the retirement age limit, they can still be employed with conditions requiring willingness, scientific competency, and physical & mental health, which are based on competency standards (Article 53 point 2). S&T workforces receive protection in the form of social security and legal assistance (Article 57). And even if the results of the research that has been conducted based on scientific methods are not aligned with hypotheses or expectations, they would not be subject to any sanctions (Article 58).

■ R&D Funding (Article 62 point 1)

Law No. 11-year 2019 explicitly states that the sources of R&D funding are not only from the government but also from private sectors and endowment funds (Article 59)²⁸. The funding is adjusted to the priority of the Master Plan for Advancing Science and Technology (Article 60). Industries or enterprises will have tax incentives (tax deduction) if they conduct R&D activities. The policy implementation using some policy instrument (e.g. fiscal incentives and financial support) is important to stimulate R&D in the private sector (Aminullah, 2020). Such a fiscal incentive has been arranged under the Ministry of Finance Regulation No. PMK 153/PMK.010/2020.

Law No. 11-year 2019 has introduced a research endowment fund scheme as one of the alternative sources of R&D funding. This endowment can come from the education budget or non-education budget allocations (Article 62 point 3). The research endowment fund needs to be regulated through a presidential regulation which has not been stipulated until now. The existing endowment fund mostly focuses on education-related programs as mandated in Presidential Decree No. 12-year 2019 ((STEPI, 2020) / Survey Research STEPI-LIPI, 2020 – K Innovation ODA Program with Indonesia Chapter 5).

Besides the endowment fund (Article 62), the creation of BRIN (Article 48) could influence the architecture of research and development funding in Indonesia. The integration of four government R&D agencies into BRIN leads to the integration of their R&D budget allocation under the control of one single body (Nature, 2021). Therefore, R&D governance programs should be redesigned and assessed continuously (Aminullah, 2020). BRIN's accountability is also important to ensure that the autonomy of staff and grant recipients is protected (Nature, 2021).

■ Infrastructure (Articles 65 and 67)

Provision and improvement of S&T facilities and infrastructure are carried out by improving the number of research buildings and maintaining and operating; research laboratories, research areas, science & technology training centers, innovation centers, incubation

28 The endowment fund here refers to "dana abadi"

centers, and other S&T facilities & infrastructure centers (Article 65). Research institutes for the development of S&T assessment and applications that produce inventions and innovations are required to collect data and record the facilities and infrastructure of S&T resources (Article 66). Additionally, S&T workforces have the right to use and utilize S&T facilities and infrastructure (Article 67 point 3).

4.2.5 Network

According to Law No. 11-year 2019 and Law No. 18-year 2002, every element of S&T institutions is required to have a partner with other parties such as foreign researchers or foreign institutions to develop S&T networks. Such networks are evaluated to benefit all involved parties in the form of having access to information, R&D infrastructure, and S&T human resources (Article 72). Technology transfers are also expected to occur when such networks are already established (Article 73). The exchange of science and technology between elements of science and technology institutions is facilitated by the central government (Article 74 point 1). However, there is no information mandated in the law concerning which official agency would facilitate technology transfers.

■ Partnership (Articles 71, 72, & 73)

Law No. 11-year 2019 describes the science and technology network as an interactive fabric of an S&T workforce which combines elements of S&T institutions to produce great performance and benefits than those produced by each element of S&T institutions (Article 71). Elements of S&T institutions are required to conduct partnerships both locally and abroad (Article 72 point 3), and thereby develop science and technology by providing easy access to information, science and technology facilities & infrastructure, and mobility of S&T, according to Indonesia's free and active foreign policy (Article 72 point 4). Technology transfers must occur if research, study, development, and application that result in inventions and innovations are fully or partially funded by the central government or regional governments (Article 73 point 1).

■ Human Resource Mobility (Research Permit) (Articles 75 and 76)

The activities of research, development, and application conducted by foreign institutions or foreign individuals must obtain a permit from the central government (Article 75 points 1 & 2). Such a permit is given by looking at the ethical feasibility assessed by an ethics committee (Article 75 point 3). If research funding is obtained from foreign financing, it is obligatory to produce outputs that benefit the Indonesian people and to involve Indonesian scientists with equal rights such as ascribing Indonesian scientist names in each output produced from joint research. The involved parties are also required to do a technology transfer, primary data submission, proportional profit-sharing, and make written agreements regarding materials transfer, both physical and digital materials (Article 76).

Regarding the partnerships with foreign institutions, Law No. 11-year 2019 states the requirements related to ethical feasibility for foreign science & technology institutions before obtaining permission to conduct research and development (R&D) in Indonesia. Such an ethical feasibility will be assessed by a national ethics commission, an ad hoc ethics commission formed by the government. If research funding is obtained from foreign financing, it is obligatory to produce outputs that benefit the Indonesian people and to involve Indonesian scientists with equal rights such as ascribing Indonesian scientist names in each output produced from joint activities. The involved parties are also required to do a technology transfer, primary data submission, proportional profit-sharing, and make written agreements regarding the material transfer, both physical and digital materials (Article 76). There is a prohibition on transfer materials if the materials are a part of national biodiversity, local Indonesian specimens, social wealth, culture, and local wisdom, in both physical and digital form, and as long as testing for such materials can be carried out in Indonesia (Article 77 point 1).

The existing ethics committees are still in a sectoral domain, such as the Health Research Ethics Commission and National Health Research & Development Ethics Commission which are responsible for issuing guidelines and ethical standards for national health research and development (Rachmawaty, 2017). Before the arrival of Law No. 11-year 2019, the requirements of research permits were arranged in Government Regulation No. 41-year 2006 governing licensing for research and developmental activities conducted by foreign universities, foreign research and development institutions, foreign business entities, and

foreigners.

■ National S&T Information System (Article 78)

Law No. 11-year 2019 also arranges the S&T information system. According to this law, all science and technology institutes are obligated to spread information related to the results of their invention and innovation and establish a unit that manages all issues related to IPR. To support the use of such information, Law No. 11-year 2019 mandates the central government to create a national information system that is nationally integrated (Article 78 (1)). A national S&T information system is needed to support the implementation of the science and technology network (Article 78 point 1). This science and technology information system stores and displays basic data on nationally integrated research operations (Article 78 point 2). This S&T information system will be developed and managed by the central government or an agency appointed by the central government (Article 78 point 3).

4.2.6 Guidance and Supervision

Law No. 11-year 2019 arranges several STI policy instruments to support the national S&T system by providing guidelines and supervision in doing research, development and application of STI in Indonesia. The central and local governments maintain the implementation of science and technology policies by creating incentives, providing stimulation and facilitation, and creating a conducive ecosystem for the development of the S&T national system (Article 79). Additionally, local governments guide to create a synergy of institutional elements, resources, and S&T networks as an integral part of the national science and technology system (Article 79 point 2). To create a conducive climate for S&T development and develop S&T institutions, the central and local governments need to develop policy instruments including resource support, institutional strengthening support, incentives, and science & technology programs (Articles 80 & 82). Meanwhile, to develop resources, the government applies certification programs of an S&T workforce, providing research incentives, and improving science & technology facilities and infrastructure (Article

83 point 2). To develop S&T networks, the government facilitates S&T institutional partnerships in cooperation with regional governments, business entities, communities, foreign S&T institutions, foreign institutions, and international institutions (Article 83 point 3). The government also provides guidelines and determination of S&T-related policies through periodic measurement of national science and technology indicators (Article 84).

According to Law No. 11-year 2019, the supervision of S&T is carried out to monitor the planning and implementation of S&T development following the Master Plan for Advancing S&T (Article 86 point 1). Supervision by the central government is also carried out on the mandatory storage of primary data and R&D outputs, transfer of materials, research activities, and inventions & innovations (Article 86 point 3).

■ Sanction (Articles 93 until 96)

Law No. 11-year 2019 also regulates sanctions for those who violate research permits and other obligations arranged in this law including the obligations to disseminate research results and to submit and store primary data, and the results of research, development, assessment, & application activities (litbangjirap)²⁹. According to this law, sanctions are not only imposed on those who violate research permit requirements but also on those who transfer Indonesian local specimen abroad without permission and those who inflict damage to other objects or serious injury or death to people. The following table shows the sanctions imposed on those who violate the research permit requirements and other obligations according to Law No. 18-year 2002 which has been changed to Law No. 11-year 2019.

[Table 5-4] Research Permit Related Sanctions According to Law No. 18-year 2002 & Law No. 11-year 2019.

Sanctions	Law No. 18-year 2002	Law No. 11-year 2019
-----------	----------------------	----------------------

²⁹ Litbangjirap stands for penelitian, pengembangan, pengkajian, dan penerapan

For foreigners who carry out R&D in Indonesia without permission	N/A	Administrative sanctions: Placed on the blacklist of foreigners (Article 92) or a criminal fine of Rp 4 bil. if they commit a violation again and a ban on obtaining permits for 5 years (Article 93)
Violation of the transfer of Indonesian local specimens abroad without permission	N/A	Receives a 2-year prison sentence or a fine of IDR 2 bil. (Article 94) and a permit ban
For foreigners who carry out dangerous and high-risk research activities without having permission	Six-month imprisonment or a criminal fine of IDR 50 mil. (Article 30)	1-year imprisonment or a fine of IDR 2 billion (Article 95)
Violation by causing damage to goods or objects	N/A	3-year imprisonment or a fine of Rp. 3 bil. (Article 95)
Violation by causing serious injury to people	N/A	4-year imprisonment or a fine of Rp. 4 bil. (Article 95)
Violation resulting in the death of a person	N/A	7-year imprisonment or a fine of Rp. 7 bil. (Article 95)
Criminal acts committed by business entities	N/A	Businesses are fined such as Article 93, Article 94, Article 95 plus 1/3 (Article 96), and the business entity's license is revoked (Article 96).

Source: Authors

5. Major Regulation Issues Related to Laws of Science & Technology in Korea

When examining the historical development of the Korean legislations of science and technology, we can find that the Korean legislature, considering continuous changes of various policy environments, has provided specific regulations and laws that fit into certain circumstances and satisfy public demands. In particular, the number of legislations of science and technology has increased notably since 1990s because of the discussions on the development of administrative structure and framework related to science and technology.

This section provides five major regulatory issues on the Korean science and technology system, and thereby analyzes the distinctive features in the laws covering certain problems. This study is categorized into five major topics of (i) oversight of R&D funding, (ii) reshaping current R&D environment, (iii) control of safety & ethics, (iv) technology transfers & commercialization, and (v) protection of intellectual property rights.

Among others, we will focus on the National Research & Development Innovation Act that went into effect in 2021 because this Act is deeply related to the main three topics of regulatory issues. Before this Act was enacted, each ministry had managed its state-run R&D innovation businesses under various and fragmented laws and enforcement decrees that permitted its jurisdiction. The National Research & Development Innovation Act that was enacted in 2020 provides the legal measures of generally managing all R&D innovation businesses funded by the Korean government. This Act contains the measures for the management of state-run R&D innovation businesses.

[Figure 5-3] below explains the overall contents of the provisions in the National R&D Innovation Act which became effective from January 2021. Chapter 2 of the Act related to the procedure of national R&D projects provides the rules on R&D budget management. Chapter 3 of the Act, which involves the enhancement of R&D innovative environment,

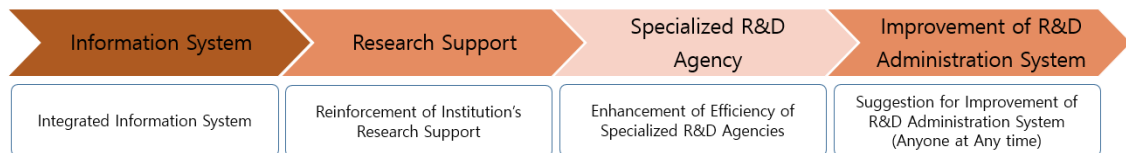
deals with the issue of improving the R&D support environment. Chapter 4 of the Act, which covers the research code of ethics and sanctions related to national R&D, deals with the management of safety and ethics in R&D.

[Figure 5-3] Structure of the National R&D Innovation Act (2020)

Research Procedure(Chapter 2)



Research Environment(Chapter 3)



Research Ethics(Chapter 4)



Source: Ministry of Science & ICT (2020). pp. 3-5.

5.1 Oversight of R&D Fund

There are two major legislations related to Korea's national R&D system, consisting of the Framework Act on Science & Technology and the National Research & Development Innovation Act. The Framework Act on Science & Technology of 2001 aims to establish the foundation for science & technology development, thereby improving innovation development and national competitiveness. This Act covers national R&D budget management and deals with the implementation of R&D, utilization of R&D outcomes (Chapter 3), and expansion of R&D investment & human resources (Chapter 4). Specifically, it contains various matters, such as national R&D budget management, the assessment of national R&D projects & financial support, expansion of science & technology investment,

and establishment of a science & technology fund. The most important part of national R&D budget management is the implementation direction and objective of the national R&D system, priority of investment sectors, and R&D budget management, etc.

The Act articulates the tasks of the government, R&D institutes, and researchers related to the issues of R&D projects in utilizing allocated resources. Most of all, for effective management, it clarifies the details of the R&D project process. Compared with other legal provisions, this Act indicates its aim of management in a micro-perspective. The details of the Act include (i) the public announcement of an R&D project fund, etc., (ii) selection of an R&D subject and R&D institutes by considering the project's creativity and content, research institute capacity, research outcome implementation, consistency of national R&D projects, (iii) agreement of an R&D subject, including objectives, budget, period of R&D implementation, (iv) conducting an assessment of each level, and submitting R&D implementation and the report on research outcomes, etc., as well as (v) R&D-related expenses, etc.

5.2 Reshaping Current R&D Environment

It is very important to establish the implementation system for R&D budgets in the area of utilizing national R&D budgets while it is essential to provide measures to control the R&D process. It is important to monitor the national R&D project process, provide the competencies, establish the R&D supports, and auditing the R&D institute. The Act aims to improve the effectiveness of R&D by providing various legal measures for a national R&D environment.

The major part of the measure is the establishment of the Integrated R&D Information System (IRIS). Firstly, through IRIS, it is possible to manage the R&D topic, research budgets, researcher information, and offer a web portal service which is a one-stop-shop method.

Secondly, it stipulates that the R&D institute needs to establish supporting human resources and structure when the institute conducts national R&D businesses. This requirement is

clarified in order to make researchers focus on their tasks by means of dividing the duties of administration and research, which is a more systemic structure. The R&D institute should satisfy the requirements of R&D support that are set by the Minister of Science & Technology, and they need to be evaluated by the authority.

Thirdly, it aims to make the research administration service structure more efficient, and avoid inefficient outcomes arising from too many overlapping R&D management agencies by appointing a representative of each ministry. The dedicated or competent institutes, which conduct planning, management and evaluation of national R&D, are appointed by various legal measures such as an enforcement decree and the Act on the Management of State-Run Institutions, etc. The R&D management agency plays a role of formulating national R&D businesses, etc.

Fourthly, it articulates the task of central government related to formulating a national R&D system. The central government may operate various national R&D policies and relevant laws. In addition, it can deal with various issues specific to the national R&D system, improvement of R&D agencies and institutes' internal measures, and administration of R&D subjects.

5.3 Control of Safety and Ethics

Because of the increasing concerns about environment and safe usage and ethics of science technology, numerous legislations have been instated, and it is possible to categorize three types of the laws as follows.

Firstly, the rise of concerns of environmental problems during the time of economic development has made the adoptions of various regulations on environment protection. In the 1990s, the Natural Environment Conservation Act, Water Environment Conservation Act, etc. were enacted. These regulations have the objectives of conserving water, ecosystem, and environment, thereby improving sustainable usage of the natural environment and ensuring people's healthy and daily life safety. In 2006, the government enacted the

Environmental Testing & Inspection Act which articulates the standard of assessment of polluted materials, contamination status, and hazard level, etc.

Secondly, the government tried to adopt the regulations preventing possible risks and harms in the R&D process and its usage. One of the examples is the nuclear and biotechnology sectors. Korea has focused on the use and improvement of nuclear power. However, the issues of preventing possible disasters such as safety rules on nuclear power have been highlighted. Accordingly, the government enacted the Nuclear Safety Act in 2011, thereby enhancing the safety of using nuclear power.

Due to the fast development of biotechnology since the 2000s, Korea instated the regulation of the sector biotechnology. Above all, the goal of the law is to ensure the public safety. This Act applies to the use of genetically modified products, the management of cord blood for curing inveterate diseases. Moreover, the topic of research code of ethics that is related to human life became one of the critical issues in the area of biotechnology. The 2004 Bioethics and Safety Act contains several provisions to prevent any infringement of human rights when handling embryos & genes and to prohibit human cloning.

Thirdly, the government also provides regulations for researchers' ethics in the implementation of national R&D. The National Research & Development Innovation Act contains rules on research code of ethics in the national R&D system. It prohibits any fake outcomes from R&D, misuse of R&D funds, and security leaks. The central government may limit any participation of violators of the Act, which is fraudulence-related, in applying for national R&D projects within a ten-year period. In addition, the government may redeem the R&D funds and impose further sanctions, and the law provides an appeals process for allowing alleged violating researchers to request a review of sanctions and thus protect their rights.

5.4 Technology Transfer and Commercialization

There are legislations of technology transfers which are about performance and mainly

transfers. Korea instated the Act on the Performance Evaluation & Management of National Research & Development Projects, etc. in 2005. This Act aims to evaluate the R&D activities in the state-led science and technology sector and effectively manage and utilize R&D outcomes, thereby improving the efficiency and responsibility of R&D investment. It provides a set for clear goals and indicators for the performance management of R&D results. It also aims to provide a management plan of R&D outcomes and factual surveys on utilization of R&D outcomes. For commercialization, the Act articulates the rules for supports on the actual cost of evaluating technology values of R&D outcomes and the patent costs, etc.

The Technology Transfer & Commercialization Promotion Act of 2006 covers the overall ecosystem of commercialization of technologies. This Act was enacted (i) to enhance commercialization by transferring the technologies that are developed by the public sector to the private sector, (ii) to establish and formulate the relevant policies for effective trade of technologies developed by the private sector, and (iii) to improve the competitiveness of technology by establishing a foundation for the transfer, evaluation, and distribution of technology. The technology transfer regulation includes five major substantive provisions, namely (i) the rule on formulation of government's technology transfer and commercialization, (ii) expansion of technology transfer and commercialization, (iii) detailed rules on implementing commercialization, (iv) financial supports, and (v) technology evaluation and technology trust management.

5.5 Protection of Intellectual Property Rights

Korea built up the system of recognizing the importance of intellectual property rights even in the 1960s when Korea established its legal system of science and technology. In particular, the Trademark Act, Utility Model Act, Design Act, and Patent Act have been enacted. In 2006, the Act on Confidentiality & Protection of Industrial Technology Secrets was enacted to protect industrial technologies.

In 2011, the Framework Act on Intellectual Property was also enacted, which has aimed to

provide the environment for creating, protecting and utilizing national intellectual property and new knowledge property. This Act provides various measures for protecting creators of intellectual properties, such as authors, inventors, scientists, and artists through (i) the establishment of intellectual property policy and its formulation, (ii) creation, protection, and improvement of utilizing intellectual property, and (iii) a foundation for creating, protecting, and utilizing intellectual property. Particularly, the Act contains the measure stating that it is necessary to support the sector, and thereby incorporate the R&D outcomes with the creation of intellectual property.

6. Summary and Implications

S&T laws are the institutional basis for technological and industrial development. Korea has achieved rapid industrialization in a few decades, and S&T has played an essential role in economic development. For the successful operation of the S&T system, it is necessary to build effective governance in terms of administrative structure, budget allocation, R&D programs, public research institutions, performance evaluation, etc. The S&T laws have contributed significantly to the formation of effective S&T governance through regulating various governance elements. Indonesia has been undergoing a significant period of governance reform through Law No. 11-year 2019 on the national system of science and technology.

Accordingly, this chapter will review the following areas of 1) the development of Korean S&T laws, 2) Indonesian Law No. 11-year 2019, 3) future direction of Indonesian STI laws & regulations, 4) S&T for industrial development, and 5) regulatory issues contained in Korea's R&D and Innovation Act of 2020.

6.1 Historical Development of S&T Laws in Korea: from the Perspective of S&T Governance

Indonesian Law No. 11-year 2019 on the national science and technology system has set the framework for S&T governance by introducing new institutional elements. In this context, we focused on the laws closely related to the formation of S&T governance, among the Korean S&T laws presented in section 3. The following laws were selected based on applying the governance elements in Law No. 11-year 2019.

[Table 5-5] Governance Elements in Korean S&T Laws

Period	Major S&T laws	Features / contents
1960s	- Science and Technology Promotion Act (1967) - KAIST Act (1970)	- Early S&T Framework Act - The source of S&T human resources
1970s	Technology Development Promotion Act (1972)	Accumulation of technology development reserve in the private sector (Financial & tax benefits)
1980s	Technology Development Promotion Act (Amended, 1981)	The first National R&D Program (1982)
1990s	- Special Act for S&T Innovation (1997) - Act on the Establishment, Operation, & Fostering of State-Funded Research Institutes (1999)	- National S&T Council, 5% rule for R&D in government budget - Reform of GRIs (Separation from ministerial supervision by introducing five research councils)
2000s	- Framework Act on Science and Technology (2001) - Technology Transfer & Commercialization Promotion Act (2006)	- National S&T Council - Master Plan - Technology Forecasting - Pre-studies for budget allocation (Survey, analysis, assessment) - R&D Management Institute (KISTEP) for - Evaluation & planning
2010s	Act on the Establishment, Operation, and Fostering of state-Funded Science & Technology Research Institutes (2014)	National Research Council of Science & Technology (NST) customized for S&T GRIs
After 2020	National Research and Development Innovation Act (2020)	- Micro-management process of R&D projects (Selection, agreement, conduct, etc.) - Integrated information system - R&D management agencies - Researcher's ethics

Source: Authors

In the above table, the following elements have important meanings in Korea's evolution of S&T governance. Firstly, the policy/ budget coordination body coordinates S&T policies and budget allocation among government ministries. The National S&T Council, established by the Special Law for S&T Innovation (1997) and Framework act on Science and Technology (2001), is the highest-level policy coordination body for the government's policymaking process. It coordinates the policies and budgets of R&D-related ministries and has the final decision-making authority on national R&D budget allocation.

Secondly, it is the governance structure of public research institutes. The current governance of state-funded research institutes (GRIs) has been formulated by the Act on Establishment, Operation, & Fostering of State-Funded Research Institutes (1999) and its amendment Promotion Act on Establishment, Operation, & Fostering of State-Funded Science & Technology Research Institutes (2014). According to the latter Act (2014), there are 25 S&T GRIs that are managed by the National Science and Technology Research Council (NST). These GRIs has independent authority over government intervention with increased autonomy, but the National Science and Technology Research Council (NST) supervises their operation and performance.

Thirdly, there are the acts on policy instruments. The representative policy tool is the national R&D program. The first R&D program was launched by the Ministry of S&T in 1982 with the Technology Development Promotion Act amendment in 1981. Then, in the 1980s, many ministries, such as the Ministry of Industry and Ministry of Information and Communication, started to implement their own R&D programs, which led to an increase in the government's R&D investment and R&D programs conducted in various technological fields.

In addition, the Technology Development Promotion Act (1972) required private companies to accumulate funds necessary for technological development and utilization, and government provided tax and financial incentives to the companies.

Fourthly, there is the establishment of professional R&D management agencies. These agencies support dedicated or competent S&T ministries by carrying out R&D project

management tasks such as selection, contract, budget execution, ex post-evaluation, etc.

One of the most important agencies is the Korea Institute of Science & Technology Evaluation and Planning (KISTEP), which was established based on the Special Act for S&T Innovation (1997) and Framework Act on Science & Technology (2001). In addition, other R&D management agencies under many S&T-related ministries are operated based on the Act on the Management of State-Run Institutions. It is because these agencies are legally classified as quasi-state-run institutions and managed according to their legal status.

In addition to the above four points about the formation of institutions/ actors, it is necessary to consider the activities necessary for effective governance operation. The Framework Act on Science & Technology (2001) contains many activities, including a master plan, technology forecasting, and pre-studies for budget allocation (survey, analysis, assessment). The National Research & Development Innovation Act (2020) also regulates activities such as micro-management (selection, contracting, execution, etc.) of R&D projects, integrated information systems, and researchers' code of ethics. Moreover, the Technology Transfer & Commercialization Promotion Act (2006) deals with technology transfer and commercialization from public research institutes to industry. In short, since these activities are the contents of the role of institutions/ actors constituting governance, the effectiveness of governance is determined by the combination of institutional formation and activity performances.

6.2 Law No. 11-year 2019 on the National System of Science and Technology

Law No. 11-year 2019 on the National System of Science and Technology is an amendment of Law No. 18-year 2002 on the National System on Research, Development, & Application of Science and Technology. The new law consists of six strategic elements of a master plan, implementation, policies & institutions, resources, networks, and guidelines & supervision.

[Table 5-6] Structure of Law No.11-year 2019 (Related Korean S&T Laws)

Chapters	Contents	Articles
Master plan	• Master Plan for Advancing Science and Technology/ Rencana Induk Pemajuan IPTEK (RIPK)	• Articles 3 (2) – (3), • Articles 8 (1) – (4), • Article 9 (1) – (2)
	• Framework Act on Science and Technology (2001) for S&T Basic Plan	
Implementation	• Education, research, development, assessment, application ■ Education ■ Research and development ■ Assessment ■ Application	• Article 14 (1) • Article 15, 16, • Article 17, 19, 21, 22(2) • Article 23(2), 24(1) • Article 29(1), 29(3), (32)
Technology Transfer and Commercialization Promotion Act (2006)		
Policies & institutions	• Assessment & Application Agency • Establishment of BRIN • Ethics/ ethical assessment • Obligation to submit and store primary data and the output of research, development, assessment, & application • Dissemination of S&T	• Article 42, b • Article 48 (1) • Article 39 • Article 40
	• Act on Establishment, Operation, & Fostering of State-Funded Science & Technology Research Institutes (2014) for the National Research Council of Science & Technology (NST)	
Resources	• Human resources • Funding • Research Endowment Fund • Facilities/ infrastructure	• Articles 50, 51, 53, 57, 58 • Articles 59, 60 • Article 62 (1) • Articles 65, 67
	• Special Law for S&T Innovation (1997) for 5 % rule • Framework Act on Science and Technology (2001) for establishing the Evaluation and Planning Institute (KISTEP)	
Network	• Partnership • Repository and dissemination of R&D output	• Articles 71, 72, 73 • Articles 73, 74

Chapters	Contents	Articles
	• HR mobility (Research permit) • Information and access to facilities/ infrastructure • National S&T Information System	• Article 75 • Article 78, (1) • Article 78
	• National Research and Development Innovation Act (2020) for Integrated S&T Information System	
Guidelines & supervision	• Guidelines • Resource support, Institutional strengthening support, Incentives, Implementation of S&T program • Supervision • Sanctions	• Article 79 (1) • Article 80 (2), 82 • Article 83 (1-3) • Article 86 • Article 93-96
	• Technology development Promotion Act (1981) for implementing the National R&D Program • National Research and Development Act (2020) for research code of ethics	

Source: Authors

Law No.11-year 2019 has historical significance in that it lays the framework for science & technology governance in Indonesia. It is a collection of the critical elements necessary for building an effective S&T system. Moreover, it is expected to be an important opportunity to develop S&T governance in Indonesia in the future.

[Table 5-6] analyzes the main chapters of Law No. 11-year 2019 and introduces related Korean S&T laws. The characteristics of the S&T legal systems of the two countries can be compared as follows.

Firstly, compared to Law No. 11-year 2019, which comprehensively stipulates multiple elements in one law, Korea is different in that there are individual laws related to key governance elements. For instance, Korea has a different law on the governance of state-funded research institutes (GRIs). The Framework Act on Science & Technology explicitly stipulates the role of KISTEP concerning the allocation of research funds.

When multiple elements are comprehensively stipulated in a single law, it can be not easy to formulate specific regulations for each matter, so additional regulations such as the Presidential Decree should be prepared and provide details on how the institutional operation will be carried out.

Secondly, Korea has a quasi-state-run sector as a third sector, rather than the government and private sector. Many of the institutions that make up science and technology governance fall into this third category.

[Table 5-7] Quasi-State-Run Sector in Korea

	Government sector	Quasi-state-run sector	Private sector
Organization	• Central government • Local government • Ministries	• Government research institutes (GRIs) • KISTEP for Planning & Evaluation • NST for supervising GRIs • R&D management agencies	• Enterprises • Associations
Status	• Public officials	• Civilian with public duties	• Civilian
Acts	• Government Organization Act • Civil Servant Act	• Act on the Management of State-Run Institutions	-

Source: Authors

Prime examples for the quasi-state-run sector are the National Science and Technology Research Council (NST) and Korea Institute of Science & Technology Evaluation and Planning (KISTEP). NST manages government research institutes (GRIs), and KISTEP is responsible for S&T planning and evaluation. The two institutions correspond to Indonesia's cases of BRIN and research endowment fund agency (LPDP), respectively.

The advantage of a quasi-state-run institution is that it can perform administrative tasks based on expertise and autonomy, free from an inflexible ministerial bureaucracy. In other words, it is crucial to foster quasi-state-run institutions for professional administrative

execution. The assessment and application agency under Article 42, letter B of Law No.11-year 2019 seems to have a possibility to develop into a quasi-state-run sector.

Meanwhile, concerning the main provisions of Law No.11-year 2019, the following matters need to be determined. Firstly, the scope of control and role of BRIN need to be cleared. Before the establishment of BRIN, there are two types of R&D organizations in Indonesia which are: 1) public R&D institutions directly under the president, coordinated by the Ministry of Research & Technology/ BRIN and 2) R&D organizations/ units under the ministries, state-run institutions, & universities, as well as local governments. To integrate all public R&D institutions, the coordination and control mechanism concerning the working mobility of S&T human resources between BRIN and ministries should be arranged clearly since the ministries still need researchers to support their functions and to obtain scientific-based evidence, whereas the employment status of researchers is under BRIN.

Secondly, currently, the working incentives (*tunjangan kinerja*) across ministries and central government institutions (non-ministries) are different. Some ministries have already obtained 100 percent working incentives based on the central government standard for all civil service officers. Other ministries and central government institutions (non-ministries) including BRIN still have obtained working incentives less than 90 percent. The different working incentives across ministries and central government institutions may affect the willingness of an S&T workforce, notably among those who come from ministries such as Ministry of Finance, to join with BRIN. Some of researchers in ministries probably choose to change their functional position from researcher to other functional positions. If this happens, the number of researchers in the central government will decrease. In fact, the low number of researchers in Indonesia has become one of the main concerns of Indonesia's STI development.

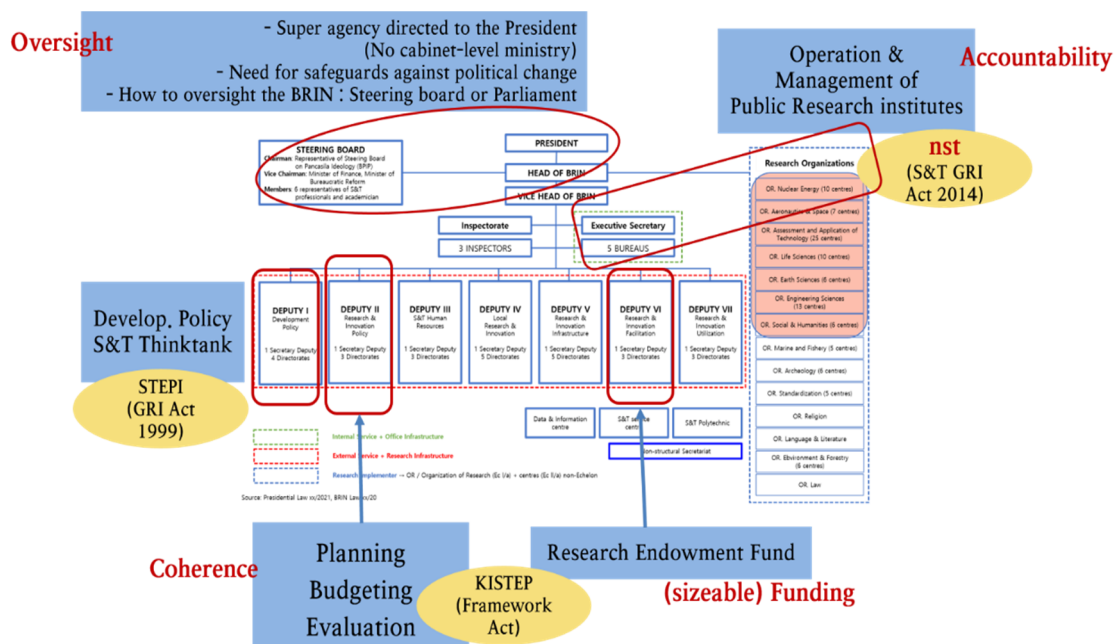
Thirdly, there is the operation of the Research Endowment Fund. According to Law No. 11-year 2019 Articles 59 and 62, one of sources of S&T activities is from endowment fund which can be obtained from education or non-education budget allocations. However, it must be decided who will manage such endowment fund's annual budget allocation and who will be responsible for its investment.

6.3 The Future Direction of Indonesian S&T Laws and Regulations

The situation in Indonesia currently is such as the situation in Korea in 1999 regarding the centralized R&D funding. In 1999, the Korean government enacted the Act of the Establishment, Operation, and Fostering of Government-Funded Research Institute. Previously, each ministry had jurisdiction over certain industrial sectors, but after instating this Act, the Prime Minister had the authority to monitor and control the state-funded institutes for R&D, thereby increasing the power to reinforce the research outcome of the institutes and evaluate the management of the institutes. In 2004, the Korean government then enacted the Act on the Establishment, Operation & Fosterage of Government-Held Research Institutions of Science & Technology that rearranged the control of the sector under the Prime Minister being shifted to MSICT. That situation also pertains to the current situation in Indonesia. The Indonesian government Enacted Law No. 11-year 2019 concerning S&T national system in which all state-funded research institutes should be integrated into one agency called BRIN (national research and innovation agency). By this agency, the funding for R&D activities is also integrated as well as the function of planning, monitoring, and evaluation of R&D funding. According to what Korea has already achieved, strengthening the ability to monitor and control R&D funding is needed to shift from a technologically catching-up country to a technologically advanced one. Perhaps this policy can increase the research outcomes in Indonesia.

Regarding the future direction of S&T laws in Indonesia, we can discuss it from two perspectives. The first perspective is concerned with the structure and function of the new National Research & Innovation Agency (BRIN), which was established in 2021. The primary purpose of Law No. 11-year 2019 is to integrate public research institutes under BRIN. The future task of BRIN can be classified into four areas of: 1) policy functions related to the planning, budgeting, & evaluation of R&D programs, 2) the management of the Research Endowment Fund, 3) balancing the principles of accountability and autonomy for integrated public research institutes, and 4) supervision of BRIN. [Figure 5-4] shows these four areas, and future S&T laws and regulations should make them work more solid and effective.

[Figure 5-4] Structure and Function of BRIN



Source: Kim (2021), p. 15

The second perspective is to review the Korean experience from 1999 to the present. Korea has seen the following policy measures develop through its advancement of science and technology governance. In consideration of the above, the future laws and regulations of Indonesia on science and technology governance should consider defining the following:

- Comprehensive STI Master Plan:
 - Mid-Term STI Master Plan: from 1st 5-year STI development plan (1962-1966) to 6th 5-year STI development plan (1987-1991)
 - Long-Term STI Master Plan: Long-term plan for STI development (1967-1986); Long-term STI development plan toward year 2000 (1987-2000); Long-term STI development vision toward year 2025 (2000-2025)
- Human Resources Development:
 - Cultivation of R&D actors: GRIs (State-sponsored research institutes): KIST among

others

- Cultivation of universities
- Cultivation of R&D institutes in private enterprises: Large firms and SMEs
- Financial Incentives:
 - Subsidies, policy loans, bank loans
 - Tax Incentives, Technology Development Reserve Fund
- Institutions:
 - National Research Foundation
 - Intermediate Agencies: Planning and evaluation agencies (KISTEP)
 - Cooperative networks among R&D actors
- National R&D Programs
 - Specific-purpose R&D program, highly advanced national program (G7 Program)
 - Industrial technology development program, ICT R&D program
 - BK-21 Program, 21st Century Frontier Program, Global Frontier Program
- Policy coordination and budget allocation
 - National Science & Technology Council
 - Survey, analysis and evaluation of program performance
- Feasibility Analysis on STI Investment
 - Cost/benefit analysis
 - Major R&D programs
 - Large-scale investment: R&D and infrastructure
- Diagnosis of Situation and Trend Analysis
 - 'Annual R&D Survey': from 1963 until now

- STI data and STI indicators
- Analysis on international STI competitiveness

* The contents in the box above are based on the ideas of Dr. Young-rak Choi, former president of STEPI. The authors would like to thank Dr. Choi.

6.4 S&T for Industrial Development

6.4.1 Promotion Acts for Strategic Industries in Korea

S&T laws and regulations in Korea follow the development of specific industrial sectors. In the 1960s, the Korean government developed the machinery, shipbuilding, and electronics sectors. To support such development, the Korean government made the general standards for industrial sectors including measuring and gauging standards. Therefore, Korea enacted the Measures Act of 1961 and Industrial Standardization Act governing the quality of mining and manufacturing products. The Korean government stipulated specific acts related to machinery, shipbuilding, and electronics sectors which are the Machinery Industry Promotion Act and Electronic Industry Promotion Act of 1969. In the Indonesian case, S&T laws and regulations during the periods of the 1960s mostly focused on the development of S&T institutions. In 1963, the Indonesian government established the National Institute of Aeronautics & Space (LAPAN) through Presidential Decree No. 236-year 1963. In 1965, the Indonesian government established the National Atomic Energy Atomic (BATAN) through Law No. 31-year 1964. In 1966, the Indonesian Institute of Sciences (LIPI) was established to replace the function of previous institutions, DURENAS (Departemen Urusan Riset Nasional) and MIPI (Majelis Ilmu Pengetahuan Indonesia). The difference in terms of S&T focuses and strategies between Indonesia and Korea may affect the development of S&T as well as economic development in each country. In general, the Korean Government focuses on the development of specific industries followed by S&T laws and regulations, whereas in Indonesia during the 1960s, S&T laws and regulations mostly were to establish S&T institutions.

Regarding the promotion of specific industries, the Korean government has enacted a series

of acts to foster individual industries. For example, the Machinery Industry Promotion Act in 1967, Shipbuilding Industry Promotion Act, and Electronics Industry Promotion Act in 1969 were enacted. The policy measures used in the Acts to foster individual industries are as follows.

<Policy Tools of the Machinery Industry Promotion Act in 1967>

- 1) Machinery industry promotion basic plan and implementation plan
- 2) Registration of machine industry business
- 3) Designation and management of specific machinery industry
 - Instructions for supplementation of old facilities, rationalization and integration of facility arrangement, establishment of facility standards and product inspection standards
 - Creation and designation of long-term low-interest funds
 - Loans securitized against company product
- 4) Financial support for nurturing technicians
- 5) Promotion of localization
 - Designation of localization ratio for machinery facilities and encouraging localization
- 6) Others: Installation of the policy deliberation committee

<The Korean government has continuously revised the Act (1967) and added the following measures.>

- 1) Regulations on cancellation of registration for machinery industry business operators (1971)
- 2) Regulations on the establishment and utilization of the Machinery Industry Promotion Fund (Issuing government bonds for fundraising, lending funds) (1971)
- 3) Creation of Machinery Industrial Complex (1971)

- 4) Tax reduction and exemptions (1971)
- 5) Regulations on the establishment of various institutions: Establishment of Mutual Aid Business Organization (1971), Korea Machinery Industry Promotion Association, Korea Production Technology Corporation (1981)

6.4.2 Promoting Technology Transfers to Industries

Korea enacted the Technology Transfer Promotion Act in 2000, whose provisions focus on technology transfers, assessment of technologies, distribution of technology information, and was revised in 2006 to the Technology Transfer & Commercialization Promotion Act. In Indonesia, there is Government Regulation (PP) No. 20-year 2005 concerning technology transfers of intellectual property and results of research and developmental activities by universities and research & development institutes.

The primary purpose of Korea's Technology Transfer Act is to transfer new technologies from public research institutes to private companies in order to promote commercialization. And in the process of enforcing the Act, the Korean government has recognized the importance of evaluating and managing the R&D performance of public research institutes before a technology transfer. Therefore, the Technology Transfer Promotion Act enacted in 2000 is closely related to the Act on Performance Evaluation & Performance Management of National R&D Projects passed in 2006. This Act is designed to improve the efficiency and accountability of R&D investment by evaluating R&D activities in the field of science and technology promoted by the government with a focus on performance and efficiently managing and utilizing research results.

<Policy Tools of the Act on Performance Evaluation & Performance Management of National R&D Projects>

- 1) Establishment of the plan: A basic outline for performance evaluation and performance management
- 2) Setting performance goals and performance indicators: Develop and disseminate standard performance indicators
- 3) Implementation of various evaluations: specific evaluation, high-level evaluation, self-evaluation, science, and technology innovation capacity evaluation, etc.
- 4) Establishment of research result database
- 5) Support for technology valuation cost
- 6) Support for specialized research and management institutions
- 7) Education and training
- 8) Others: Securing the evaluation budget and submitting the evaluation results to the National Assembly

6.5 Recent Development: National R&D Innovation Act (2020) in Korea

The Figure 5-3 shows recent policy trends in Korea related to technology regulation. The Act stipulates the general management regulations for project support, the establishment of integrated R&D information system, fostering of R&D management agencies, and details of research code of ethics.

6.5.1 Micro-process for R&D project management

The National R&D Innovation Act specifies the responsibilities of government, R&D institutions, and researchers in carrying out R&D projects. Compared to the Framework Act on Science & Technology, the new Act deals with micro-level management and explicitly

stipulates the details of the announcement, selection, agreement, conduct, and research expenses. Above all, the selection and operation procedures for national R&D projects are expressly stipulated.

6.5.2 Integrated R&D Information System

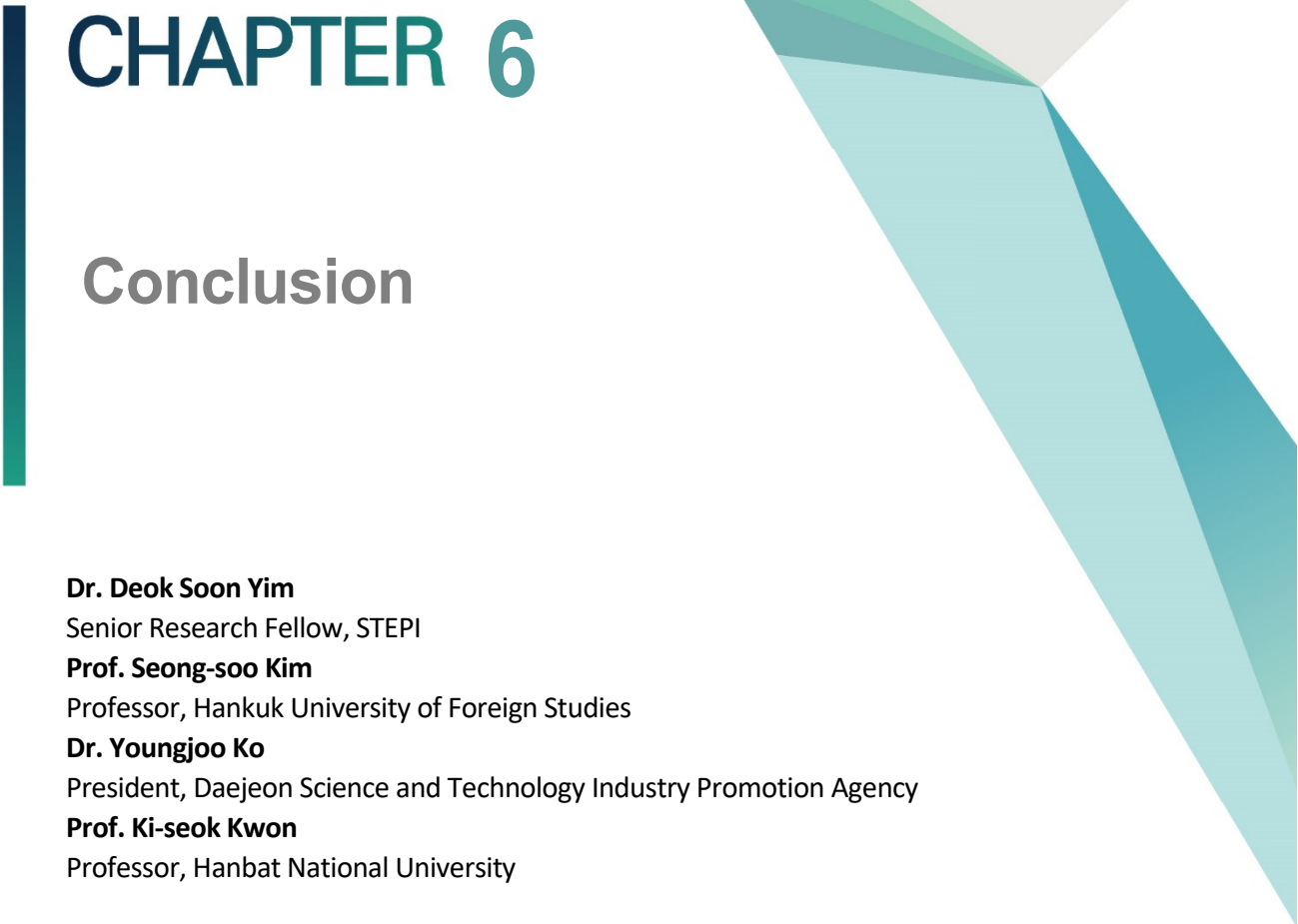
The Integrated R&D Administration System (IRIS) has been established for more efficient processing of R&D information. It integrates R&D tasks, research funds, and national researcher information through the information system and provides one-stop-shop information through the web portal service.

6.5.3 Reforming ministerial R&D management agencies

The R&D management agency is an executive agency that carries out tasks related to the implementation of national R&D projects. The R&D management agency supports the dedicated or competent S&T ministries by providing administrative services to researchers in the selection, management, and evaluation of R&D projects. The Act intends to limit the number of agencies by designating a dedicated or competent agency for each dedicated or competent S&T ministry.

6.5.4 Research code of ethics

The National R&D Innovation Act stipulates the research code of ethics in conducting national R&D projects. Acts of misconduct such as falsification of R&D performance, violation of research fund usage standards, and security leaks are subject to sanctions. In the case of research misconduct, participation in national R&D activities may be restricted within ten years for R&D institutions. It can also recover paid R&D costs and impose additional sanctions and charges. Meanwhile, a procedure to protect the researcher's rights has been prepared through an appeals process of allowing a review of sanction request.

A decorative vertical bar on the left side of the page, transitioning from dark blue at the top to teal at the bottom. On the right side, there are abstract geometric shapes in shades of teal, light blue, and grey, resembling stylized mountain peaks or a modern architectural design.

CHAPTER 6

Conclusion

Dr. Deok Soon Yim

Senior Research Fellow, STEPI

Prof. Seong-soo Kim

Professor, Hankuk University of Foreign Studies

Dr. Youngjoo Ko

President, Daejeon Science and Technology Industry Promotion Agency

Prof. Ki-seok Kwon

Professor, Hanbat National University

Chapter 6. Conclusion

1. Summary

1.1 The Evolution of STI Think Tanks in South Korea & Implications for Indonesia & Current Status

The experience of Korean think tanks provides a unique reference to other catch-up countries attributed to not only its rapid economic success but also aligned management of the STI system & process and appropriate function of STI institutional elements including STI think tanks. In Korea, we identify some stylized facts in STI process consisting of policy agenda setting, policymaking, policy implementation, and policy evaluation. Firstly, regarding agenda setting in R&D, outside initiation is limited. Secondly, generally after STI agencies prepare detailed policy alternatives, ministries take over initiatives to determine overall directions and strategies of the alternatives. Thirdly, R&D policy implementation is the duty of R&D actors such as researchers in universities and public research institutes. However, they are still under the strong control of R&D agencies and ministries. Fourthly, as opposed to project evaluation, program evaluation tends to lack in strong attention from them.

At the agenda-setting stage, a think tank is required to have holistic policy coordination, help macro-system transition and adjust the goals and needs of main actors. At the subsequent stage, think tank identifies and compares the policy alternatives and then predicts policy outcomes in advance. The think tank usually does not get highly involved in the implementation stage but needs to align the conflicts between heterogeneous organizations and interest groups. Lastly, in the evaluation stage, the main role is to collect

the evaluation results and analyze the implications for the feedback loop. Our discussion reaches an agreement that the main role is to coordinate STI policy by coordinating across technology sectors and aligning the policy actors from agencies to R&D actors. We also highlight that a think tank is required to have a more holistic approach in predicting, confronting, and planning the transition of policy systems.

Indonesia is currently undergoing a very big change in the history of its S&T institutional governance. This is a mandate from the enactment of Law No. 11-year 2019 concerning the National System of Science and Technology, especially in Article 48. This article states the need for integration of R&D institutions in Indonesia embodied in the National Research & Innovation Agency (BRIN). This integration intends to ensure the sustainability of one-way R&D activities in order to increase S&T capabilities in the national development progression. The government through the provisions of a presidential decree established the agency.

The deputy for research & innovation policy has the task of carrying out the implementation of R&D as well as inventions and innovations in the framework of the formulating policies in the field of research and innovation including a master plan for the advancement of science and technology, and roadmaps for research and innovation. This deputy has duties for the BRIN think tank in formulating national STI plans and policies.

1.2 Building Monitoring & Evaluation Systems for National Research & Development Program in Indonesia

In 2016 Indonesia launched the country's new vision on research and development (R&D) in the so-called National Research Master Plan (*Rencana Induk Riset Nasional* or RIRN) 2017-2045 and a five-year PRN as an operational plan of RIRN has been implemented since 2020. PRN has a national program for individual researchers and institutional program for public research institutes and universities. These programs consist of various kinds of projects. MoRT/ BRIN selects projects proposed by PM and evaluates research performance. The projects selection and evaluation system still do not have a systemic way in terms of structure & mechanism, evaluation team management, and evaluation index.

This study has analyzed the current evaluation system and then proposes some reform suggestions through the related survey of researchers, interviews with government relevant officers and funding managers, focus group discussions, a second-stage in-depth survey, etc. A comparative analysis with Korea's evaluation system was also conducted through research team meetings and seminars with some Korean experts.

Two levels of structure in proposal selection and research monitoring & evaluation can give the PMs more authority to coordinate and integrate their research projects. In addition, such a structure would also make the division of responsibility between BRIN and the PMs clearer, notably for proposal selection and research monitoring & evaluation. The proposed actors involved have represented important stakeholders that we believe can provide PMs and research teams feedback on their research programs. Meanwhile, the proposed indicators are expected to make the process of proposal selection and research monitoring & evaluation activities more organized. We have also worked to address the issue of inter-institutional collaboration in R&D programs/ projects. Accordingly, coordination and integration are expected to be improved by giving more responsibility and autonomy to PMs. These are essential in the PRN program to produce innovative products.

While the proposed design resulted in this study clearly indicates a decentralized management system, the recent reorganization of research institutions in Indonesia through the establishment of the National Research & Innovation Agency (BRIN) has raised a new issue worthy of discussion. Although more studies are required to gauge the implication of BRIN to our proposed system, there has been a tendency that BRIN has strengthened its centralized and fragmented funding management, a system that we do not recommend as it may create a new and added bureaucratic process into the existing R&D management system.

Further work will be needed to complement the findings of this study such as a new design of the next five-year PRN, co-funding and coordination between competitive and institutional projects, and better commercialization of research results based on more participation of industries, etc.

1.3 Laws & Regulations of Science & Technology

This study dealt with science and technology laws and regulations in Korea and Indonesia and analyzed the following issues.

Firstly, it presented theoretical criteria for classifying science and technology laws. These include legislative purpose, policy purpose, fields of technology, levels of the laws, ministerial jurisdiction, and time of enactment

Secondly, it analyzed the development process of science and technology laws in Korea from the 1950s to the present. It introduced the laws that have an essential impact on shaping Korea's science and technology governance, including the Science & Technology Promotion Act, the Science & Technology Framework Act, Act on the Establishment, Operation, & Fostering of Government-Funded Research Institutes, and Industrial Technology Promotion Act, etc.

Thirdly, it introduced the current status of Indonesia's science and technology laws. From the 1950s to the present, the development process of Indonesia's science and technology laws was discussed on a 10-year basis. In particular, Law No. 11-year 2019 for Indonesia's national science and technology system was intensively discussed.

Fourthly, the implications of Korea's experience for Indonesia are discussed. Law No. 11-year 2019 has a significant meaning to reorganize Indonesia's science and technology governance. The legislation's primary purpose is to integrate public research institutions that had been dispersed throughout various institutions under the newly launched National Research & Innovation Agency (BRIN). It is to enhance the performance of public research institutes through the integration. In this context, the current change in Indonesia can be seen as a situation similar to when Korea integrated state-funded research institutes in 1999 and enacted the Framework Act on Science & Technology in 2001.

In summary, the development of science and technology laws and regulations in Korea has promising implications for developing Indonesian Law No. 11-year 2019 in the future. Indonesia also needs to legislate various policy measures that have evolved through the

advancement of Korean science & technology laws and regulations.

2. Policy Recommendations

2.1 The Evolution of STI Think Tanks in South Korea & Implications for Indonesia & Current Status

The experience of Korean think tanks provides a unique reference to Indonesia attributed to not only its rapid economic success but also aligned management of the STI system & process and appropriate function of STI institutional elements including STI think tanks. Meanwhile, the pitfalls in the evolutionary path of think tanks such as microscopic approach to STI policy can provide lessons not to choose inappropriate decisions when we create a new STI think tank in Indonesia.

In particular, the initial Korean STI think tank model based on a more holistic approach could be a more preferable option for Indonesian policy practitioners. Moreover, shortcomings found in current Korean STI think tank model provides valuable lessons. For example, we observe the difficulties in coordination and integration policy goals and objectives between various STI actors, when they carry out R&D projects and program. Moreover, in order to fulfill the given myopic performance indicators, individual actors and agencies linked with various ministries have a tendency to obstruct the proper management of R&D efforts such as AI and big data.

Indonesian policy experts want their STI think tank to integrate STI functions (i.e. the science and technology ministry) with non-STI functions (e.g. ministry of industry and ministry of agriculture). Moreover, new evidence-based policy system includes policy researchers, analysts, and policy-makers who are providing more enhance policy alternatives, while the new policy function of an STI think tank in Indonesia will integrate both the top-down approach based on national STI strategies and bottom-up approach focused on the voice of the people, market, and R&D performers.

Conclusively, we have drawn successfully some policy implications for the creation of an

Indonesian STI think tank this year. Next year, we may implement a more detailed analysis on the interview data collected this year on the establishment of a think tank in Indonesia. Moreover, by applying a comparative framework on the two think tanks in Korea and Indonesia, we look forward to producing richer lessons for two countries. To this end, an intensive workshop will be helpful for the creation of policy implications from comparative analyses.

2.2 Building Monitoring & Evaluation Systems for National Research & Development Program in Indonesia

MoRT/ BRIN needs to give program managers (PM) more autonomy for coordination and integration of their research projects with more time for projects formulation with other researchers rather than strengthening bureaucratic management.

The division of responsibility between BRIN and PMs is made clearer, specifically in terms of proposal selection and research monitoring & evaluation structure.

The actors involved in selection and evaluation mechanism should have represented important stakeholders who can provide PMs and research teams with appropriate feedback on their research programs.

In order to make the process of proposal selection and research monitoring & evaluation activities more organized, proposed indicators are needed to develop in more detail aligned with the purpose and targets of PRN.

More appropriate arrangement of the R&D budget system which have performance results reflect their grades.

Various kinds of Inter-institutions collaboration in R&D programs and projects are needed to improve coordination and integration even better through more autonomous PM authority.

Industries should participate more both in selection and evaluation mechanism and research implementation to catalyze industry-centered innovation producing innovative products.

Much wider range of communication among government, universities, public research institutes and industries should be organized to expand the consensus of a new evaluation system for proposal selection and research performance

Future integrative work needs to be designed for the next five-year PRN along with a new co-funding system, coordination process between competitive and institutional projects, and better innovative commercialization of research results based on more participation of industries.

2.3 Laws & Regulations of Science & Technology

Currently, science & technology governance in Indonesia is in the process of undergoing massive changes.

- Law 11-year 2019 is a law with a significant meaning to reorganize Indonesia's science & technology governance.
- The legislation's primary purpose is to integrate public research institutes under the newly launched National Research & Innovation Agency (BRIN).

This study introduced the structure and function of BRIN, which was launched in 2021 and translated its Indonesian name into Korean for the first time.

- In this process, four functional areas of BRIN were divided, and the need for more systematic development in the function of each area was conveyed based on the experience of Korea.
- The four areas consisting of 1) policy functions related to the planning, budgeting, & evaluation of R&D programs, 2) the management of the Research Endowment

Fund, 3) balancing the principles of accountability and autonomy for integrated public research institutes, and 4) supervision of BRIN.

The Indonesian government intends to innovate its science & technology system by Law No. 11-year 2019.

- The current change in Indonesia can be seen as a situation similar to when Korea reformed its structure of state-funded research institutes in 1999 and enacted the Framework Act on Science & Technology in 2001.
- Therefore, Korea's experience has many implications for Indonesia's future development. For example, the Science & Technology Promotion Act, Science & Technology Framework Act, the Act on the Establishment, Operation, & Fostering of Government-Funded Research Institutes, and Industrial Technology Promotion Act can give useful implications.

Regarding the next study, the subject of research collaboration needs to deepen the contents of this year's project on science & technology laws and regulations.

- For example, we can discuss how various S&T policy measures have evolved through the advancement of science and technology laws in Korea.
- Research on the establishment, procedure, and content of the S&T Master Plan, which is importantly dealt with by Indonesian Law No. 11-year 2019, is also a worthwhile topic.

ABBREVIATIONS

AIPI	Akademi Ilmu Pengetahuan Indonesia (Indonesian Academy of Sciences)
APBN	Anggaran Pendapatan dan Belanja Negara (State Budget)
ARN	Agenda Riset Nasional (National Research Agenda)
ASIP	Asian Society for Innovation and Policy
BAPETEN	Nuclear Energy Regulatory Agency
BAPPENAS	Badan Perencanaan Pembangunan Nasional (National Development Planning Agency)
BATAN	Badan Tenaga Atom Nasi
Balitbangda	Badan Penelitian dan Pengembangan Daerah (Regional Research and Development Agency)
BPIS	Badan Pengelola Industri Strategis (Strategic Industry Development Agency)
BPOM	National Agency of Drug & Food Control
BPPT	Badan Pengkajian dan Penerapan Teknologi (Agency for the Assessment and Implementation of Technology)
BRIN	Badan Riset dan Inovasi Nasional (National Research & Innovation Agency)
BSN	Badan Standarisasi Nasional (National Standardization Agency)
CIPG	Centre for Innovation Policy and Governance
CSTP	Center of S&T Policy
DIPA	Entry list of budget implementation
DISTEP	Presiden, Daejeon Institute of Science & Technology for Enterprise & People
DRN	Dewan Riset Nasional (Government Regulation National Research Council)
DSN	Dewan Standardisasi Nasional (National Standardization Council)

EBPM	Evidence-based policymaking
FGD	Focused group discussions
GDP	Gross domestic product
GERD	Gross expenditure on research and development
GRIs	Government research institutes
Himpenindo	Himpunan Peneliti Indonesia (Indonesian Researchers Union)
litbangjirap	Penelitian, pengembangan, pengkajian, dan penerapan (research, development, assessment, and application)
IPTN	Industri Pesawat Terbang Nusantara (Nusantara Aircraft Industry)
IRIS	Integrated R&D Information System
Jakstranas Iptek	Kebijakan Strategis Nasional Iptek(National Policy and Strategy in S&T)
KAIST	Korea Advanced Institute of Science & Technology
KEIN	National Committee of Economics and Industry, Indonesia
KEIT	Korean Evaluation Institute of Industrial Technology
KIAT	Korea Institute for Advancement of Technology
KISED	Korea Institute of Startup & Entrepreneurship Development
KIST	Korea Institute of Science & Technology
KISTEP	Korea Institute of S&T Evaluation & Planning
KSI	Knowledge sector initiative
LAPAN	Lembaga Antariksa dan Penerbangan Nasional (National Institute of Aeronautics & Space)
LIPI	Lembaga Ilmu Pengetahuan Indonesia (Indonesian Institute of Sciences)
LPDP	Insentif Riset Sistem Nasional (Educational Fund Institute of Management Insinas)
M&E	Monitoring & evaluation

MITI	Majelis Ilmu Pengetahuan Indonesia
MOE	Ministry of Education
MOEF	Ministry of Economy & Finance, Korea
MORT	Ministry of Research, Technology
MORTHE	Kementerian Riset, Teknologi dan Pendidikan Tinggi (Ministry of Research, Technology, and Higher Education)
MOST	Ministry of Science and Technology
MOTIE	Ministry of Trade, Industry, and Energy
MSIT	Ministry of Science & ICT, Korea
MSS	Ministry of SMEs and Stratups
NABO	National Assembly Budget Office
NIS	National Innovation System
NGO	Nongovernmental organization
NRF	National Research Foundation of Korea
NST	National Research Council of Science & Technology
NSTC	National Science & Technology Council
ODA	Official development assistance
OR	Organization of research
P2KMI	Center for Policy Research & Management of Science, Technology, & Innovation
P.O.	Presidential Office
PACST	Presidential Advisory Council on Science & Technology
PCFIR	Presidential Committee on the Fourth Industrial Revolution
PEPUNAS IPTEK	Formulator of the National General Program in S&T
PMO	Project management office
PRIs	Public research institutes

PRN	Prioritas Riset Nasional (in Indonesia); National Research Priority
PROPENAS	Program Pembangunan Nasional (National Development Program)
PSHK	Pusat Studi Hukum dan Kebijakan Indonesia (Centre for Policy & Law Studies)
PTI	Pengembangan Teknologi Industri (in Indonesia); Industrial Technology Development
REPELITA	Rencana Pembangunan Lima Tahun (Five-year development plan)
RFP	Request For Proposal
RIPK	Rencana Induk Pemajuan IPTEK (Master Plan for Advancing Science and Technology)
RIRN	Rencana Induk Riset Nasional (National Research Master Plan)
RISPRO	Riset Inovatif Produktif (in Indonesia); Productive and Innovative Research
RPJMN	Rencana Pembangunan Jangka Menengah Nasional (National Medium-term development plan)
RPJPN	Rencana Pembangunan Jangka Panjang Nasional (Long-Term National Development Plan)
R&D	Research and development
SOE	State-owned enterprise
STEPI	Science & Technology Policy Institute
STI	Science, technology and innovation
S&T	Science and technology
WBS	Work breakdown structure
WP	Work packages

REFERENCES

Appendix 1. Interviewed Resource Persons (Chapter 3)

No.	Name	Agency/ affiliation	Position	Rationality	Type of resource person	Date of interview
1	Mr. MK	Research and Development (R&D) Agency in local government of Sidoarjo (Balitbang Kabupaten Sidoarjo)	Researcher	Practice of policy think tank in public research & development (R&D) institutes (public policy & regional development) at regency level	Government – regency level	August 23, 2021
2	Mr. I	Research and Development (R&D) Agency in East Java Province government (Balitbang Provinsi Jatim)	Researcher	Practice of policy think tank in public research & development (R&D) institutes (public policy & regional development) at provincial level	Government – provincial level	August 23, 2021
3	Mr. FP	University of Brawijaya	Expert of public policy/ innovation policy	Academic expert who has capability in public policymaking or STI policy, and can explain in both academic & practical way.	Academy	August 24, 2021
4	Mr. YM	Directorate of Science-Technology-Innovation (STI) Policy and Institution at the Ministry of Research & Technology/ National Innovation & Research Agency	Director	National policy think tank at ministry level (specific field of STI affairs)	Central government	September 1, 2021
5	Mr. TH	Policy Analysis Unit at the Coordinating Ministry for Economy Affairs (Kemenko-Perekonomian)	Policy analyst	National policy think tank at ministry level (field of national economy)	Central government	September 2, 2021

No.	Name	Agency/ affiliation	Position	Rationality	Type of resource person	Date of interview
6	Mr. TM	Directorate of Higher Education and Science & Technology at the National Agency for Planning & Development (Bappenas)	Director	National policy think tank at ministry level (specific field of STI development and planning)	Central government	September 2, 2021
7	Mr. RA	Directorate of Higher Education and Science & Technology at the National Agency for Planning & Development (Bappenas)	Key staff	National policy think tank at ministry level (specific field of STI development and planning)	Central government	September 2, 2021
8	Mr. AF	Directorate of Higher Education and Science & Technology at the National Agency for Planning & Development (Bappenas)	Key staff	National policy think tank at ministry level (specific field of STI development and planning)	Central government	September 2, 2021
9	Mrs. R	Directorate of Higher Education and Science & Technology at the National Agency for Planning & Development (Bappenas)	Key staff	National policy think tank at ministry level (specific field of STI development and planning)	Central government	September 2, 2021
10	Mrs. EC	Directorate of Higher Education and Science & Technology at the National Agency for Planning & Development (Bappenas)	Key staff	National policy think tank at ministry level (specific field of STI development and planning)	Central government	September 2, 2021

No.	Name	Agency/ affiliation	Position	Rationality	Type of resource person	Date of interview
11	Mr. WK	University of Gadjah Mada	Expert of public policy/ innovation policy	Academic expert who has capability in public policymaking or STI policy, and can explain in both academic & practical way.	Academy	September 13, 2021
12	Mr. YW	Centre for Specific Technology Region & Innovation System at the Agency for Assessment and Application of Technology (BPPT)	Policy analyst	Practice of policy think tank in public research & development (R&D) institutes (Technology development and engineering)	Central government	September 16, 2021
13	Mr. TT	Fiscal Policy Agency at the Ministry of Finance (Kementerian Keuangan)	Vice director	National policy think tank at ministry level (field of finance)	Central government	September 17, 2021
14	Mr. MN	Research Centre for National Industry Development at the Ministry of Industry (Kemenperin)	Policy analyst	National policy think tank at ministry level (field of national industry)	Central government	September 17, 2021
15	Mr. SO	Research Centre for National Industry Development at the Ministry of Industry (Kemenperin)	Key staff	National policy think tank at ministry level (field of national industry)	Central government	September 17, 2021
16	Mr. TW	Public Administration Agency (LAN)	Deputy	National policy think tank at the special agency for public administration	Central government	September 22, 2021
17	Mrs. MG	Centre for Policy & Performance at the National Agency for Planning & Development (Bappenas)	Vice director/ policy analyst	National policy think tank at ministry level (Field of macro development and planning)	Central government	September 27, 2021

No.	Name	Agency/ affiliation	Position	Rationality	Type of resource person	Date of interview
18	Mr. FS	Centre for Policy & Performance at the National Agency for Planning & Development (Bappenas)	Key staff	National policy think tank at ministry level (field of macro development and planning)	Central government	September 27, 2021
19	Mr. JH	Centre for Policy & Performance at the National Agency for Planning & Development (Bappenas)	Key staff	National policy think tank at ministry level (field of macro development and planning)	Central government	September 27, 2021
20	Mr. RR	Centre for Policy & Performance at the National Agency for Planning & Development (Bappenas)	Key staff	National policy think tank at ministry level (field of macro development and planning)	Central government	September 27, 2021
21	Mr. AW	Centre for Aeronautics and space Policy at the National Aeronautics and Space Agency (LAPAN)	Director	Practice of policy think tank in public research & development (R&D) institutes (Aeronautics)	Central government	September 27, 2021
22	Mr. FM	Centre for Partnership & Dissemination at the National Nuclear Agency (BATAN)	Policy analyst	Practice of policy think tank in public research & development (R&D) institutes (Nuclear)	Central government	September 28, 2021
23	Mr. MW	Research Centre for Education Policy at the Ministry of Education (Kementerian Pendidikan)	Policy analyst	National policy think tank at ministry level (field of education)	Central government	October 14, 2021
24	Mrs. MU	Centre for Innovation & Governance (CIPG)	Researcher	NGO agency in science-technology-innovation (STI) policy	NGO	October 27, 2021

No.	Name	Agency/ affiliation	Position	Rationality	Type of resource person	Date of interview
25	Mr. FS	Centre for Innovation & Governance (CIPG)	Key staff	NGO agency in science-technology-innovation (STI) policy	NGO	October 27, 2021
26	Mrs. BP	Knowledge Sector Initiative (KSI)	Director	International NGO agency in public policymaking & evidence-based policy	NGO	October 27, 2021
27	Mr. FN	Centre for Law & Policy Studies	Director/Researcher	NGO agency in public affairs	NGO	October 27, 2021
28	Mr. RP	University of Padjadjaran	Expert of public policy/innovation policy	Academic expert who has capability in public policymaking or STI policy, and can explain in both academic & practical way.	Academy	November 12, 2021
29	Mrs. VW	Secretariat of the Vice President (Setwapres)	Researcher	Independent researcher who has capability in public policymaking or STI policy.	Government	November 13, 2021
30	Mr. TS	CV. Nusantara Jaya (Small-medium enterprise)	Director	Beneficiaries or participants in STI & Industrial Policymaking (on small-medium enterprise scale)	Industry	November 24, 2021

Appendix 2. Interview Questions (Chapter 3)

No.	Type of key info. providers	Interview questions	
1	NGO	a.	What is the role of your agency in supporting the science and technology policy advocacy process and evidence-based policymaking?
		b.	Does your agency have a channel to communicate/ disseminate the results of policy research or evidence-based policies?
		c.	How does your agency influence the process of making science & technology policies in Indonesia on a national, regional, sectoral level?
		d.	If your agency is often involved in the science and technology policymaking process, who is the most dominant actor in influencing the science and technology policymaking process?
		e.	What form of policy advocacy has your agency conducted to the government so far related to the issue of science and technology?
2	Academy	a.	What is public policy, and how is it related to science and technology policy?
		b.	Why are policy think tanks needed?
		c.	What is the practice of evidence-based policymaking in Indonesia?
		d.	How can policy think tanks play their role?
		e.	Are there differences between policy think tanks in government organizations and non-governmental organizations?
		f.	When and where is a policy advocacy appropriate to do?
		g.	Have think tanks in the field of science and technology been running as expected in Indonesia?
		h.	What are the criteria and how does the appropriate policy think tank (science, technology, and innovation) mechanism work in Indonesia?
3	Government	What is the role of your agency in formulating science & technology policies so far?	

No.	Type of key info. providers	Interview questions	
4	Industry	a.	Is your type of business related to technology development or innovation?
		b.	Have you been involved in the process of formulating science & technology policies by the government so far (local, sectoral, national level)?
		c.	Do you accept the impact of the current science and technology policy implementation, either directly or indirectly?
		d.	Has your business been following government regulations in several respects? If so, what are the rules, and if not, why?
		e.	Are you often involved in the process of making science & technology policies by the government/ academic experts/ non-governmental organizations?
		f.	Are you often involved in policy advocacy (science, technology, and innovation) to the relevant government?
		g.	Do you need government intervention in encouraging/developing your business better? If so, what kind of intervention needs to be done?

APPENDIX 3. Feedback Survey of PRN 2020-2024 Monitoring & Evaluation System Design (Chapter 4)

RESPONDENT PROFILE

1. Name:
2. Institution:
3. Position in PRN:
 - ☐ PMO BRIN
 - ☐ LPDP
 - ☐ Agency research fund manager (eg BPK LIPI, BPPT, LAPAN, University)
 - ☐ Program Manager (including substance and administration coordinator)
 - ☐ WBS Coordinator
 - ☐ WP Coordinator
 - ☐ Research member
 - ☐ Others
4. PRN Focus:
5. End product target (WP, WBS, or program level):
6. Scientific classification:
 - ☐ Science and Life Science
 - ☐ Engineering
 - ☐ Social and Humanities
7. Source of funds (all that apply)
 - ☐ RISPRO Mandatory
 - ☐ RISPRO other than Mandatory
 - ☐ DIPA (Internal funding) institution
 - ☐ Other:

PROPOSAL SELECTION

8. In your opinion, what is the ideal timeframe between the announcement of the call for proposal and deadline for submission?
 - ☐ < 1 pekan
 - ☐ 1-2 pekan
 - ☐ 1 bulan
 - ☐ > 1 bulan

MASTER PROPOSAL

In your opinion, how important are the following actors involved in the master proposal selection process?

Notes: 1: very unimportant; 2: not important; 3: neutral; 4: important; 5: very important

9. BRIN

Very unimportant 1	2	3	4	5 very important
--------------------	---	---	---	------------------

10. LPDP

Very unimportant 1 2 3 4 5 very important

11. BAPPENAS

Very unimportant 1 2 3 4 5 very important

12. Independent committee/body specially formed for proposal selection

Very unimportant 1 2 3 4 5 very important

13. Experts/Academic experts

Very unimportant 1 2 3 4 5 very important

14. In your opinion, which other actors are very important to be involved in the selection of integration proposals? (if none, please write NA)

PROJECT PROPOSAL

In your opinion, how important are the following actors involved in the project proposal selection process?

15. Program Manager

Very unimportant 1 2 3 4 5 very important

16. BRIN

Very unimportant 1 2 3 4 5 very important

17. LPDP

Very unimportant 1 2 3 4 5 very important

18. BAPPENAS

Very unimportant 1 2 3 4 5 very important

19. Independent committee/body specially formed for proposal selection

Very unimportant 1 2 3 4 5 very important

20. Experts/Academic experts

Very unimportant 1 2 3 4 5 very important

21. In your opinion, which other actors are very important to be involved in the selection of individual proposals? (if none, please write NA)

PROPOSAL SELECTION METHOD

22. In the proposal selection process, both master and project proposals, which method do you think is the most appropriate?

- Double blind review (reviewer dan reviewee tidak saling mengetahui)
- Single blind review (reviewer mengetahui reviewee, sedangkan reviewee tidak mengetahui reviewer)
- Reviewer dan reviewee saling mengetahui
- Other

PROPOSAL SELECTION INDICATORS

In your opinion, how important are the criteria below as an assessment component in the proposal selection process (both master and project proposals)?

1: very unimportant; 2: not important; 3: neutral; 4: important; 5: very important

23. Relevance of product targets to country needs/problems (integration proposal)/contribution to product targets (individual proposals)
Very unimportant 1 2 3 4 5 very important
24. Output contribution to academia
Very unimportant 1 2 3 4 5 very important
25. Contribution of output to the economy
Very unimportant 1 2 3 4 5 very important
26. Contribution of output to social life and government policy
Very unimportant 1 2 3 4 5 very important
27. Kontribusi output terhadap kualitas lingkungan hidup
Very unimportant 1 2 3 4 5 very important
28. Novelty
Very unimportant 1 2 3 4 5 very important
29. Research design (problem formulation, objectives, data collection procedures, analysis methods)
Very unimportant 1 2 3 4 5 very important
30. HR (Human Resource) track record on the proposed topic
Very unimportant 1 2 3 4 5 very important
31. HR education level
Very unimportant 1 2 3 4 5 very important
32. Potential for dissemination and commercialization of research results
Very unimportant 1 2 3 4 5 very important
33. Potential for collaboration between researchers (multi-institutional/multi-discipline)
Very unimportant 1 2 3 4 5 very important
34. Involvement of potential adopters (industry, government, community) from the start of the research
Very unimportant 1 2 3 4 5 very important
35. Involvement of potential users (industry, government, community) at the end of the research (when the product is (near) finished)
Very unimportant 1 2 3 4 5 very important
36. Budget (efficiency and effectiveness)
Very unimportant 1 2 3 4 5 very important

ISSUES AND INPUT SUGGESTIONS

37. In your opinion, how important is the transparency of the selection process and the criteria for acceptance of proposals?
Very unimportant 1 2 3 4 5 very important
38. In your opinion, what problems exist in the current PRN proposal selection process? (if none, write NA)

39. What are your suggestions for improving the PRN proposal selection process? (if none, write NA)

MONITORING AND EVALUATION

This section contains questions regarding the monitoring and evaluation stages of PRN research activities that have been passed both by internal monitoring & evaluation by member institutions, internal programs, and by BRIN.

In your opinion, how important are the following actors to be involved in the monitoring and evaluation process?

1: very unimportant; 2: not important; 3: neutral; 4: important; 5: very important

40. Program Manager

Very unimportant 1 2 3 4 5 very important

41. BRIN

Very unimportant 1 2 3 4 5 very important

42. LPDP

Very unimportant 1 2 3 4 5 very important

43. BAPPENAS

Very unimportant 1 2 3 4 5 very important

44. Independent committee/body specially formed for monitoring and evaluation

Very unimportant 1 2 3 4 5 very important

45. Experts/Academic experts

Very unimportant 1 2 3 4 5 very important

46. In your opinion, which other actors are very important to be involved in the monitoring and evaluation? (if none, please write NA)

MONITORING AND EVALUATION FREQUENCY

47. In your opinion, in one year, how many times should monitor and evaluation be carried out?

- ☐ 1 time at the end
- ☐ 2 times in the middle of the year and at the end
- ☐ More than 2 times

TRANSPARENCY IN MONITORING AND EVALUATION

48. In your opinion, how important is the transparency of the monitoring and evaluation process and results

Very unimportant 1 2 3 4 5 very important

INDICATORS FOR MONITORING AND EVALUATION

Based on the results of the literature reviews, research monitoring and evaluation indicators can be classified into inputs, processes, outputs and impacts.

INPUT INDICATORS

In your opinion, for the input component, how important are the following indicators in monitoring and evaluation?

1: very unimportant; 2: not important; 3: neutral; 4: important; 5: very important

49. Allocating researcher time for PRN

Very unimportant 1 2 3 4 5 very important

50. Infrastructure (eg laboratories, tools and materials, internet connection, journal access, database access)

Very unimportant 1 2 3 4 5 very important

51. Budget realization

Very unimportant 1 2 3 4 5 very important

52. Engagement of potential users (eg industry, government, community)

Very unimportant 1 2 3 4 5 very important

PROCESS INDICATORS

In your opinion, for the process component, how important are the following indicators in monitoring and evaluation?

1: very unimportant; 2: not important; 3: neutral; 4: important; 5: very important

53. The suitability of the research plan and implementation

Very unimportant 1 2 3 4 5 very important

54. Coordination process in the WP/WBS/program team

Very unimportant 1 2 3 4 5 very important

55. Communication with potential users (industry, government, community)

Very unimportant 1 2 3 4 5 very important

OUTPUT AND IMPACT

This section contains questions regarding the output and impact of a study.

OUTPUT

In your opinion, for the output component, how important are the following indicators in monitoring and evaluation?

1: very unimportant; 2: not important; 3: neutral; 4: important; 5: very important

56. Outputs that contribute to academia (journals, proceedings, scientific books, research databases)

Very unimportant 1 2 3 4 5 very important

57. Output that contributes to the economy (innovative products, processes, methods)

Very unimportant 1 2 3 4 5 very important

58. Outputs that contribute to society (policy recommendations, intervention methods)

Very unimportant 1 2 3 4 5 very important

59. Output contribution to environmental quality (environmental policy recommendations, new methods, adoption by society and industry)

Very unimportant 1 2 3 4 5 very important

IMPACT

The impact of the research is evaluated several years after the output is produced or after the project/program ends

60. In your opinion, how important is the evaluation of the impact of research after the end of the project/program? (up to 5 years after the end of the project/program)

Very unimportant 1 2 3 4 5 very important

In your opinion, for the impact component, how important are the indicators below in monitoring and evaluation?

61. Impact on scientific (citations, international research collaborations, obtaining international funding, world-class reputation, international networks)

Very unimportant 1 2 3 4 5 very important

62. Impact on the economy (royalties from intellectual property, adoption of output by industry, mass production by industry, economic growth)

Very unimportant 1 2 3 4 5 very important

63. Impact on society (recommendations are known by the government, adoption of recommendations by the government, research results are known to the community, adoption of new methods by the community, changes in society)

Very unimportant 1 2 3 4 5 very important

64. Impact on environmental quality (recommendations are known by the government, adoption of recommendations by the government, research results are known to the public and industry, changes in environmental quality)

Very unimportant 1 2 3 4 5 very important

ISSUE AND INPUT SUGGESTIONS

65. In your opinion, what are the problems found in the current monitoring and evaluation mechanism for PRN activities? (if none, write NA)

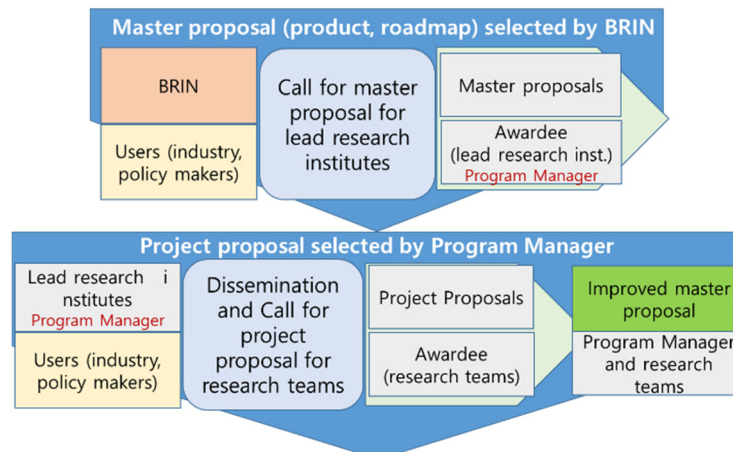
66. What are your suggestions for improving the monitoring and evaluation of PRN activities? (if none, write NA)

Appendix 4. Feedback Survey for the Proposed Research Evaluation System of PRN (Chapter 4)

I. Proposal Selection

Proposal Selection Structure

There are two stages of proposal selection: 1) selection of master proposals (proposal integration) by BRIN in collaboration with potential users. From this process, proposals and program managers (PM) will be selected. Next: 2) selection of individual proposals by the PM in collaboration with potential users. From this process, individual proposals and their research implementation team will be selected.



- What do you think about the structure of the proposal selection?
 Strongly disagree 1 2 3 4 5 agree
- Do you have any suggestions for the structure of the proposal selection? (if none, write NA)

Actors Involved in Proposal Selection

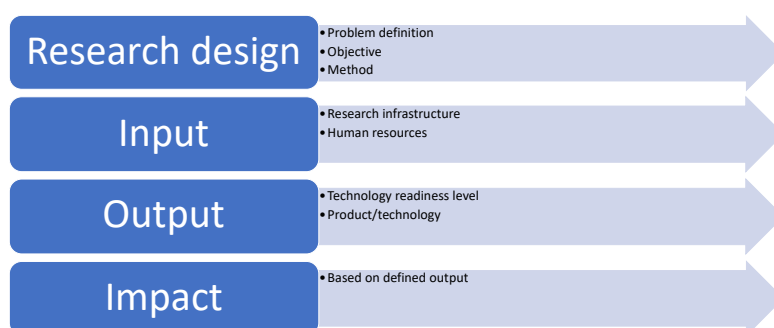
Master proposal selection	Project proposal selection
- Regulators: BRIN, Ministry of National development Planning (Bappenas) - Users: Industries, Policy Makers, society - Academic experts	- Program manager - Independent committee: Academics - Users: Industries, Policy Makers, society

3. What do you think about the actors involved in the selection of this proposal?

Strongly diasagree 1 2 3 4 5 agree

4. Do you have any suggestions for actors involved in proposal selection? (if none, write NA)

Indicator for Proposal Selection



5. What do you think about the indicators for the selection of this proposal?

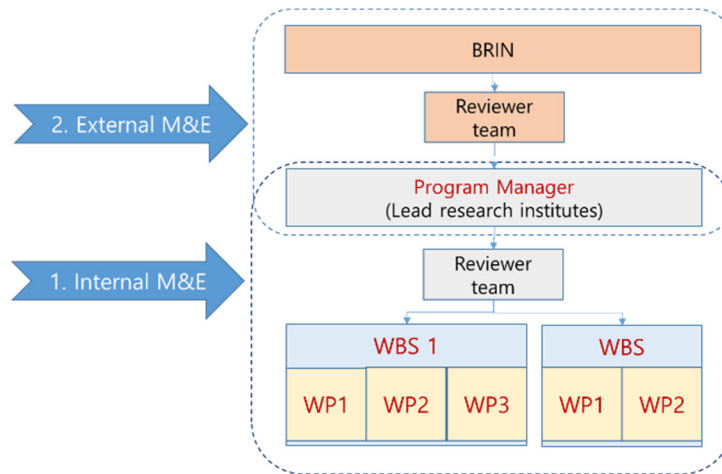
Strongly diasagree 1 2 3 4 5 agree

6. Do you have any suggestions for this proposal selection indicator? (if none, write NA)

II. Monitoring and Evaluation

Monitoring & Evaluation Structure

There are two stages of monitoring and evaluation: 1) monitoring and evaluation at the WBS/ WP level by the program manager (PM) (internal **monitoring & evalution**); and 2) monitoring and evaluation at the PM level by BRIN (external monitoring & evalution).



7. What do you think about the structure of this **monitoring & evaluation**?

Strongly disagree 1 2 3 4 5 agree

8. Do you have any suggestions for this monitoring & evaluation structure? (if none, write NA)

Actors Involved in Monitoring and Evaluation

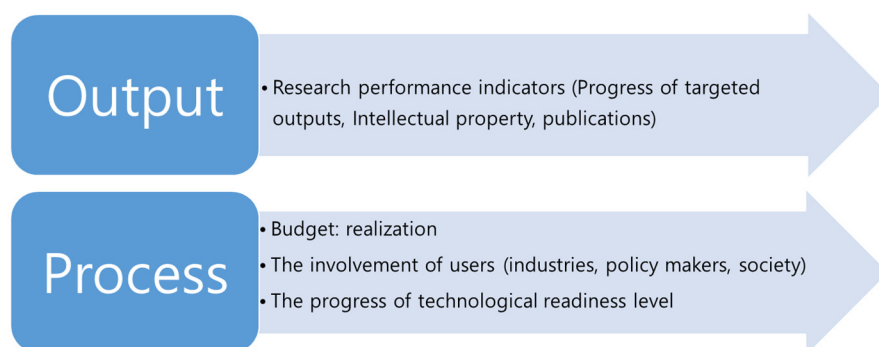
<u>Monev Internal:</u> Tim reviewer yang dibentuk oleh Manajer Program yang terdiri dari: 1. Akademisi 2. Pengguna (industri, pembuat kebijakan, publik) 3. Manajer Program 4. Petugas administrasi dari lembaga manajer program (terutama untuk keuangan)	<u>Monev Eksternal:</u> 1. BRIN (termasuk petugas administrasi) 2. Akademisi 3. Pengguna (industri, pembuat kebijakan, publik)
--	--

9. What do you think about the actors involved in this monitoring & evaluation

Strongly disagree 1 2 3 4 5 agree

10. Do you have any suggestions for actors involved in this monitoring & evaluation? (if none, write NA)

Monitoring and Evaluation Indicators



11. What do you think about this M&E indicator?

Strongly disagree 1 2 3 4 5 agree

12. Do you have any suggestions for these indicators for monitoring & evaluation? (if none, write NA)

Appendix 5. Advocacy Activities through Media Outlets

3.1. Kick-Off Workshop

■ ET News on April 15, 2021

전자신문

과기정책연, '인도네시아 국제기술협력사업' 착수세미나 개최

김영준 입력 2021.04.15. 18:42 수정 2021.04.16. 08:46



과학기술정책연구원(원장 문미옥)은 15일 '2021년도 한-인도네시아 국제기술협력사업 착수세미나'를 온라인으로 개최했다.

이번 착수세미나에서는 그간 협력 성과를 점검하고 금년도 사업의 추진방향 및 추진 체계 등을 모색하기 위해 양국의 과학기술혁신정책 전문가 70여 명이 참석하여 열띤 토론을 가졌다.

한-인도네시아 국제기술협력사업은 과기정책연이 2018년부터 과학기술 ODA 사업의 일환으로 인도네시아 정부 및 과학기술혁신 연구기관의 정책 수립과 실행역량강화를 지원하는 사업이다.

올해로 4년차를 맞이한 이 사업은 인도네시아의 국가과학기술기본법 제정을 지원한 성과를 냈으며, 금년에는 △임무지향형 과학기술혁신 연구기관 관련 법·제도 개선 방안 도출 △중기 R&D 정책 평가시스템 설계와 이행 평가 실시 △과학기술 관련 법 규정 개편 방안을 도출하기 위한 정책 컨설팅, 공동연구, 초청 역량강화 워크숍 등을 추진할 예정이다.

문미옥 과기정책연구원장의 개회사와 두디 히다얏 인도네시아국립과학원 부원장 대리 축사로 시작된 이번 착수세미나에서는 임덕순 과기정책연 선임연구위원이 '2021년도의 인도네시아 국제기술협력사업의 사업 추진 계획 및 방법'을, 권기석 한발대 교수가 '인도네시아의 과학기술혁신 싱크탱크 관련 법·제도 개선을 위한 자문 계획'을 각각 발표했다.

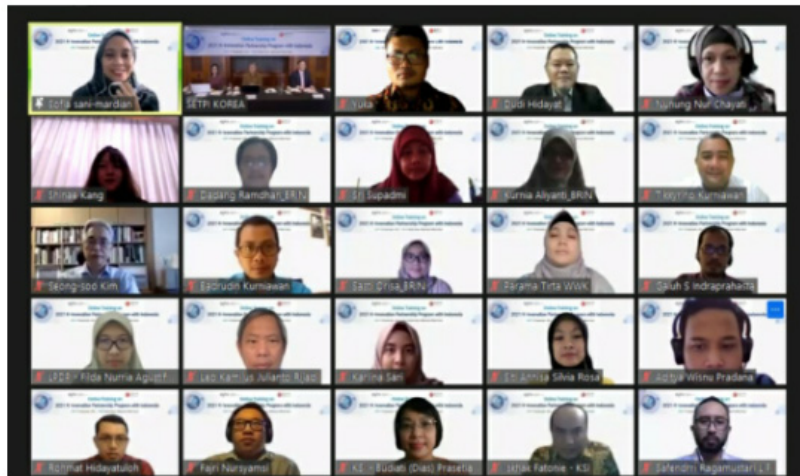
3.2 Online Training Workshop

■ BRIN Website on September 8, 2021



HIGHLIGHT ON 8 SEP 2021 1669

Kerja Sama Training of Trainer (ToT) Kebijakan 'Iptekin' Indonesia dan Korea Selatan 2021



SIARAN PERS

BADAN RISET DAN INOVASI NASIONAL

Nomor : 119/SP/HM/BPUK/IX/2021

Jakarta – Kebijakan ilmu pengetahuan-teknologi dan inovasi (iptekin) telah menjadi salah satu faktor kunci dalam mendorong dan mengakselerasi pertumbuhan sosial dan ekonomi di banyak negara. Korea Selatan merupakan negara dengan kemajuan perekonomian cukup cepat di dunia yang muncul dari belahan dunia timur. Belajar dari praktik kebijakan iptekin di Korea Selatan, Indonesia melalui Badan Riset dan Inovasi Nasional (BRIN) pada 2021 melanjutkan kerja sama *Science and Technology Policy Institute (STEPI) – Korea Selatan* dalam kegiatan *training of trainer (ToT)* di bidang kebijakan iptekin.

STEPI, 인도네시아에 'K-과학기술혁신정책' 알려

최아름 기자 | 승인 2021.09.13 11:24

| 2021년도 인도네시아 국제기술협력사업 초청연수 성료



임덕순 과기정책연 선임연구위원이 7일 열린 2021년도 한-인도네시아 국제기술협력사업 초청 연수에서 인사말을 하고 있다. [사진=STEPI]

[정보통신신문=최아름기자]

한국의 과학기술 발전 과정과 경험을 토대로 현재 국가혁신시스템 거버넌스를 새롭게 구축 중인 인도네시아 정부의 과학기술혁신정책 역량강화를 위한 온라인연수가 마무리됐다.

과학기술정책연구원(STEPI)은 지난 9월 7일부터 8일 양일간, 인도네시아 과학기술혁신 관계부처 공무원 및 전문가 35명을 대상으로 「2021년도 한-인도네시아 국제기술협력사업」초청 연수를 온라인 세미나 형식으로 개최했다.

인도네시아 과학기술정책 역량 강화를 위해 진행된 이번 초청연수는 △한국의 STI 주요 정책 및 정책 방법론 △싱크탱크 기관의 역할과 거버넌스 △국가 연구개발 프로그램 평가 시스템 △과학기술 관련 법·제도 등 다양한 프로그램이 진행됐다.

특히, 과학기술혁신 거버넌스(조정 메커니즘, 법제도 및 금융시스템) 관련 한국의 경험을 공유함으로써 협력국인 인도네시아 정책 기획과 실행 역량 강화 및 과학기술 경쟁력 제고를 목표로 진행됐다.

3.3 STEPI Special Session at the 2021 ASIP Conference

■ Herald Corporation on October14, 2021



아시아기술혁신학회 컨퍼런스에서 문미옥 과학기술정책연구원장이 기초연설을 하고 있다.[STEPI 제공]

[헤럴드경제=구본혁 기자] 과학기술정책연구원(STEPI)은 지난 7~9일 제주대학교에서 열린 '아시아기술혁신학회의 컨퍼런스'에서 '인도네시아 STI 시스템 강화'를 주제로 특별세션 진행을 통해 대한민국 성과를 아시아로 확산하는 장을 마련했다.

인도네시아 과학기술혁신 거버넌스 자문사업을 추진하고 있는 STEPI는 2년차를 맞은 현재 ▷한국의 과학기술정책체제의 진화와 싱크탱크의 역할 ▷인도네시아 국가연구개발 프로그램의 모니터링 및 평가시스템 구축 방안 ▷과학기술관련법·규정의 분석 및 개편 방안 등의 주제로 연구를 진행 중이다.

7일 컨퍼런스에서 문미옥 STEPI 원장은 '대한민국과 아시아의 변영'을 주제로 한 기초연설을 통해 산업발전의 변화와 과학기술혁신이 인간의 삶에 기여한 영향을 설명하며, 지속가능발전목표에서 과학기술혁신이 가지는 역할의 중요성을 강조했다.